

DSA0605- DATA HANDLING AND VISUALIZATION

Lab Experiments

Name: P Kavya
Reg no: 192424148

Set1:MonthlySalesData Dataset:

Monthly Sales

1. Create a **line chart** to visualize the monthly sales. Label the axes and title the chart appropriately.
2. Generate a **bar chart** to display the top-selling products for the year. Label the chart elements.
3. Develop a **scatter plot** to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.
4. Build an **interactive dashboard** combining the line chart and bar chart to allow users to explore sales data interactively.

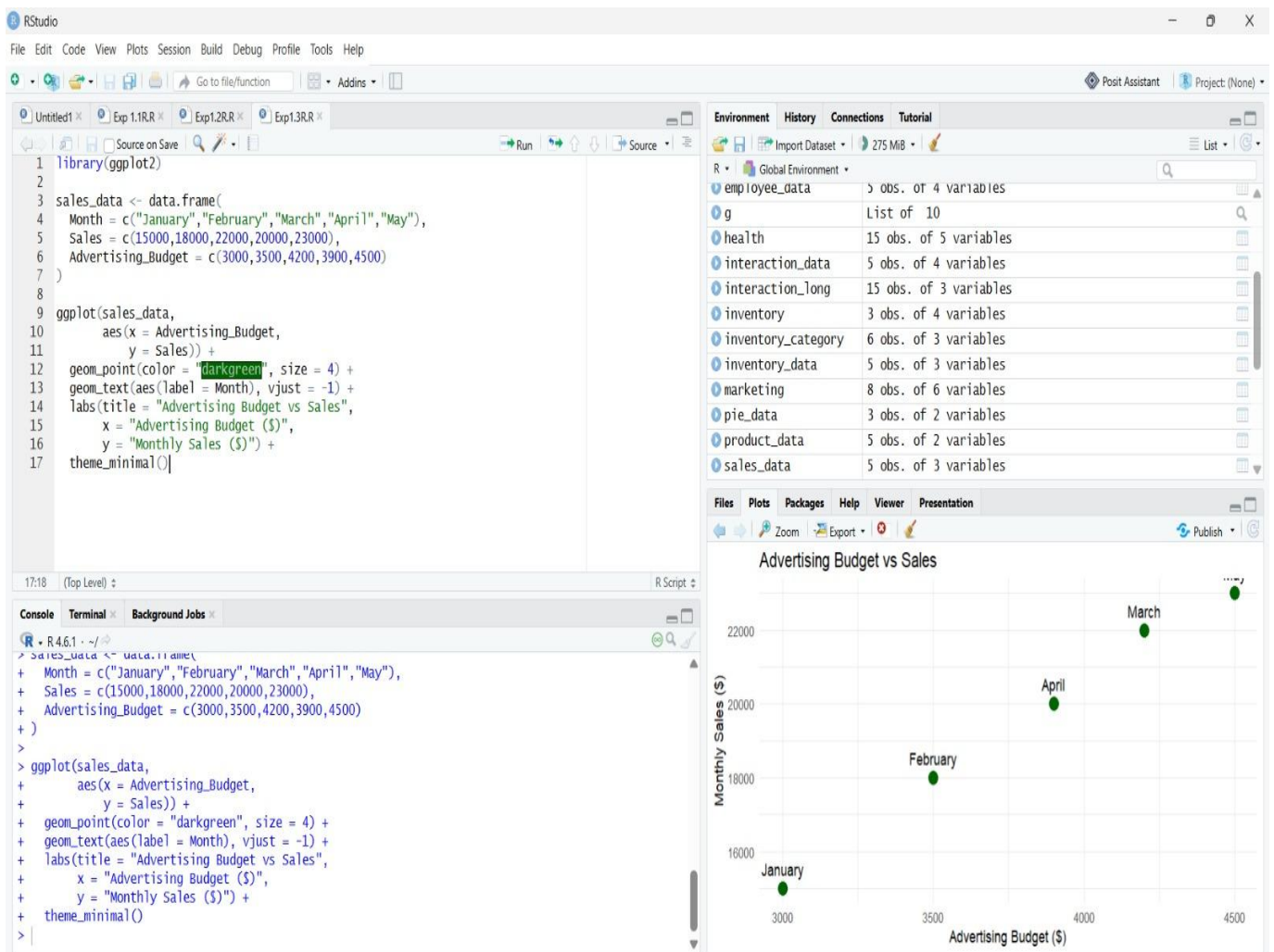
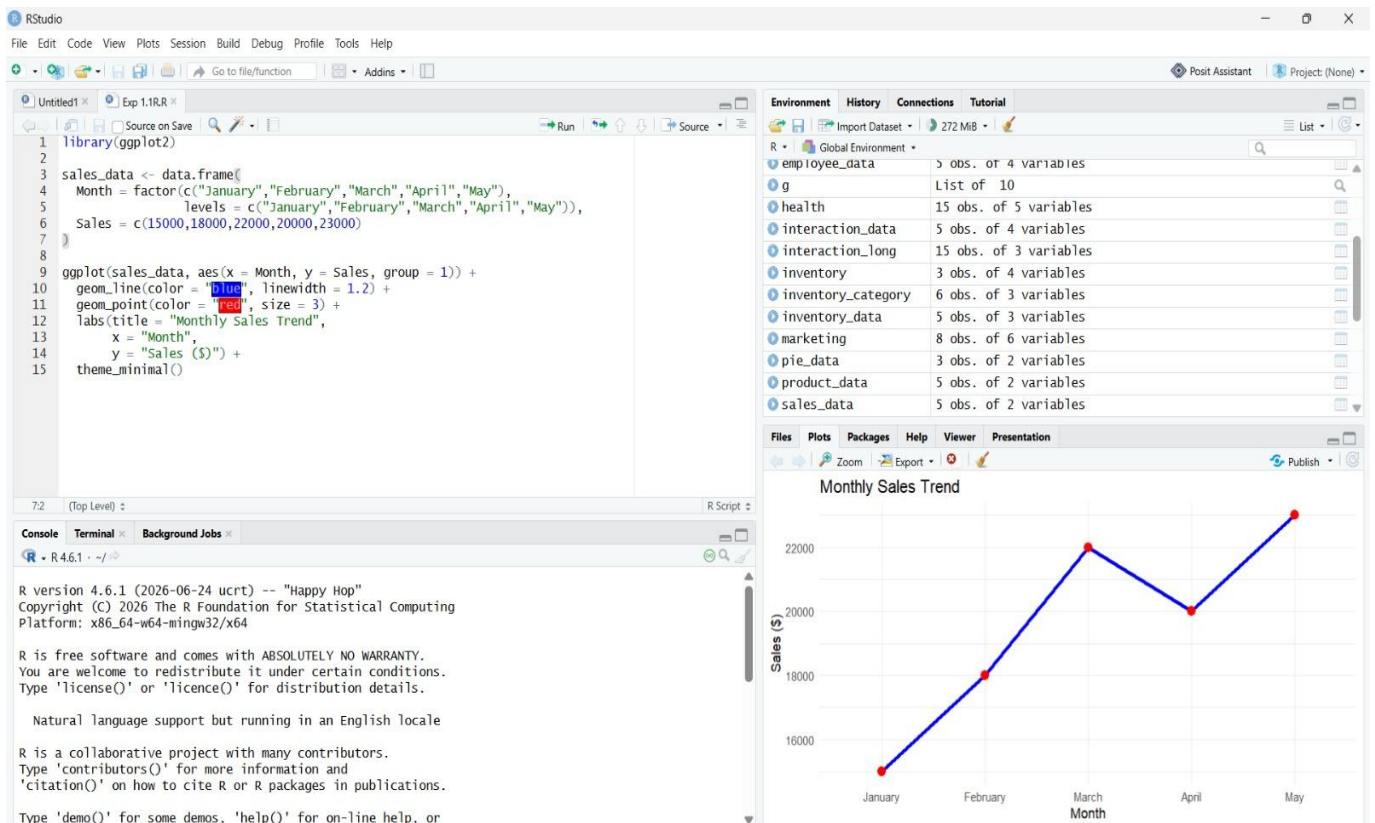
Month	Sales(in\$)
January	15,000
February	18,000
March	22,000
April	20,000
May	23,000

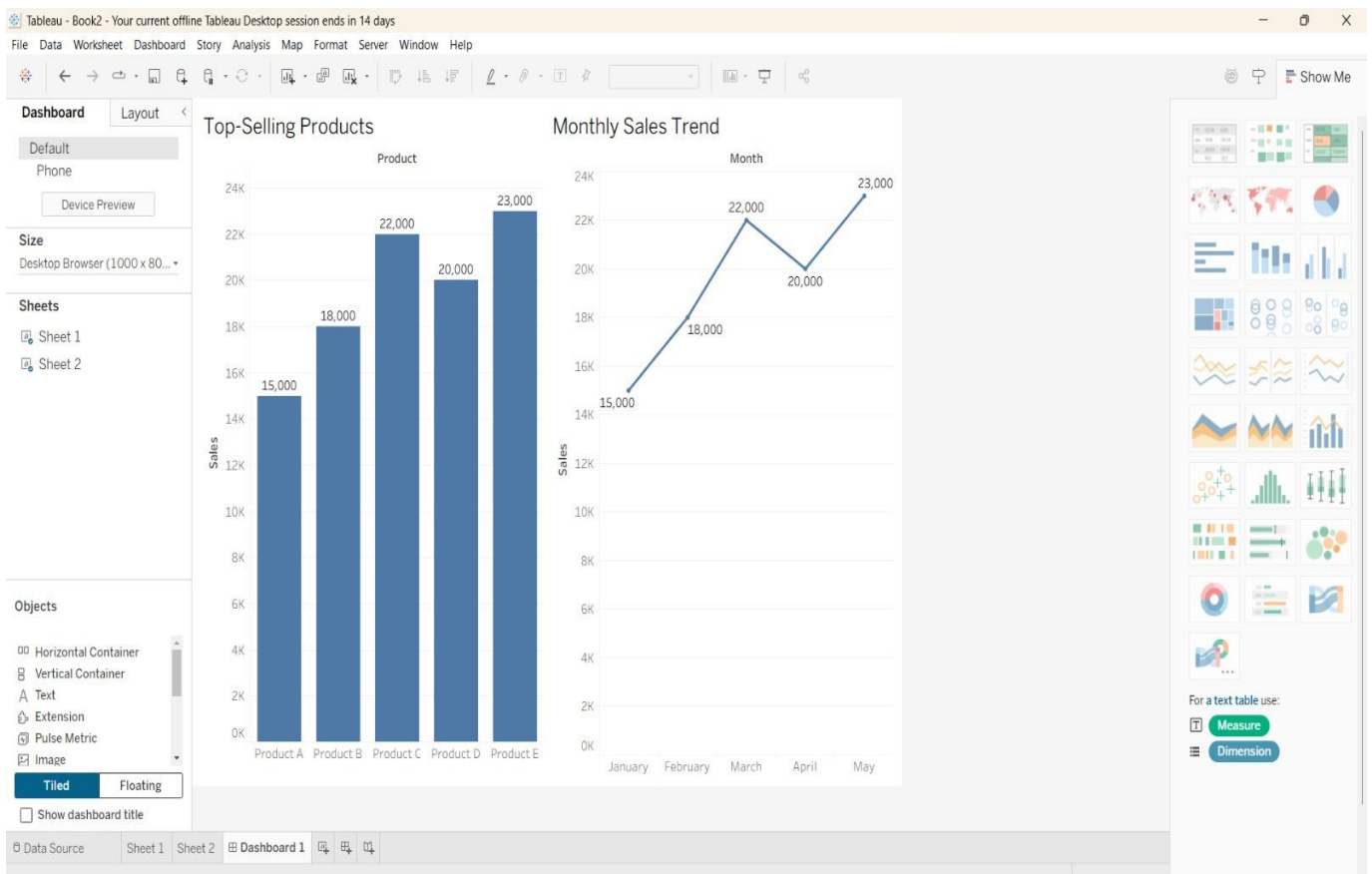
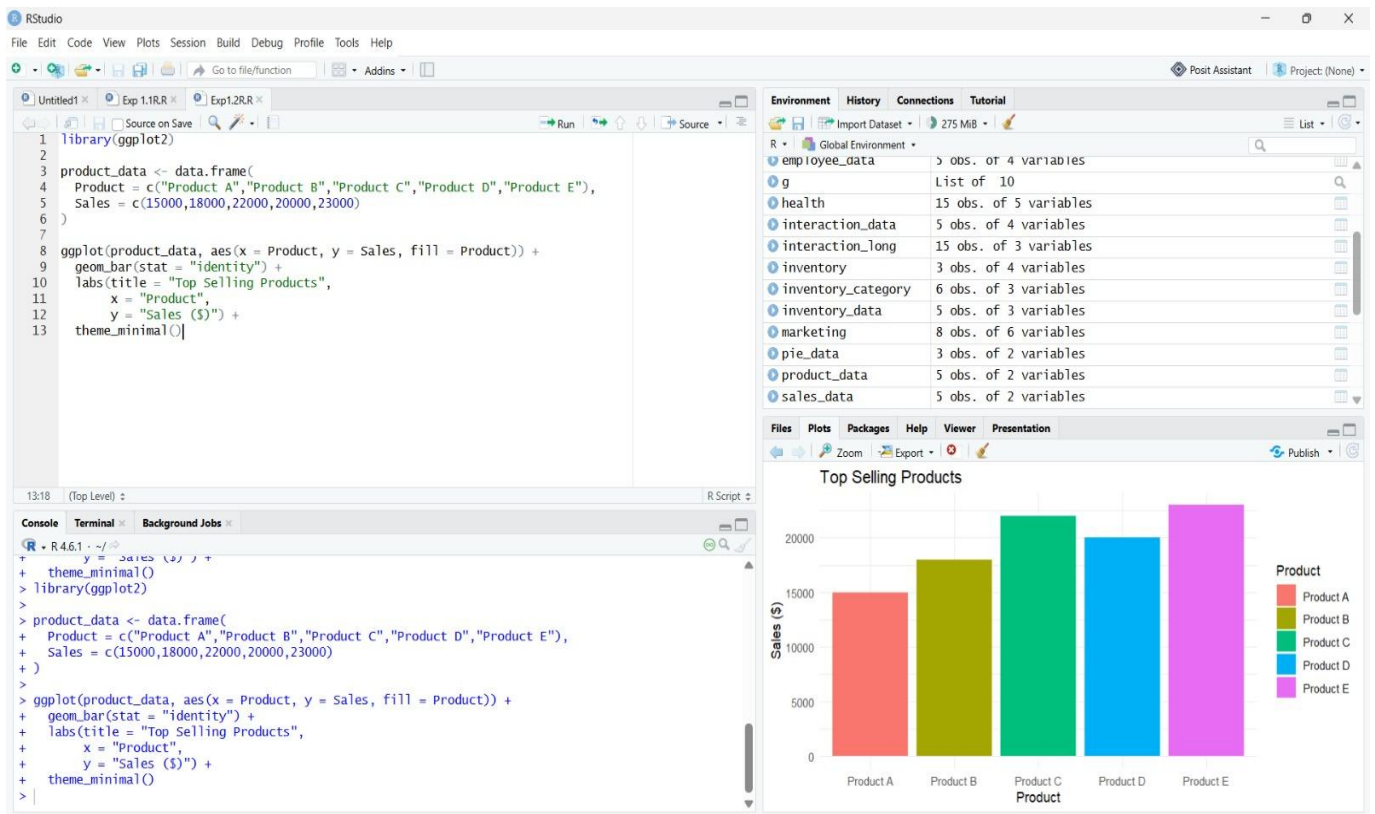
Aim:

To visualize monthly sales data using different charts and create an interactive Tableau dashboard for better analysis.

Algorithm:

1. Open Tableau and import the Monthly Sales dataset.
2. Create a Line Chart for monthly sales.
3. Create a Bar Chart for top-selling products.
4. Create a Scatter Plot for advertising budget vs. monthly sales.
5. Add titles, labels, and legends.
6. Combine the visualizations into an interactive dashboard.
7. Analyze the charts and save the workbook.





Result:

Successfully created the required visualizations and an interactive dashboard. The charts clearly displayed monthly sales trends, product performance, and the relationship between advertising budget and sales.

SET2.CustomerFeedbackAnalysis

Dataset: Customer Satisfaction

CustomerID	Age	SatisfactionScore
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

Tasks

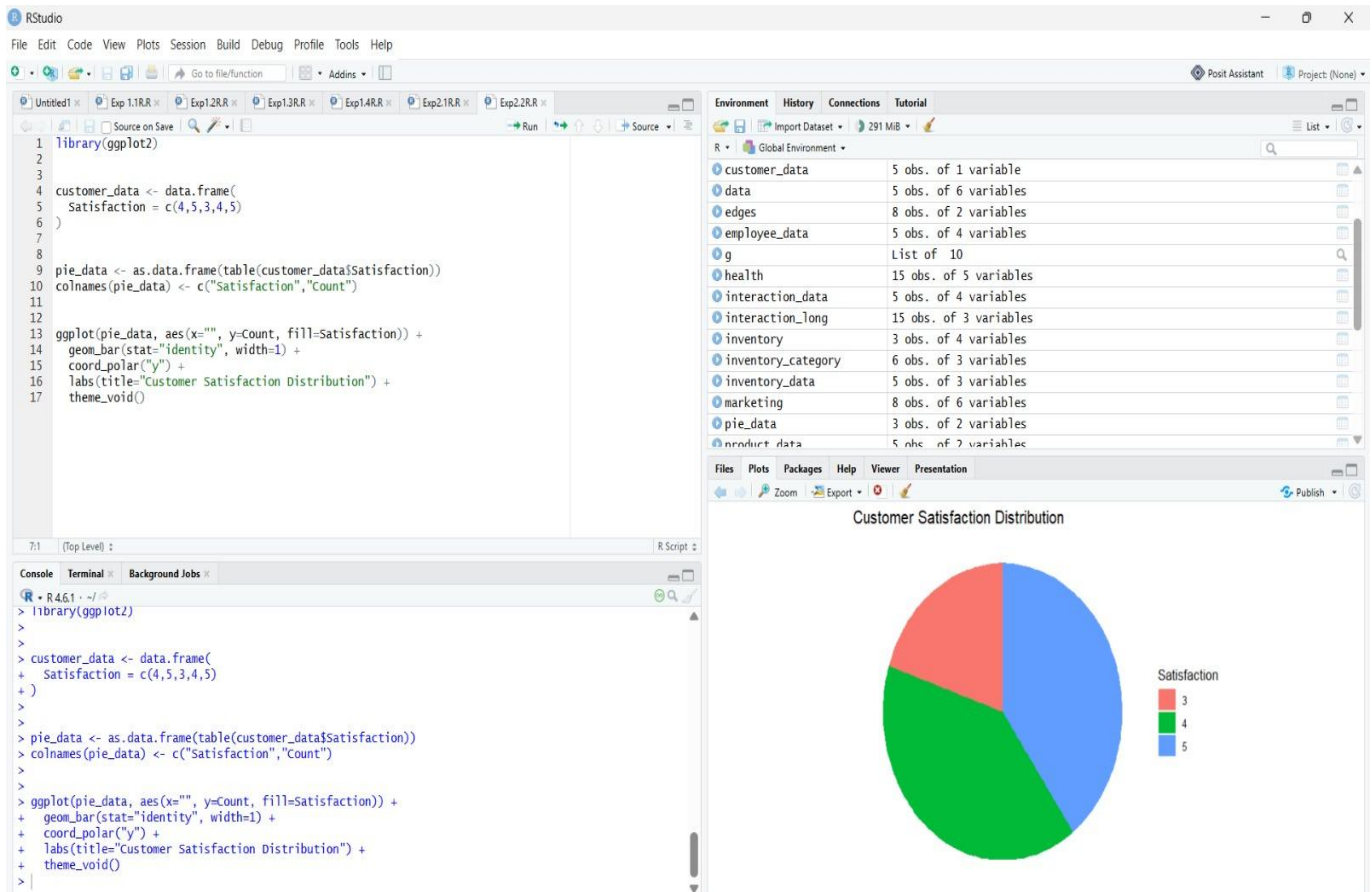
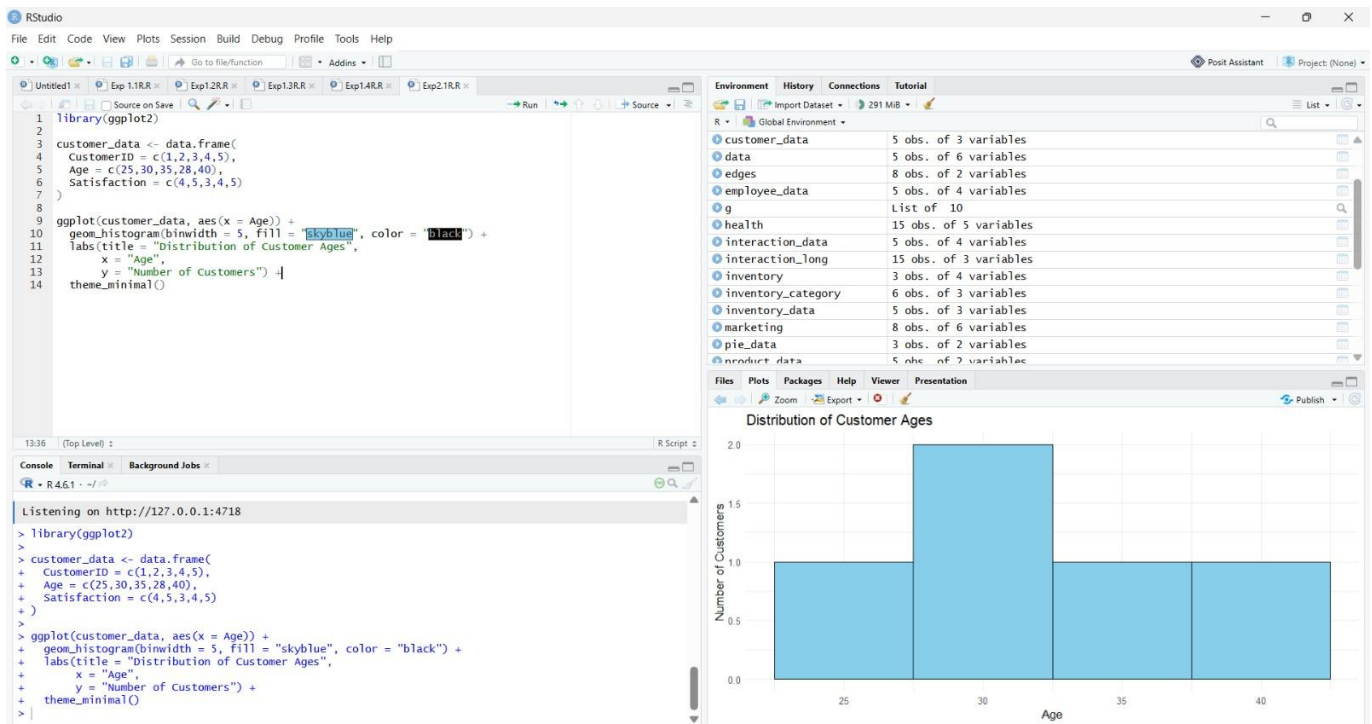
1. Create a **histogram** to represent the distribution of customer ages. Label the axes and title the chart.
2. Generate a **pie chart** to display the overall distribution of customer satisfaction scores. Include appropriate labels.
3. Build a **stacked bar chart** to visualize the distribution of customer satisfaction scores by age group.
4. Develop a **word cloud** from open-ended customer feedback to identify prevalent customer sentiments.

Aim:

To analyze customer satisfaction data using Tableau visualizations and identify customer trends.

Algorithm:

1. Import the Customer Satisfaction dataset.
2. Create a histogram for customer ages.
3. Create a pie chart for satisfaction scores.
4. Create a stacked bar chart for satisfaction by age group.
5. Generate a word cloud from customer feedback.
6. Format the charts and analyze the results.



Set 3: Employee Performance Evaluation

Dataset:EmployeePerformance

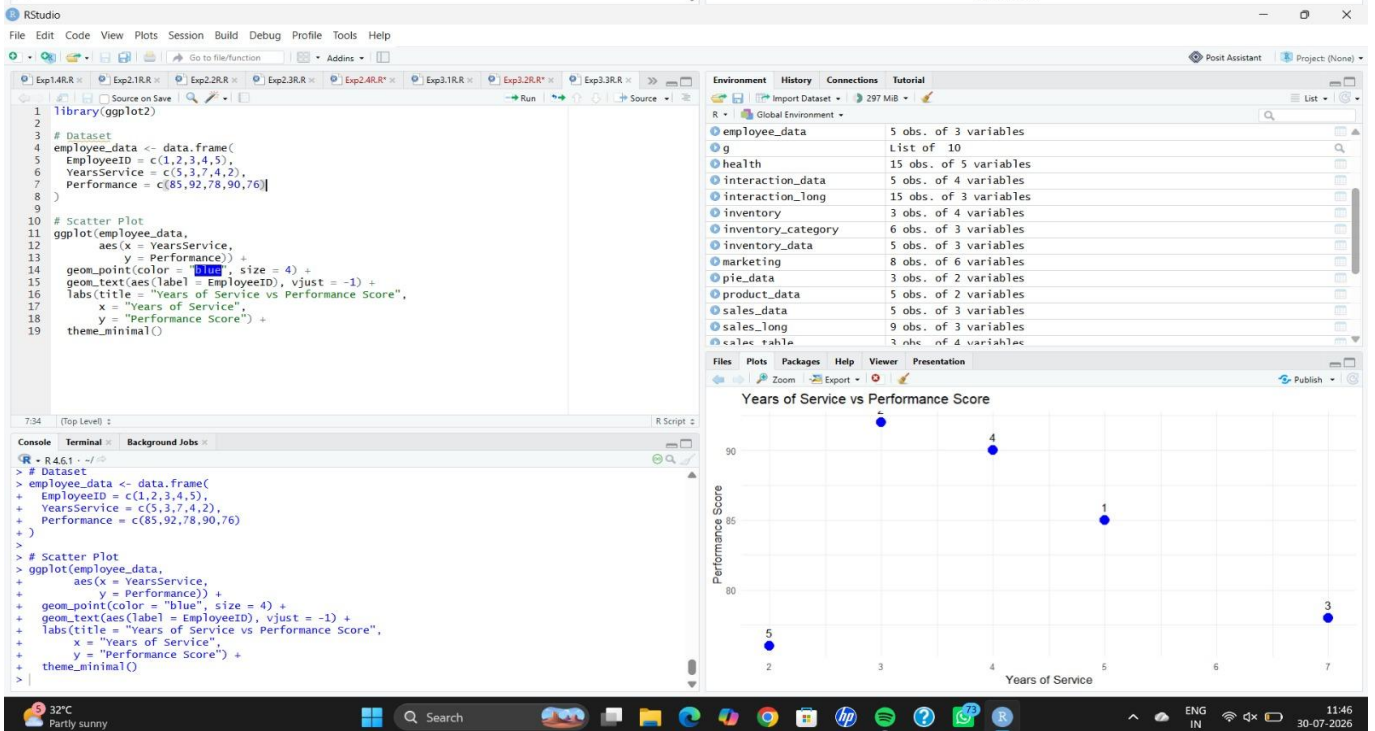
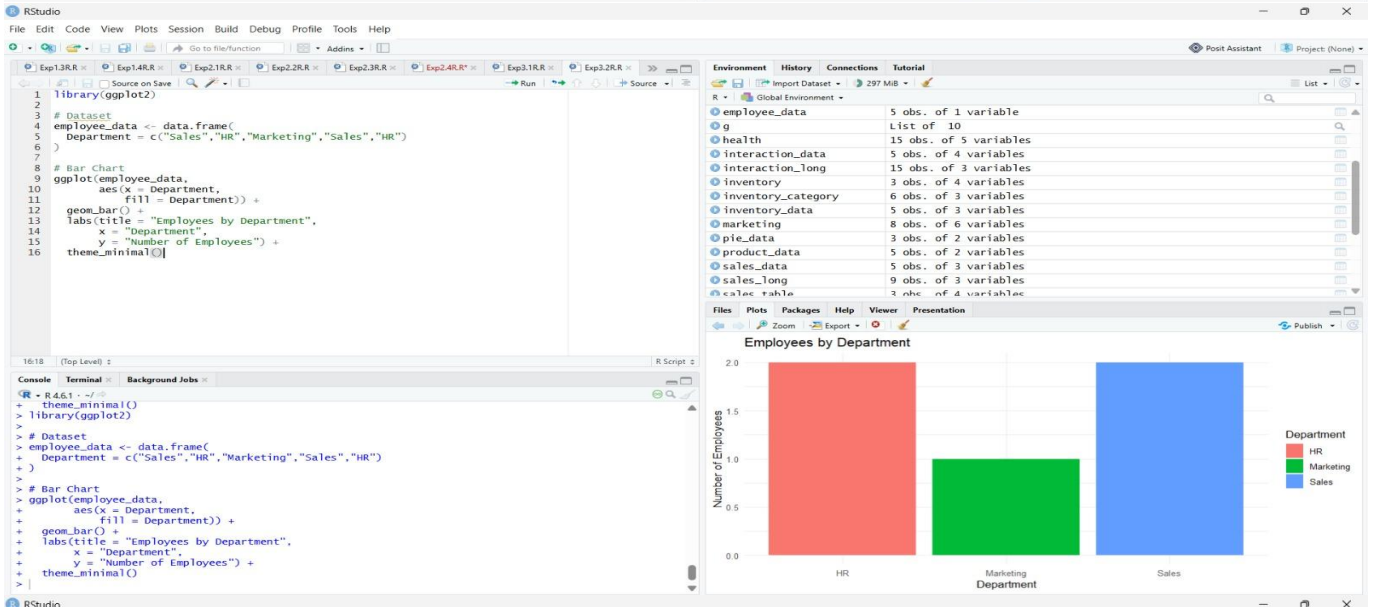
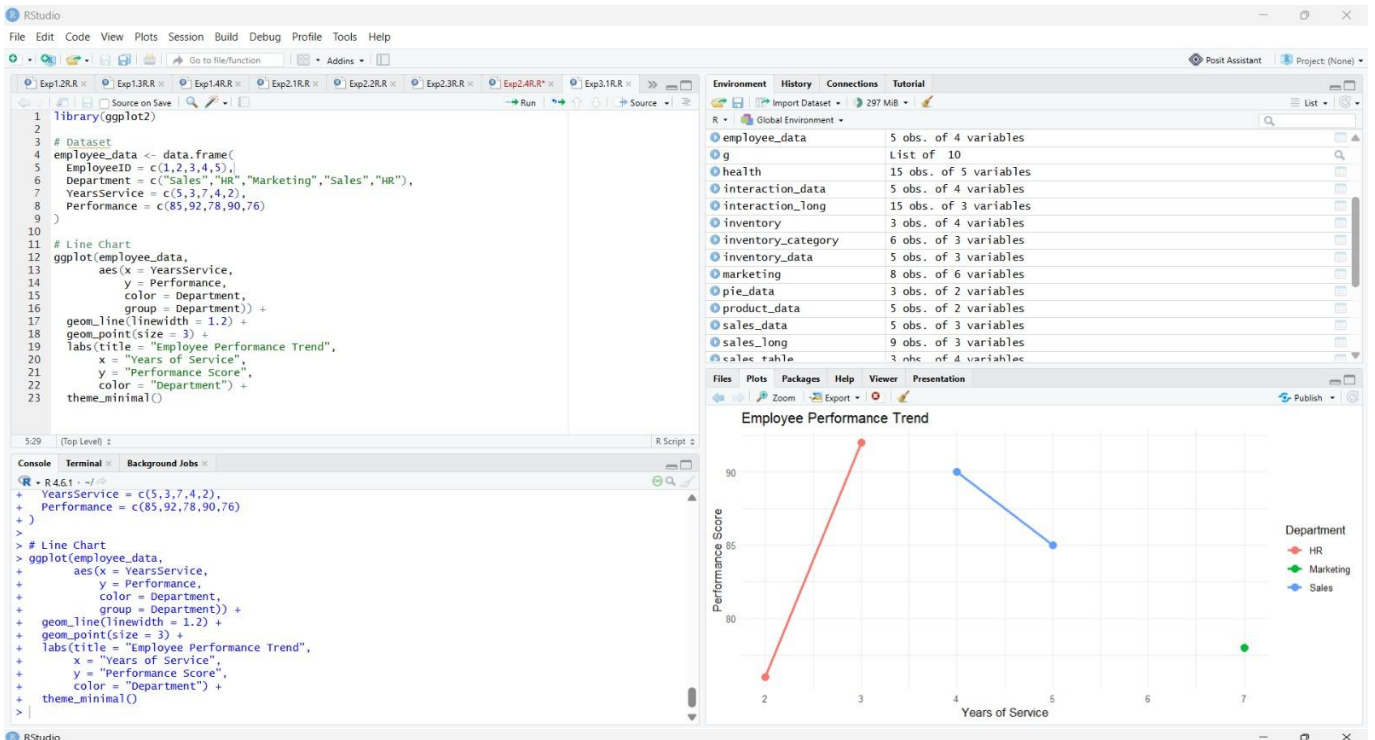
EmployeeID	Department	YearsofService	PerformanceScore
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

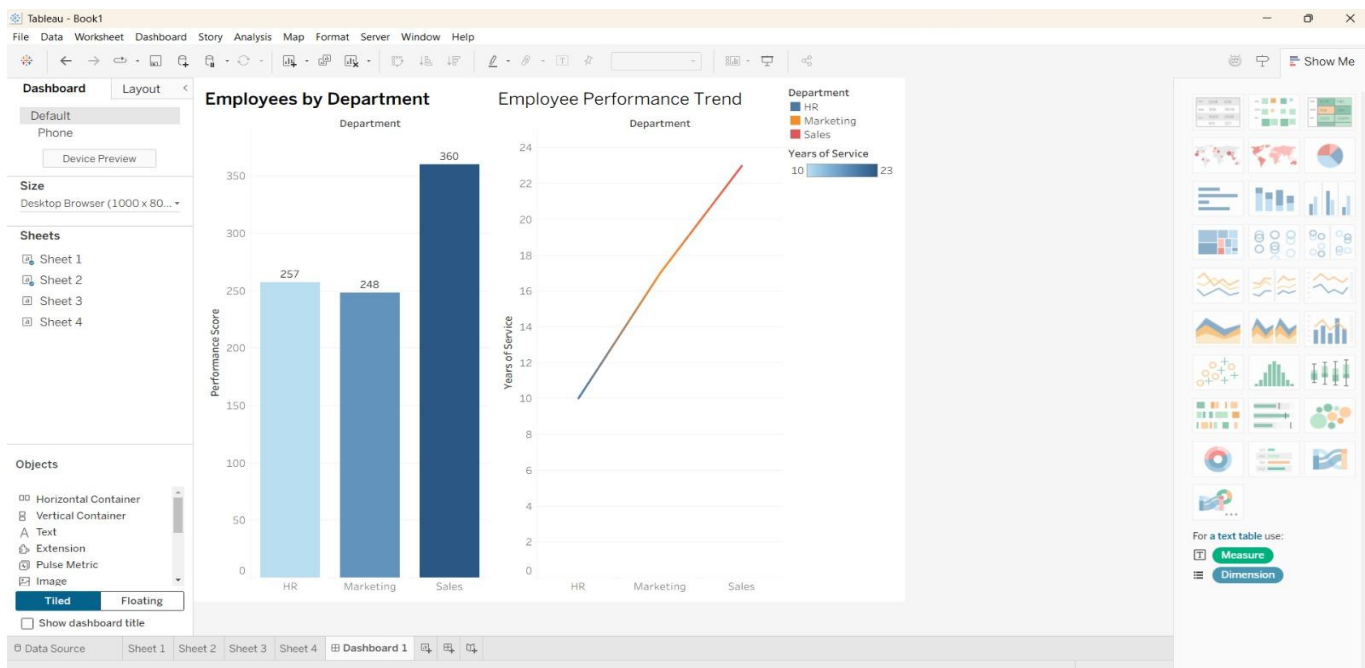
Tasks

1. Create a **line chart** to visualize the performance trend of employees over time. Include a legend and labels.
2. Generate a **bar chart** showing the distribution of employees across different departments. Label the chart elements.
3. Build a **scatter plot** to analyze the correlation between years of service and performance scores. Explain any insights.
4. Develop a **Tableau dashboard** with the line chart and bar chart for interactive exploration.

Aim:
To visualize employee performance data using Tableau by creating charts and an interactive dashboard for analyzing performance trends and departmental distribution.

- Algorithm:**
1. Open Tableau and import the Employee Performance dataset.
 2. Verify the imported fields and data types.
 3. Create a line chart to visualize employee performance trends.
 4. Generate a bar chart showing the distribution of employees across departments.
 5. Build a scatter plot to analyze the relationship between years of service and performance score.
 6. Add appropriate titles, legends, filters, and labels.
 7. Combine the line chart and bar chart into an interactive dashboard.
 8. Save and publish the Tableau workbook.





Result:

The required visualizations and dashboard were successfully created. The charts clearly displayed employee performance trends, department-wise distribution, and the relationship between years of service and performance scores.

Set4:ProductInventoryManagement

Dataset: Product Inventory

ProductID	ProductName	QuantityAvailable
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

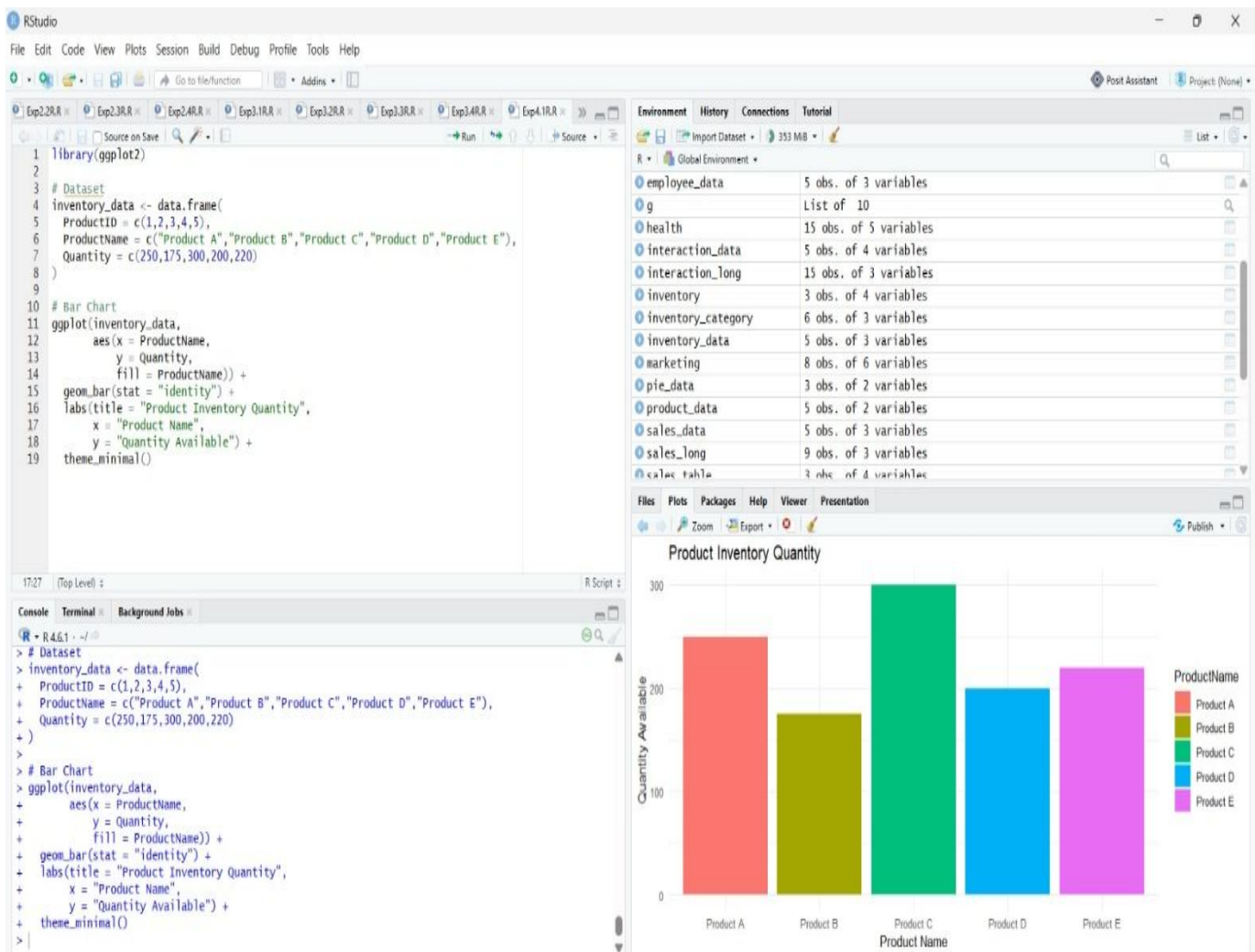
1. **Create a bar chart** to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. **Generate a stacked bar chart** to show the quantity of each product within different product categories.
3. **Build a scatter plot** to explore the relationship between product price and quantity available. Explain the findings.
4. **Develop a Tableau dashboard** with the bar chart and stacked bar chart to allow users to interact with the data.

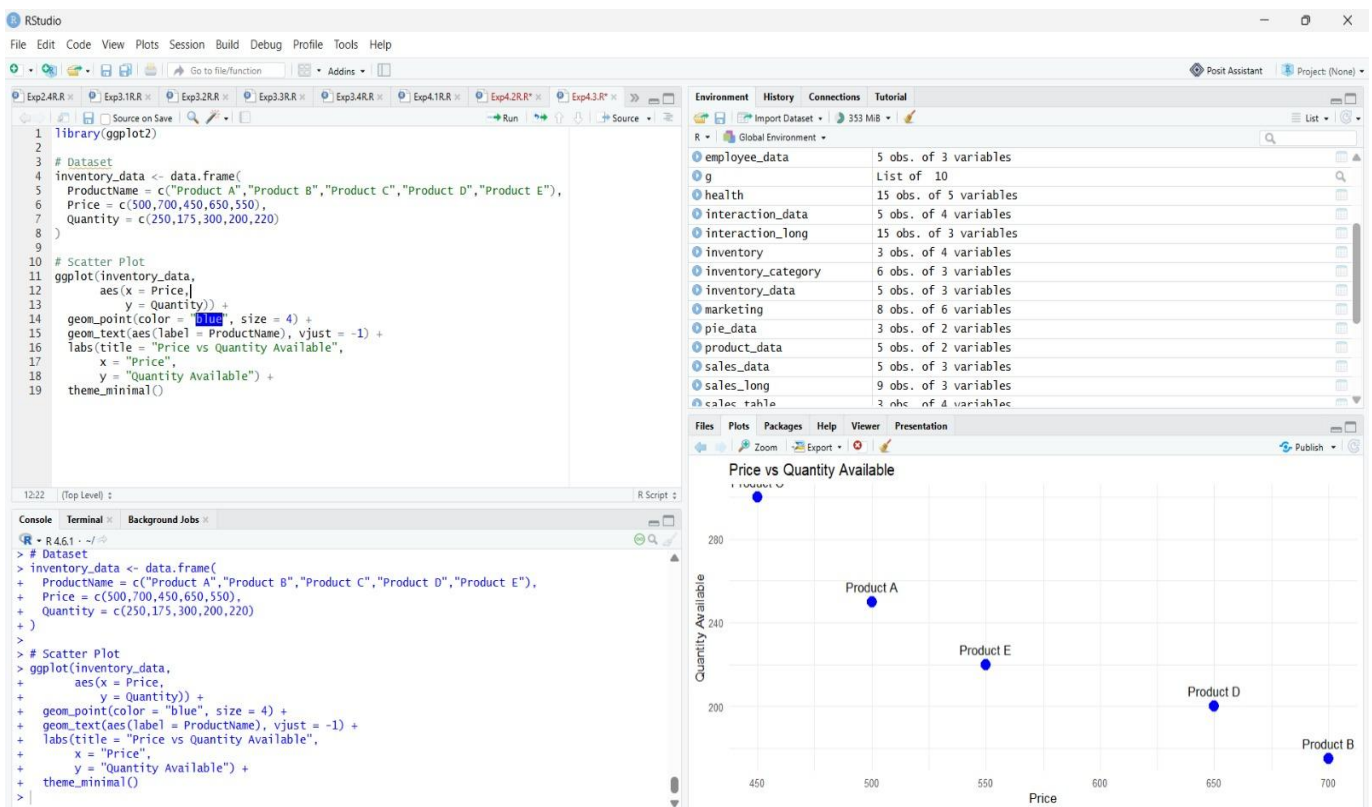
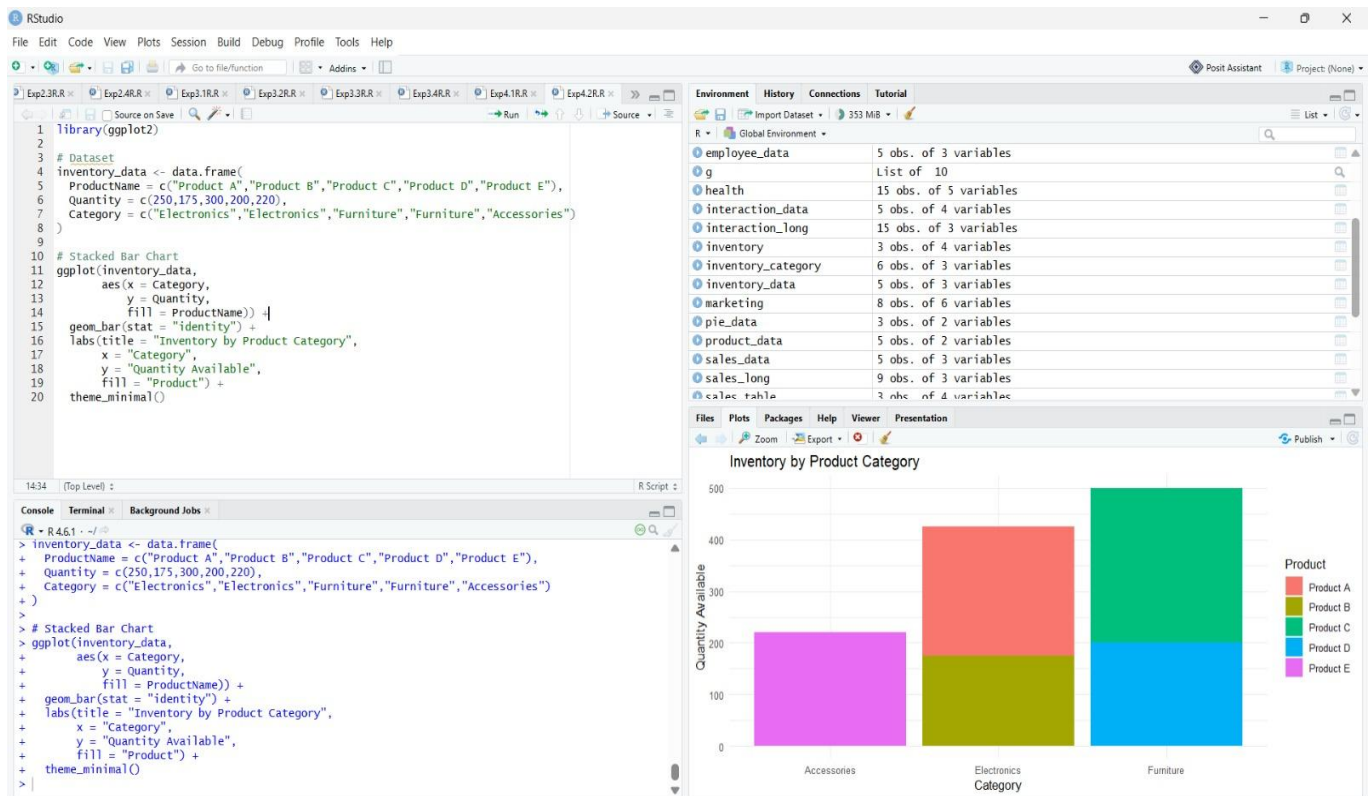
Aim:

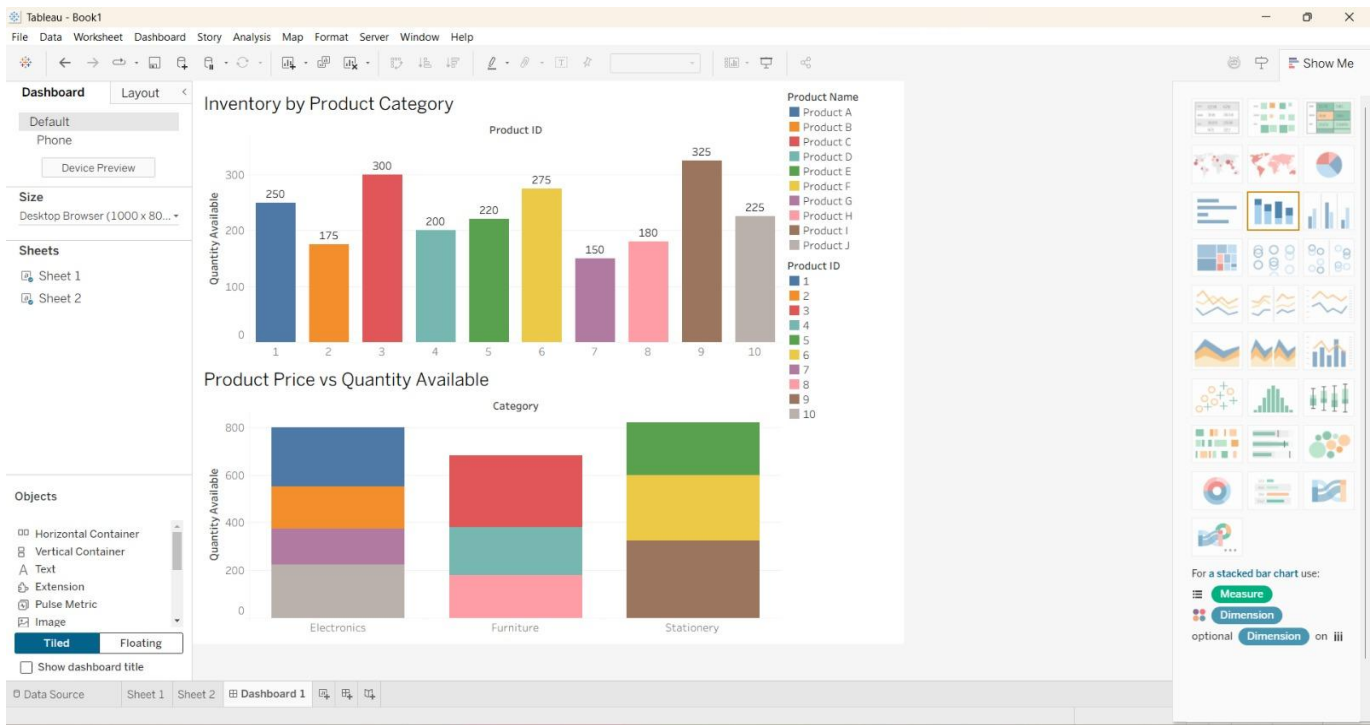
To analyze product inventory data using Tableau by creating visualizations and an interactive dashboard.

Algorithm:

1. Open Tableau and import the Product Inventory dataset.
2. Check and organize the imported data fields.
3. Create a bar chart showing the quantity available for each product.
4. Generate a stacked bar chart representing product quantities by category.
5. Build a scatter plot to analyze the relationship between product price and quantity available.
6. Add suitable titles, labels, and legends.
7. Combine the bar chart and stacked bar chart into an interactive dashboard.
8. Save the workbook.







Result:

The inventory data was successfully visualized. The dashboard effectively displayed product quantities, category-wise inventory, and the relationship between product price and available quantity.

Set5:WebsiteAnalytics

Dataset:WebsiteTraffic

Date	PageViews	Click-throughRate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

1. **Create a line chart** to visualize the trend in daily page views over time. Label the axes and title the chart.
2. **Generate a bar chart** showing the top N days with the highest click-through rates. Label the chart elements.
3. **Develop a stacked area chart** to display the distribution of user interactions (likes, shares, comments) on a website.
4. **Build a Tableau dashboard** with the line chart and bar chart for interactive exploration of website traffic data.

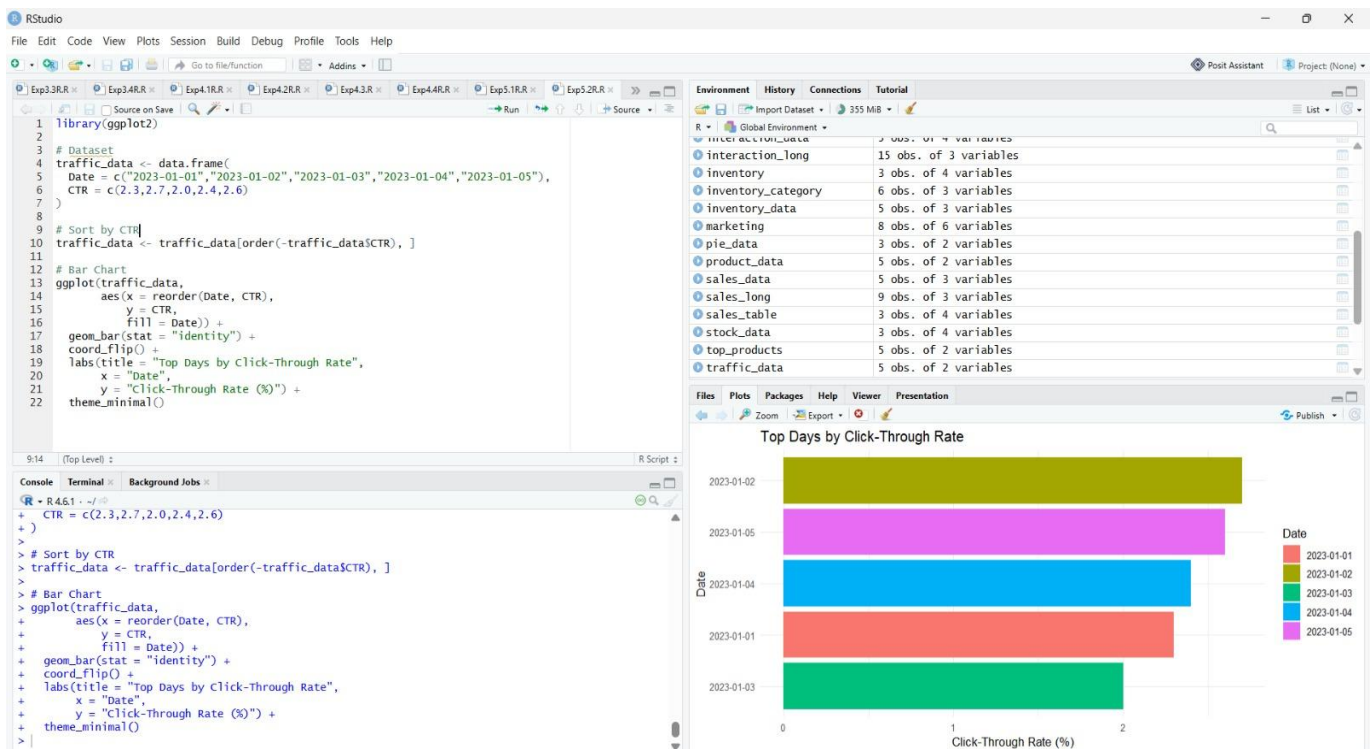
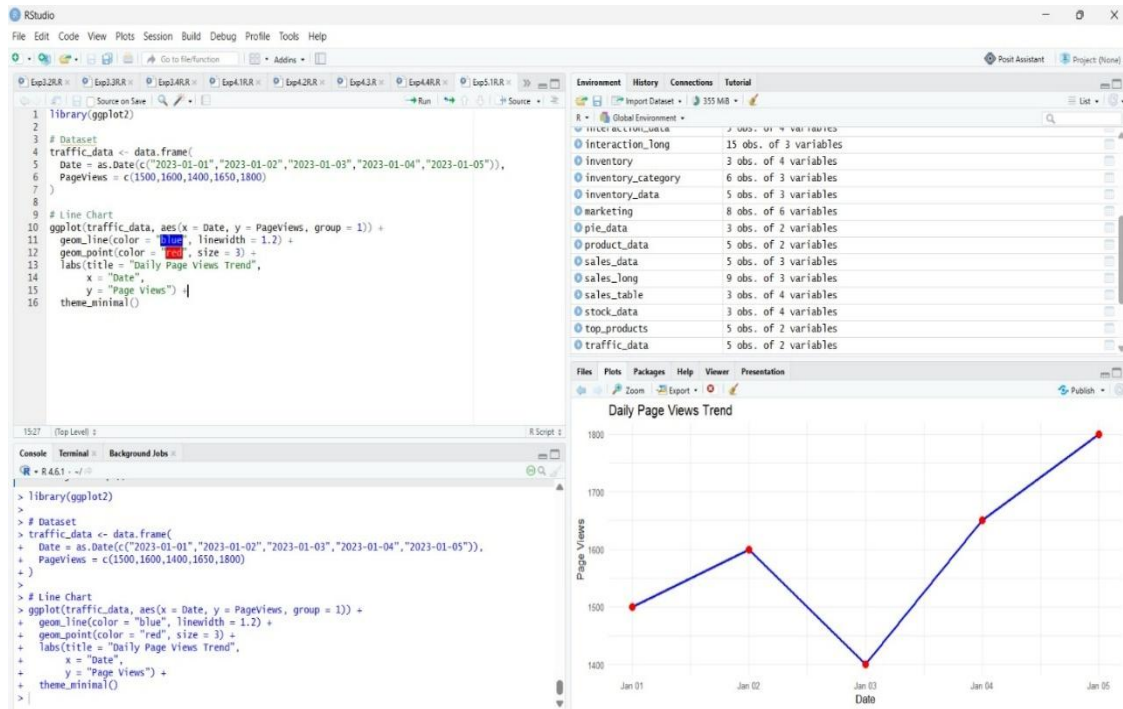
Aim:

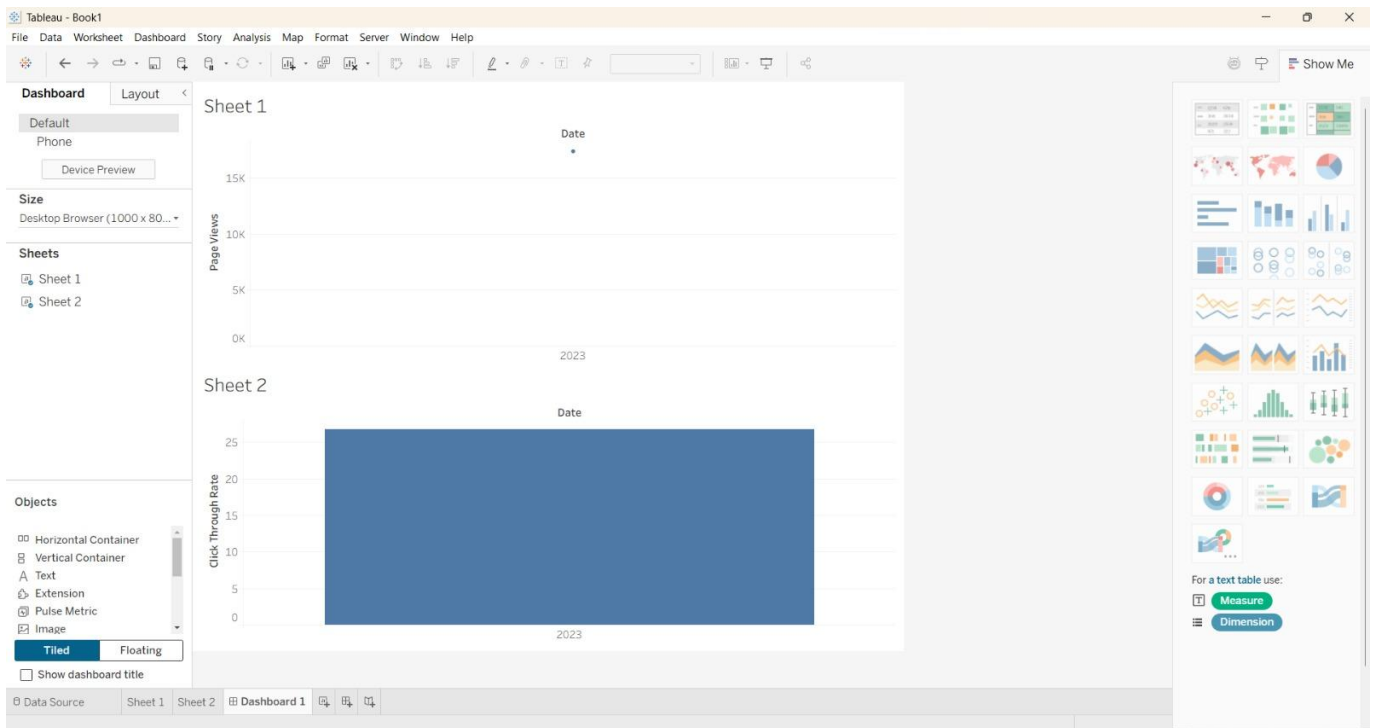
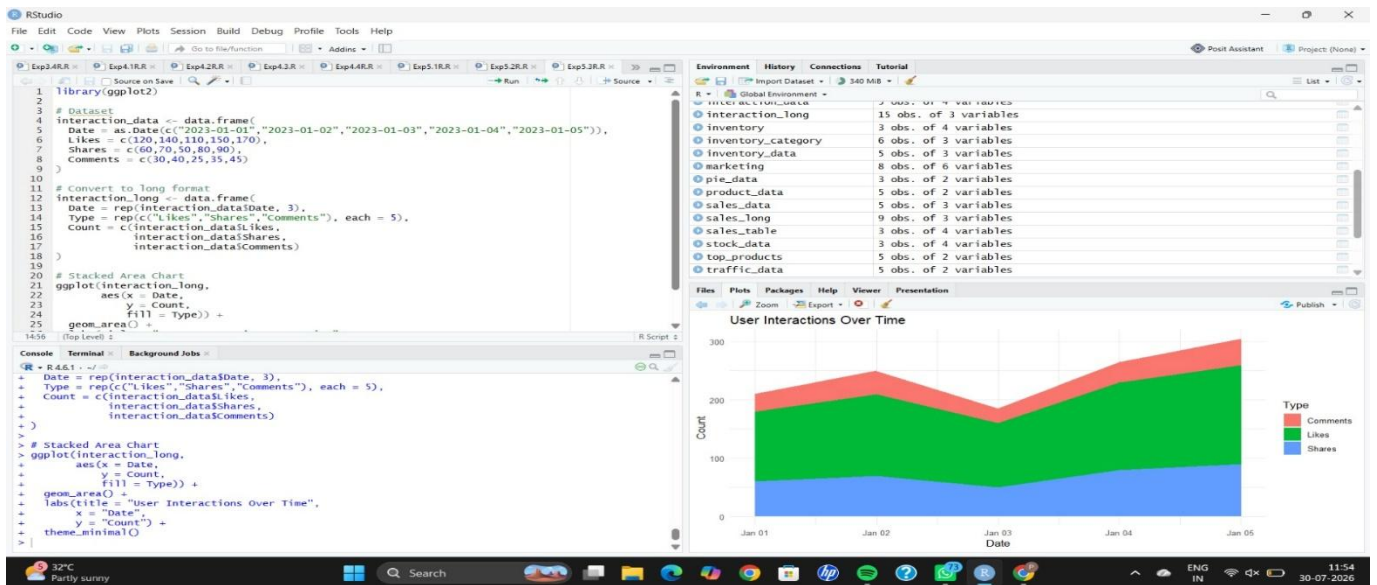
To visualize website traffic data using Tableau for analyzing page views and user interaction trends.

Algorithm:

1. Import the Website Traffic dataset into Tableau.

2. Verify the imported fields.
3. Create a line chart showing daily page views.
4. Generate a bar chart displaying the top days with the highest click-through rates.
5. Create a stacked area chart representing user interactions.
6. Add chart titles, labels, and legends.
7. Design an interactive dashboard containing the visualizations.
8. Save the Tableau workbook





Result:

The visualizations successfully represented website traffic trends, click-through rates, and user interactions through an interactive dashboard.

Set 6: Product Sales Analysis

Dataset: MonthlyProductSales

ProductID	ProductName	JanuarySales	FebruarySales	MarchSales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600

3	Product C	1200	1400	1100
---	-----------	------	------	------

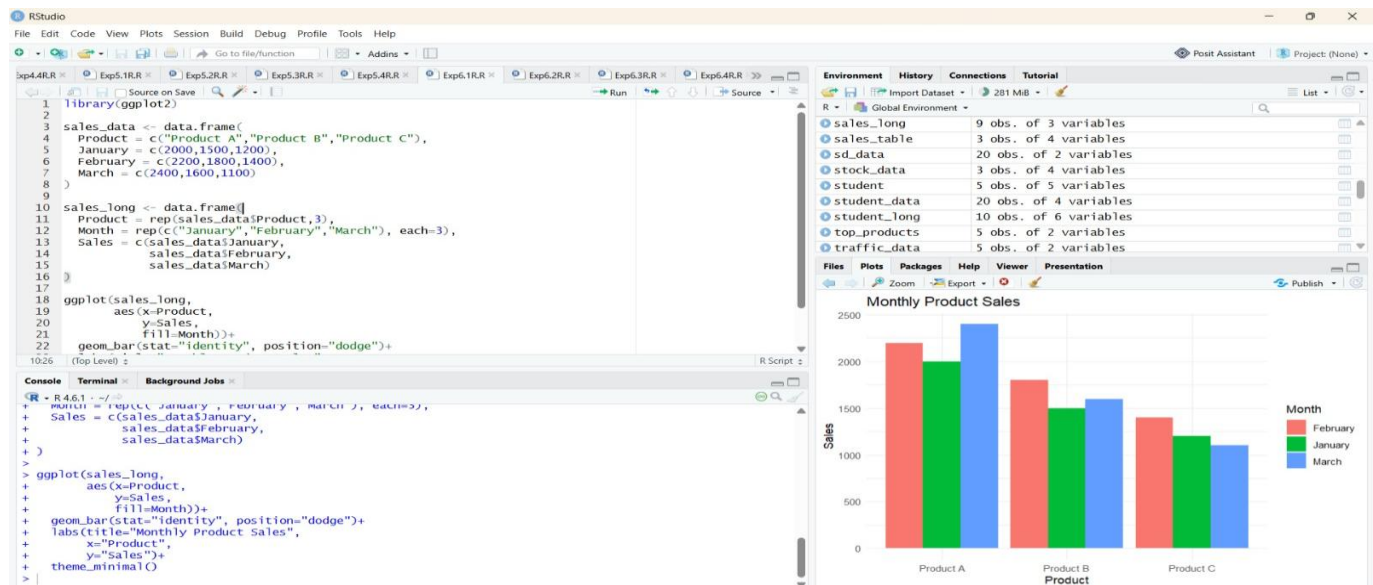
1. **Create a grouped bar chart** to visualize the sales of each product for the first quarter. Label the chart elements.
2. **Generate a stacked area chart** to represent the overall sales trend for all products over the three months.
3. **Build a table** to show the monthly sales data for each product. Label the table elements.
4. **Develop a Tableau dashboard** combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

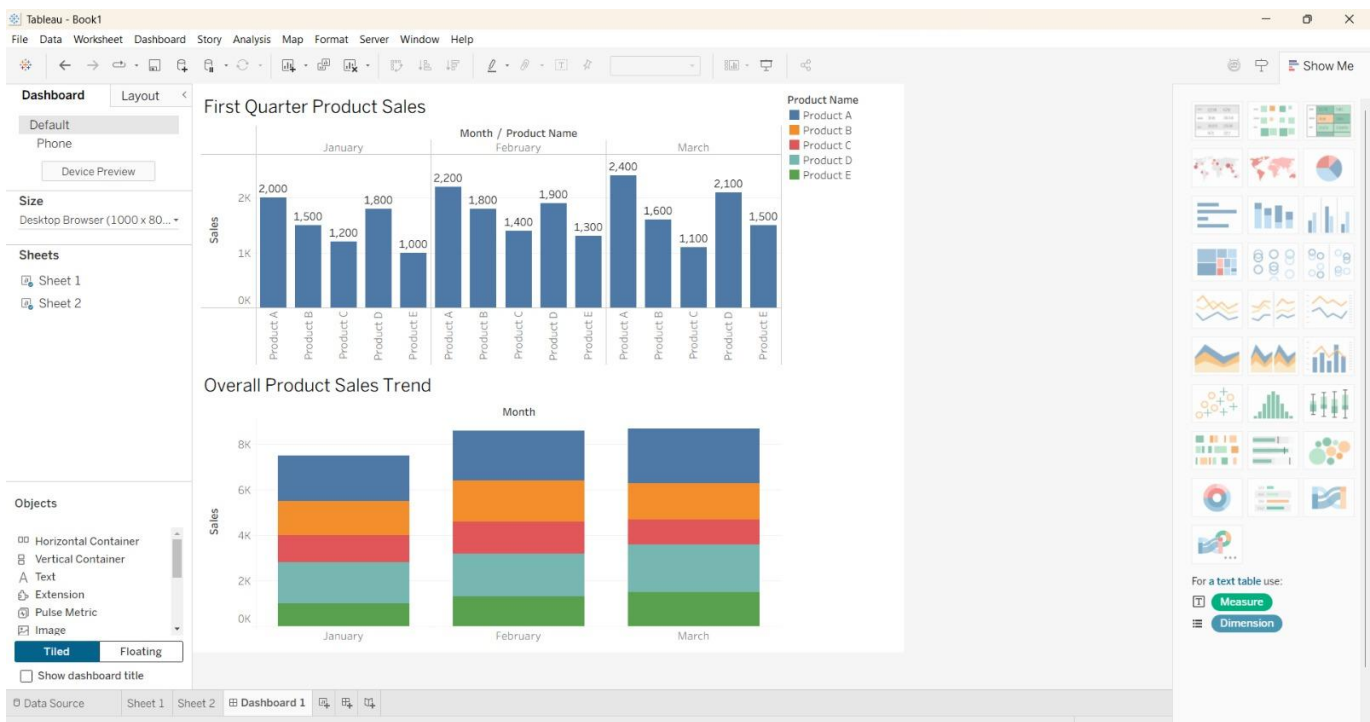
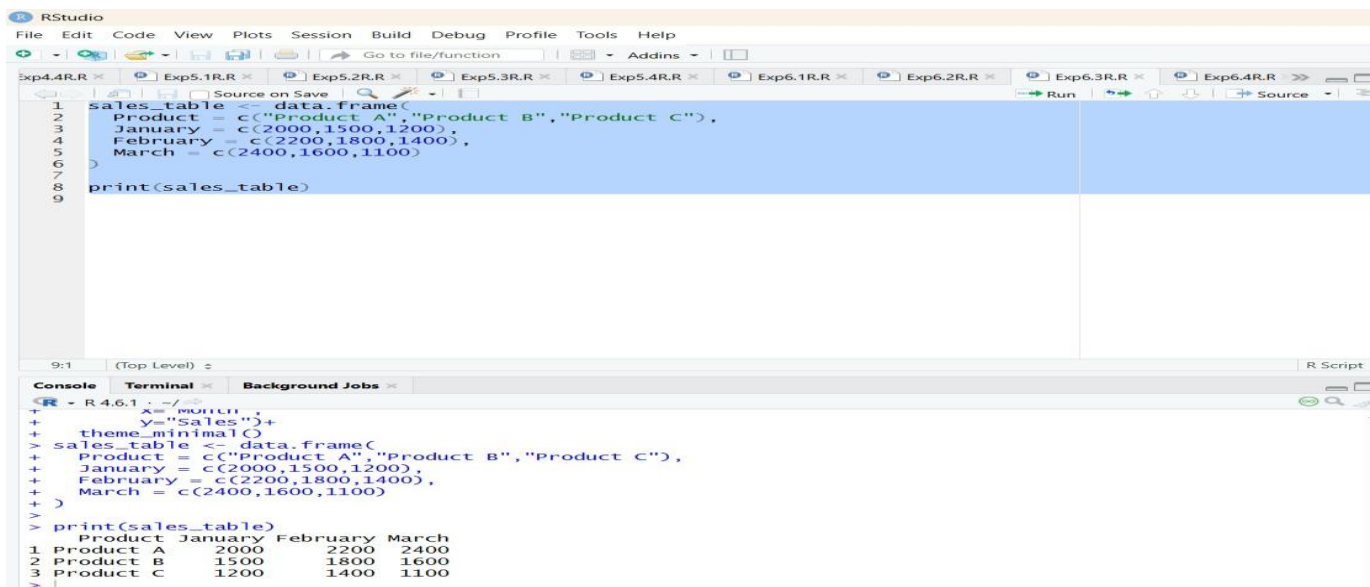
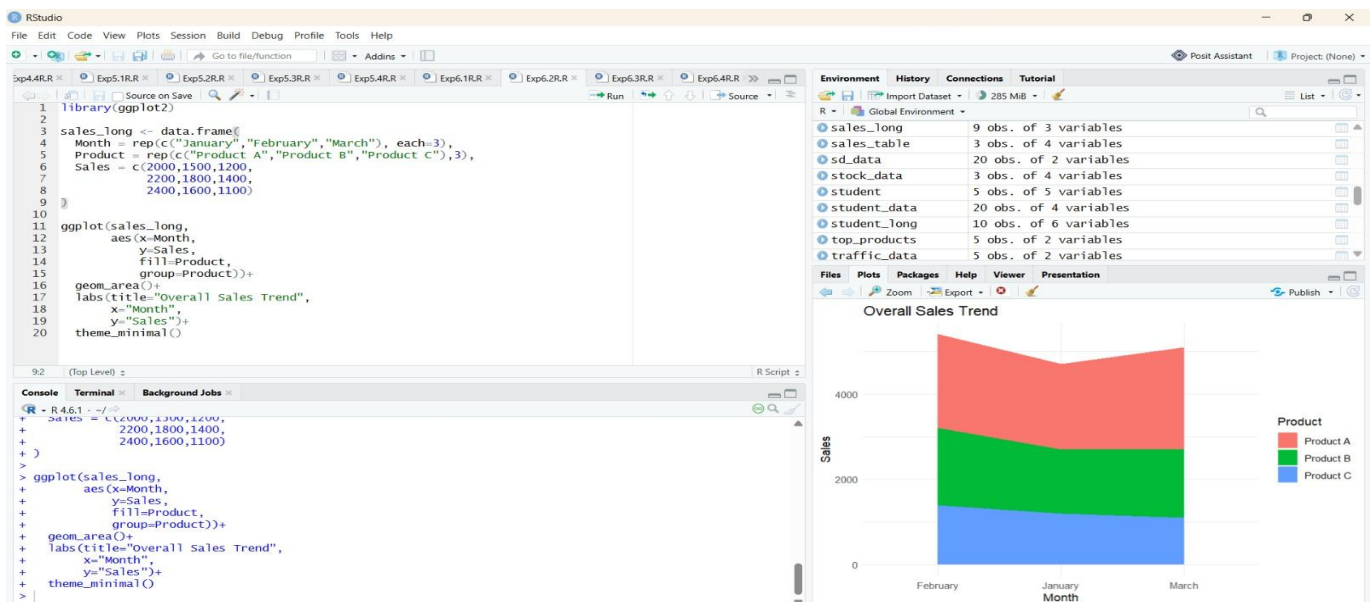
Aim:

To analyze quarterly product sales using Tableau by creating charts, tables, and an interactive dashboard.

Algorithm:

1. Open Tableau and import the Monthly Product Sales dataset.
2. Verify the imported data.
3. Create a grouped bar chart showing sales for each product during the first quarter.
4. Generate a stacked area chart representing overall sales trends.
5. Build a table displaying monthly sales for each product.
6. Add appropriate formatting, titles, and labels.
7. Combine the visualizations into an interactive dashboard.
8. Save the workbook.





Result:

The dashboard successfully displayed quarterly product sales, overall sales trends, and monthly sales details for each product.

Set7:CustomerDemographicsAnalysis

Dataset: Customer Demographics

CustomerID	Age	Gender	Income(in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

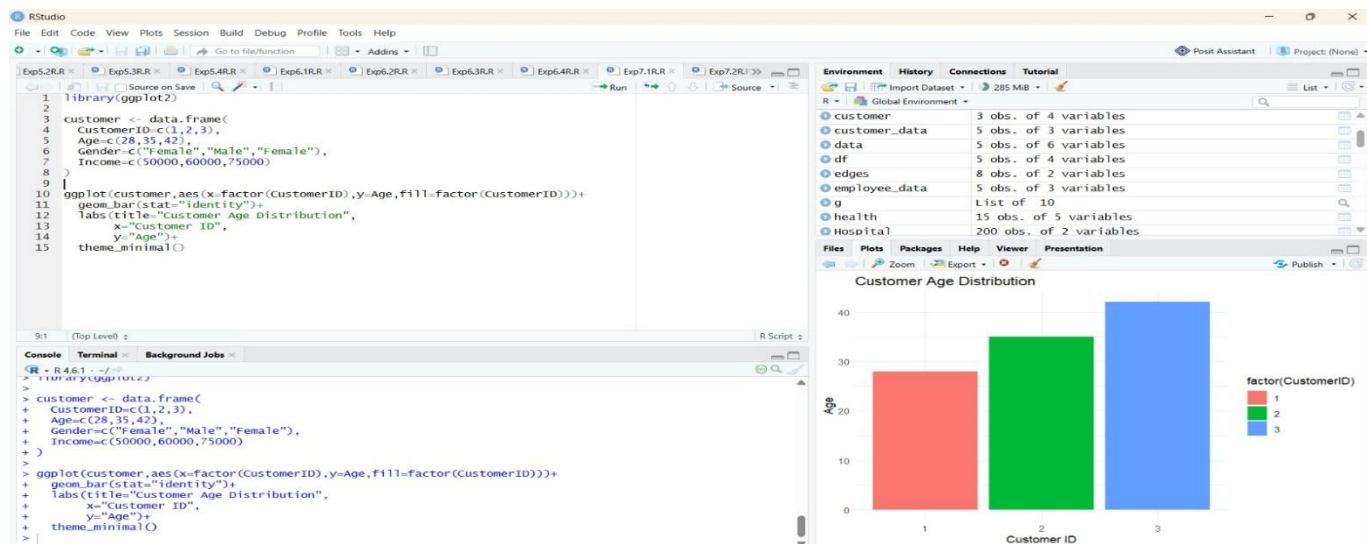
1. **Create a bar chart** to visualize the distribution of customer ages. Label the axes and title the chart.
2. **Generate a pie chart** to represent the distribution of customers by gender.
3. **Build a table** to show the demographic information for each customer. Label the table elements.
4. **Develop a Tableau dashboard** combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

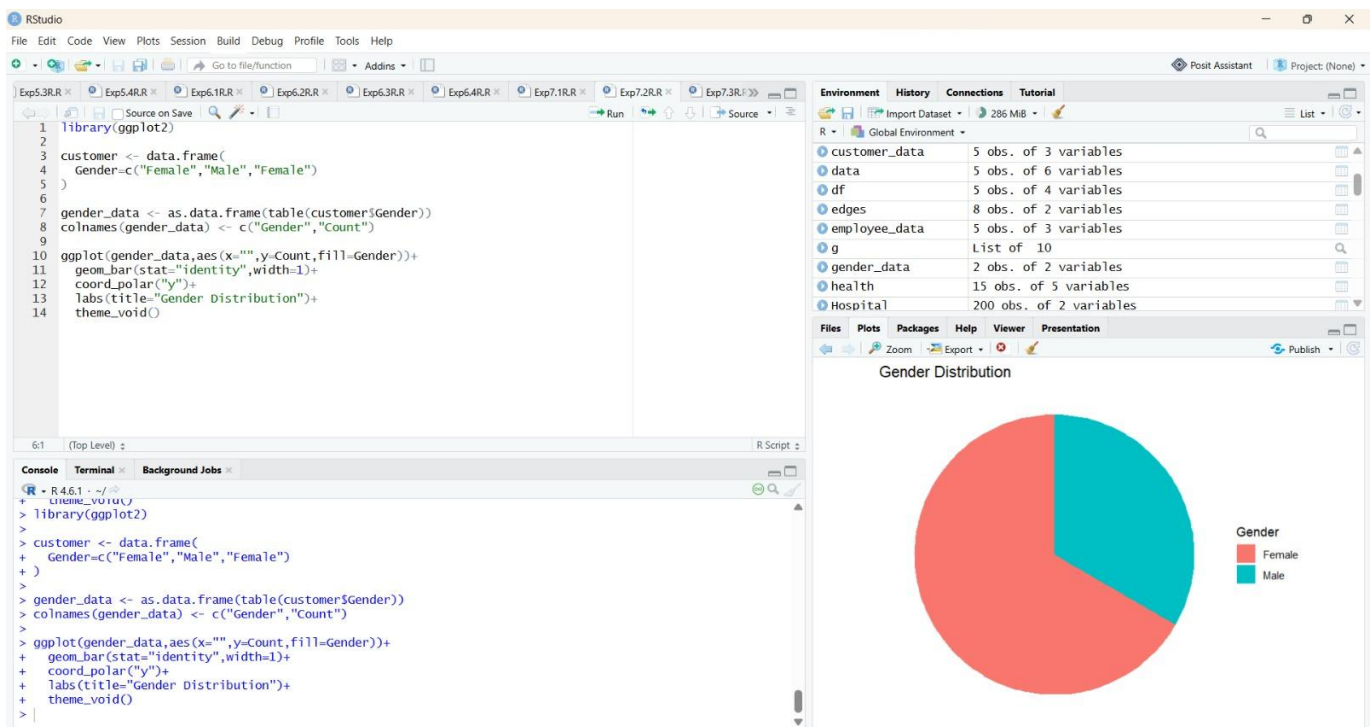
Aim:

To analyze customer demographic information using Tableau by creating charts, tables, and an interactive dashboard.

Algorithm:

1. Import the Customer Demographics dataset into Tableau.
2. Verify the imported fields.
3. Create a bar chart showing customer age distribution.
4. Generate a pie chart representing gender distribution.
5. Build a table displaying demographic details.
6. Add titles, labels, and legends.
7. Combine the charts and table into an interactive dashboard.
8. Save the workbook.



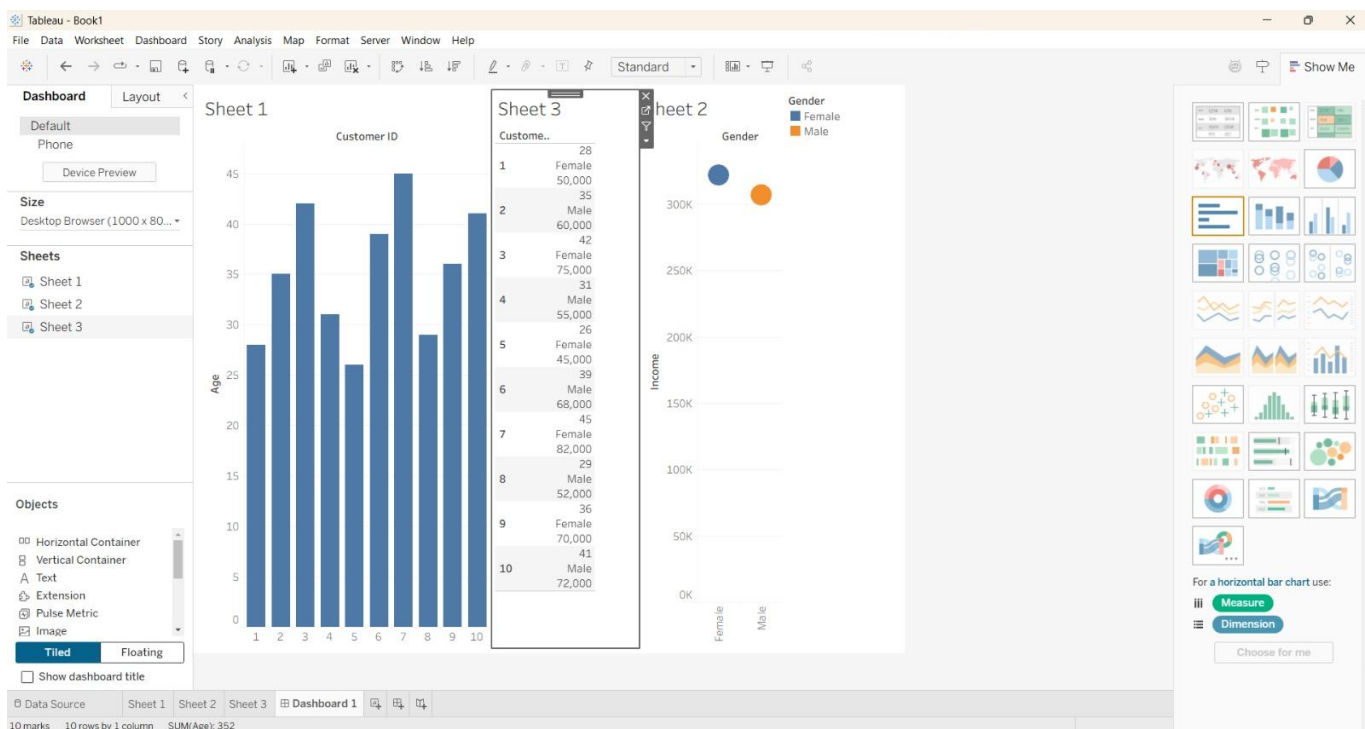


The screenshot displays the RStudio environment. The top menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. Below the menu is a toolbar with icons for file operations and a search bar. The main editor window shows an R script with the following code:

```
1 customer <- data.frame(
2   CustomerID=c(1,2,3),
3   Age=c(28,35,42),
4   Gender=c("Female","Male","Female"),
5   Income=c(50000,60000,75000)
6 )
7
8 print(customer)
```

The bottom panel shows the Console output, which includes the R version (R 4.6.1) and the execution of the script:

```
> customer <- data.frame(
+   CustomerID=c(1,2,3),
+   Age=c(28,35,42),
+   Gender=c("Female","Male","Female"),
+   Income=c(50000,60000,75000)
+ )
>
> print(customer)
  CustomerID Age Gender Income
1          1  28 Female 50000
2          2  35   Male 60000
3          3  42 Female 75000
```



Result:

The dashboard successfully visualized customer demographic information, including age distribution, gender distribution, and customer details.

Set8:OnlineLearningActivityAnalysis

Dataset: Online_Learning_Activity.csv

Student_ID	Gender	Age	Course	Study_Time(hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

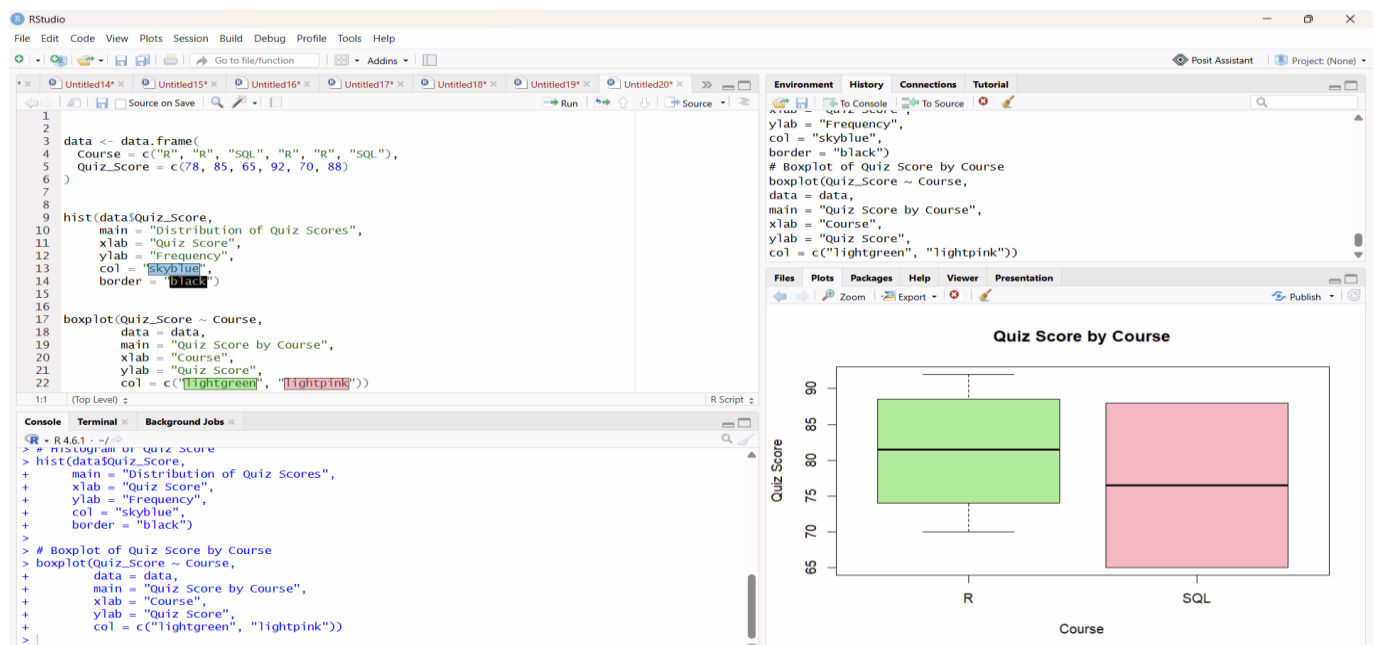
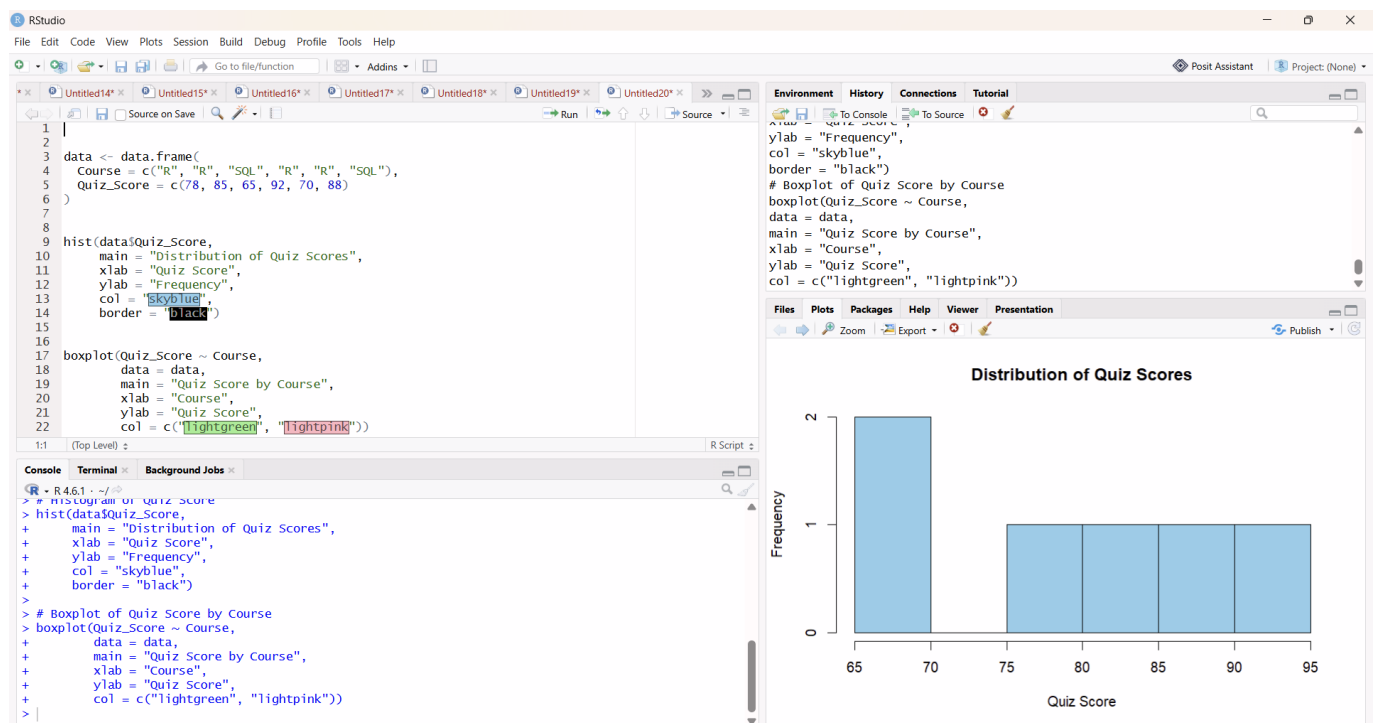
1. **Import the dataset in R.** Plot a histogram for **Quiz_Score**. Draw a boxplot of **Quiz_Score** by **Course**. Add title and labels. Interpret student performance.
2. **Create a scatter plot using R programming** between **Study_Time** and **Quiz_Score**. Represent **Videos_Watched** using bubble size. Apply transparency. Explain the learning pattern.
3. **Using R programming**, convert **Login_Date** into **Date** format. Calculate the average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.
4. **Import Online_Learning_Activity.csv into Tableau.** Create a bar chart showing average quiz score by **Course**. Create a scatter plot (**Study_Time** vs **Quiz_Score**). Add filters for **Gender** and **Course**. Design a mini interactive dashboard.

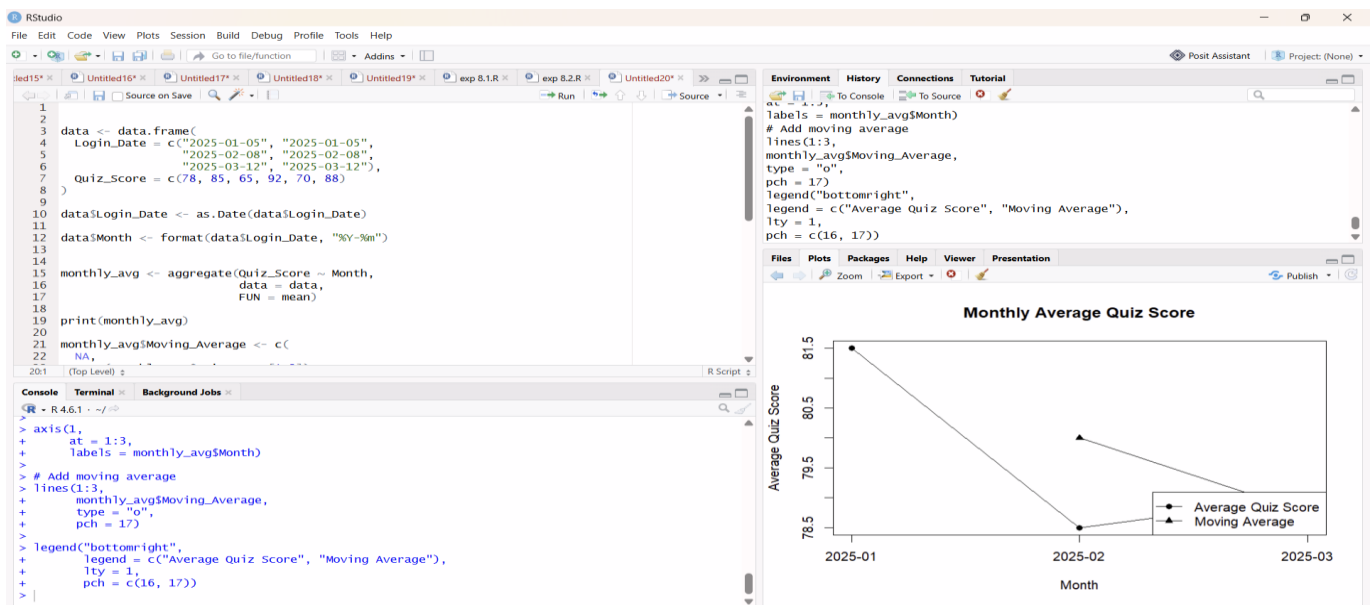
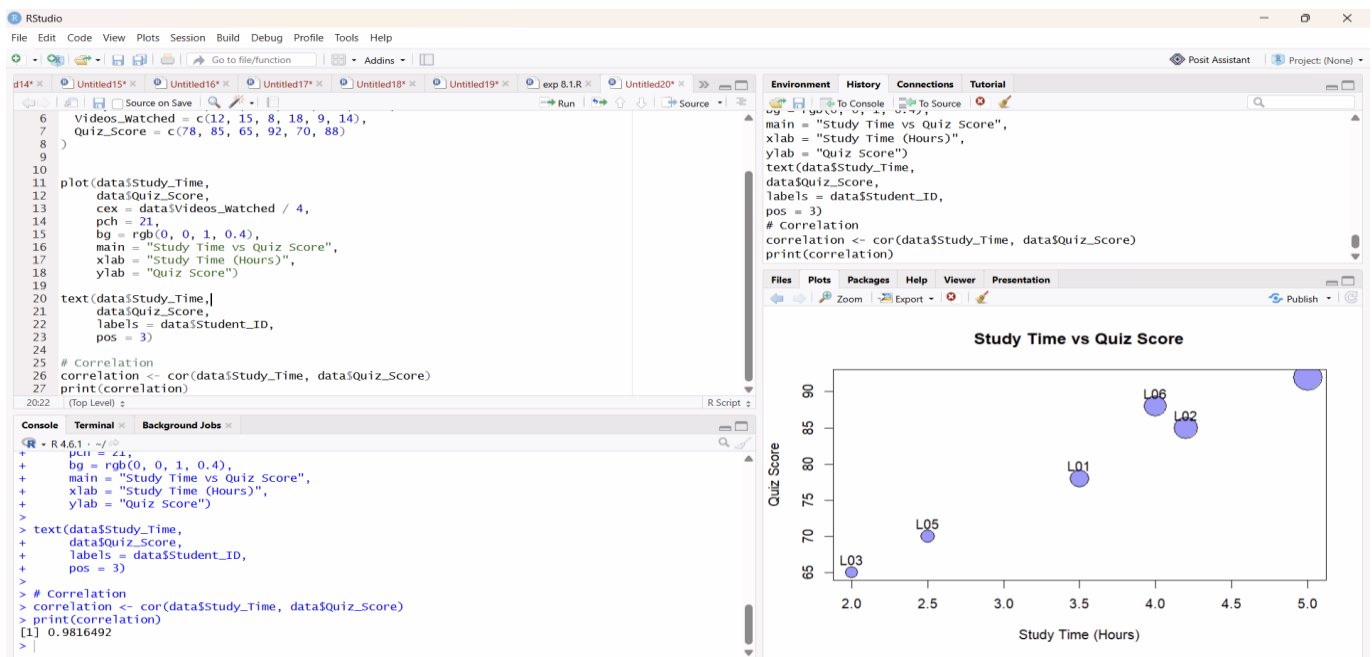
Aim:

To analyze online learning activity using R programming and Tableau through statistical visualizations and an interactive dashboard.

Algorithm:

1. Import the Online Learning Activity dataset into R.
2. Create a histogram of Quiz Score and a boxplot by Course.
3. Generate a scatter plot between Study Time and Quiz Score with bubble size representing Videos Watched.
4. Convert Login Date into Date format.
5. Calculate monthly average Quiz Scores and create a line chart with moving average smoothing.
6. Import the dataset into Tableau.
7. Create a bar chart for average quiz score by course and a scatter plot for Study Time vs Quiz Score.
8. Add Gender and Course filters and design an interactive dashboard.





Result:

The analysis successfully visualized student performance, learning behavior, monthly quiz score trends, and course-wise comparisons using R and Tableau.

Set9:SurveyResponsesAnalysis Dataset:

Survey Responses

SurveyID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.
2. Generate a stacked bar chart to represent the overall distribution of responses for all three

questions.

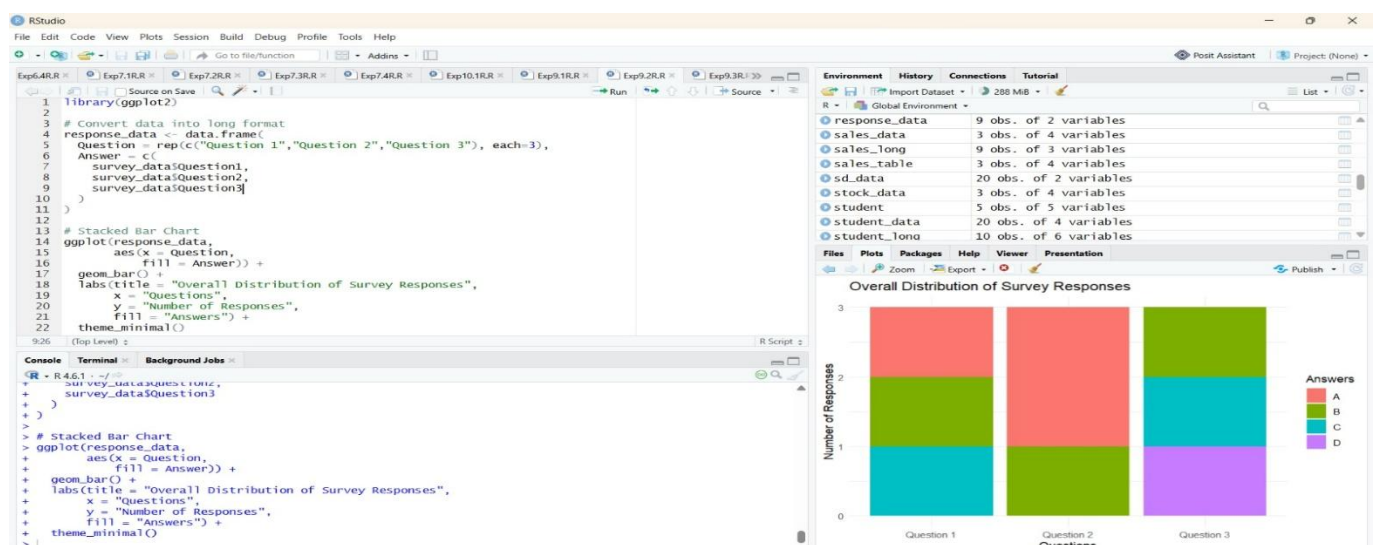
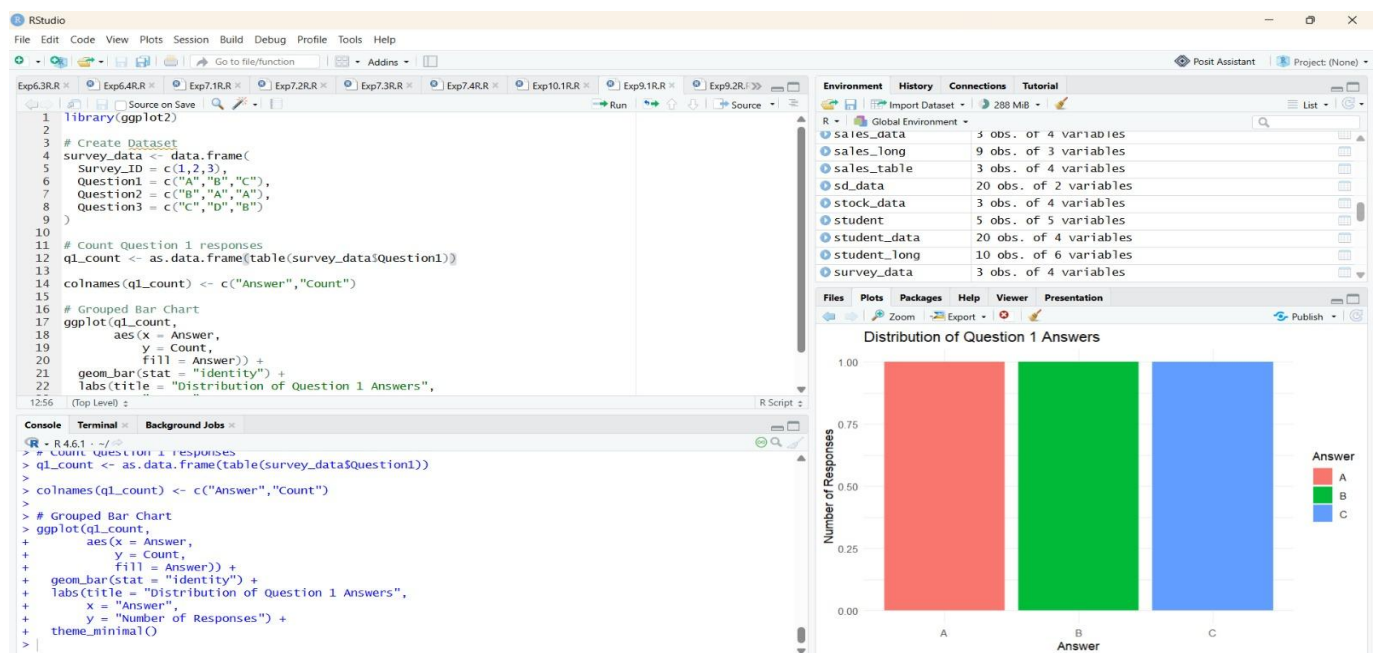
3. **Build a table** to show the survey response data for each survey. Label the table elements.
4. **Develop a Tableau dashboard** combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Aim:

To analyze survey response data using Tableau by creating charts, tables, and an interactive dashboard.

Algorithm:

1. Import the Survey Responses dataset into Tableau.
2. Verify the imported fields.
3. Create a grouped bar chart for Question 1 responses.
4. Generate a stacked bar chart showing responses for all three questions.
5. Build a table displaying survey responses.
6. Add titles, labels, and legends.
7. Combine the charts and table into an interactive dashboard.
8. Save the workbook.



```

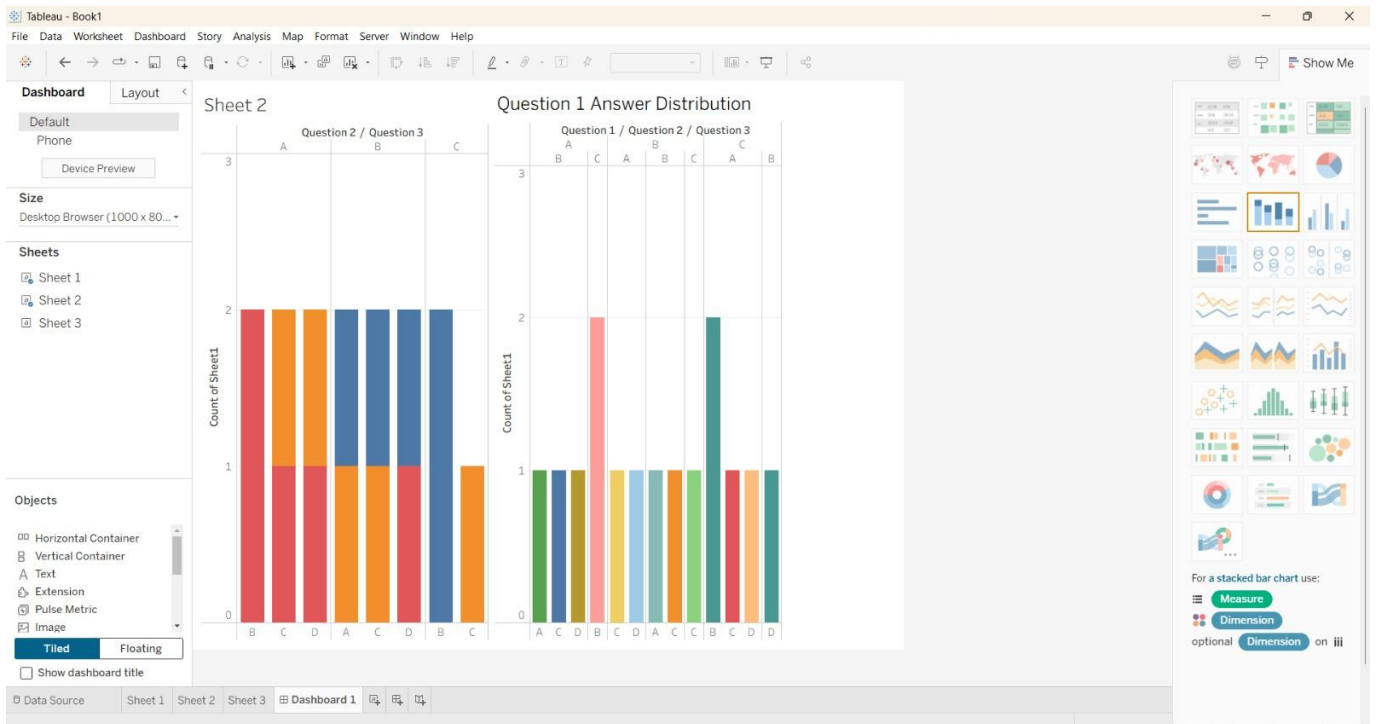
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function
Exp7.1R.R x Exp7.2R.R x Exp7.3R.R x Exp7.4R.R x Exp10.1R.R x Exp9.1R.R x Exp9.2R.R x Exp9.3R.R x Exp9.4R.R x
Source on Save
1 survey_table <- data.frame(
2   Survey_ID = c(1,2,3),
3   Question1 = c("A","B","C"),
4   Question2 = c("B","A","A"),
5   Question3 = c("C","D","B")
6 )
7
8 print(survey_table)

```

```

R - R 4.6.1 ~ /
+ y = number of responses,
+ fill = "Answers" +
+ theme_minimal()
> survey_table <- data.frame(
+   Survey_ID = c(1,2,3),
+   Question1 = c("A","B","C"),
+   Question2 = c("B","A","A"),
+   Question3 = c("C","D","B")
+ )
> print(survey_table)
  Survey_ID Question1 Question2 Question3
1         1         A         B         C
2         2         B         A         D
3         3         C         A         B

```



Result:

The dashboard successfully displayed the distribution of survey responses, enabling easy comparison of responses across different survey questions.

Set10

Product Category Analysis

Dataset: ProductCategories

Category	Sales(in\$)
Electronics	50000
Clothing	35000
Appliances	40000

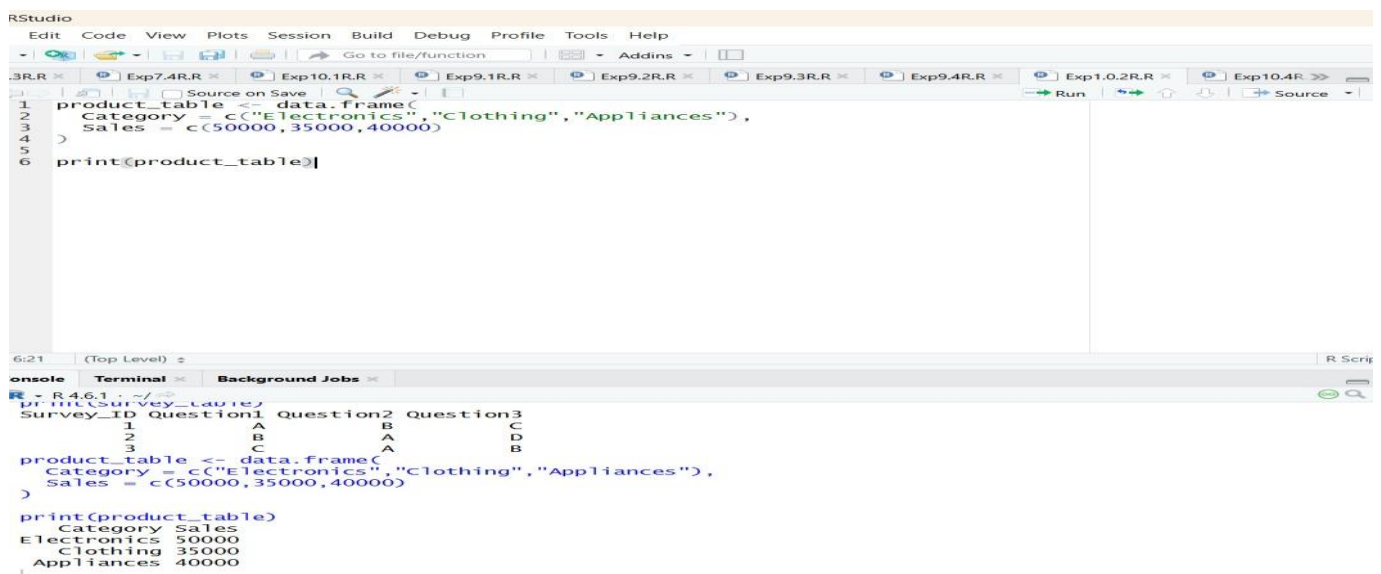
1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.
2. Build a table to display the sales data for each product category. Label the table elements.
3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.
4. Create a pie chart to represent the distribution of sales across product categories. Include labels.

Aim:

To analyze product category sales using Tableau by creating charts, tables, and an interactive dashboard.

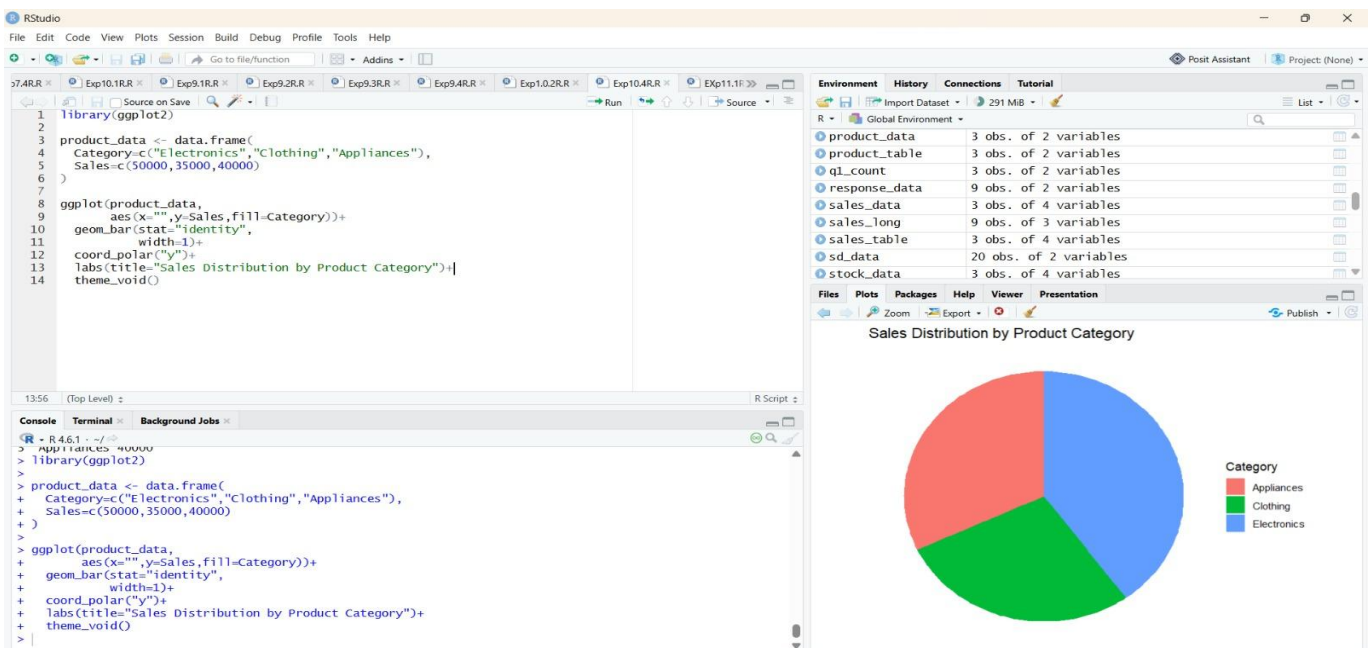
Algorithm:

1. Import the Product Categories dataset into Tableau.
2. Verify the imported data.
3. Create a funnel chart showing the sales conversion process.
4. Generate a pie chart representing category-wise sales distribution.
5. Build a table displaying product category sales.
6. Add titles, labels, and legends.
7. Combine the funnel chart, pie chart, and table into an interactive dashboard.
8. Save the Tableau workbook.



```
RStudio
Edit Code View Plots Session Build Debug Profile Tools Help
- | Go to file/function | Addins |
3RR x Exp7.4RR x Exp10.1RR x Exp9.1RR x Exp9.2RR x Exp9.3RR x Exp9.4RR x Exp10.2RR x Exp10.4R >>
1 product_table <- data.frame(
2   Category = c("Electronics", "Clothing", "Appliances"),
3   Sales = c(50000, 35000, 40000)
4 )
5
6 print(product_table)

6:21 (Top Level) R Script
onsole Terminal Background Jobs
R - R4.6.1 ~ /
print(Survey_Table)
Survey_ID Question1 Question2 Question3
1 A B C
2 A A D
3 C A B
product_table <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
print(product_table)
  Category Sales
Electronics 50000
Clothing 35000
Appliances 40000
```



Result:

The dashboard successfully visualized product category sales, conversion process, and category-wise distribution, providing an interactive overview of sales performance.

11. Dataset:GeographicData

City	Population	Avg.Temperature	Elevation
CityA	500000	75	1000
CityB	700000	68	800
CityC	600000	80	1200

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.
2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.
3. Build a table to display the geographic data for each city. Label the table elements.
4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

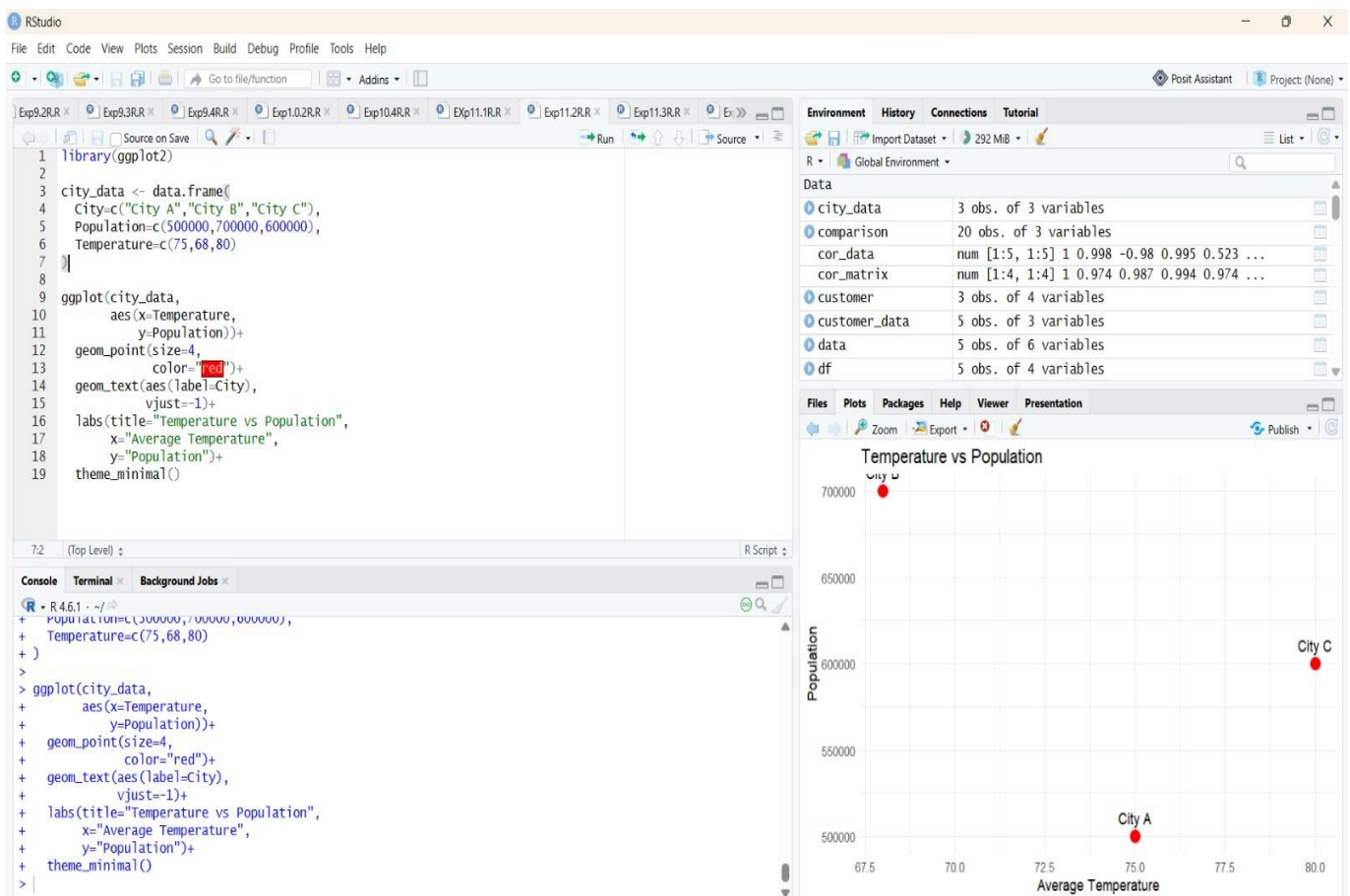
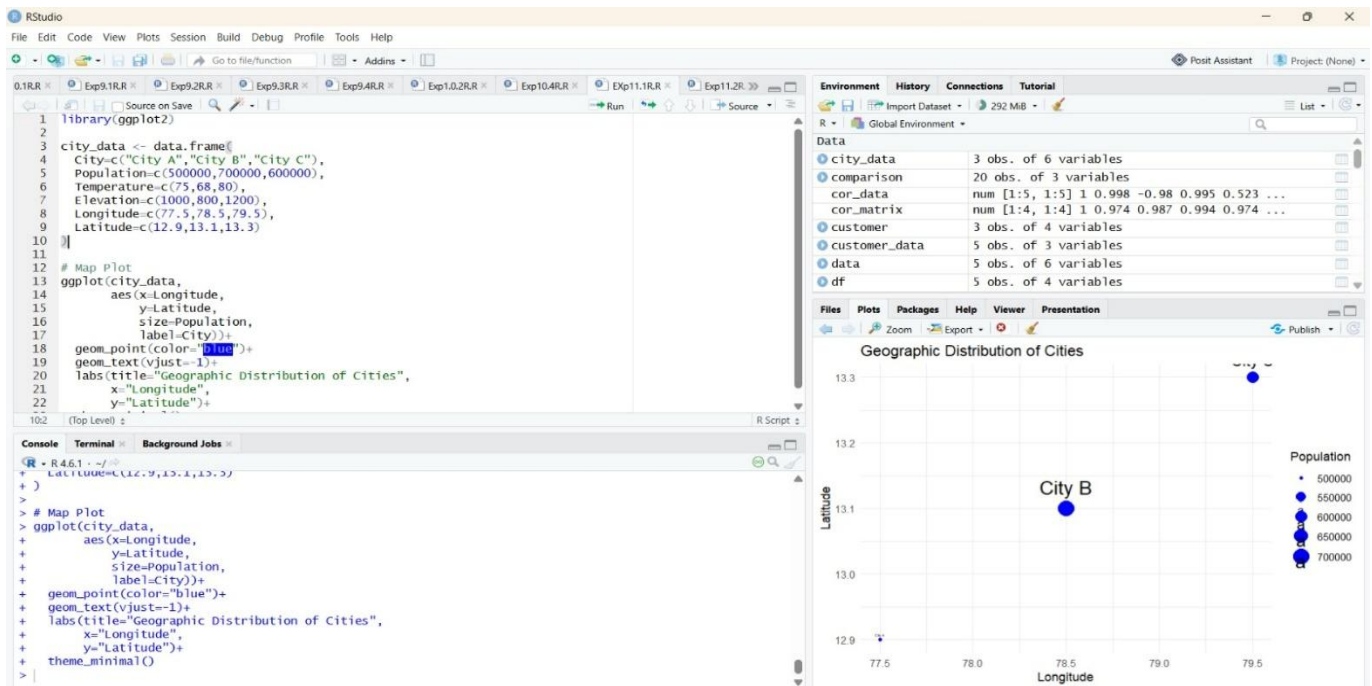
Aim:

To visualize geographic data using Tableau by creating a map, scatter plot, table, and an interactive dashboard.

Algorithm:

1. Open Tableau and import the Geographic Data dataset.
2. Verify the imported fields and assign the geographic role to the City field.
3. Create a map chart showing the distribution of cities.
4. Generate a scatter plot between Average Temperature and Population.
5. Build a table displaying the geographic data.

6. Add suitable titles, labels, legends, and tooltips.
7. Combine the map, scatter plot, and table into an interactive dashboard.
8. Save the Tableau workbook.



```

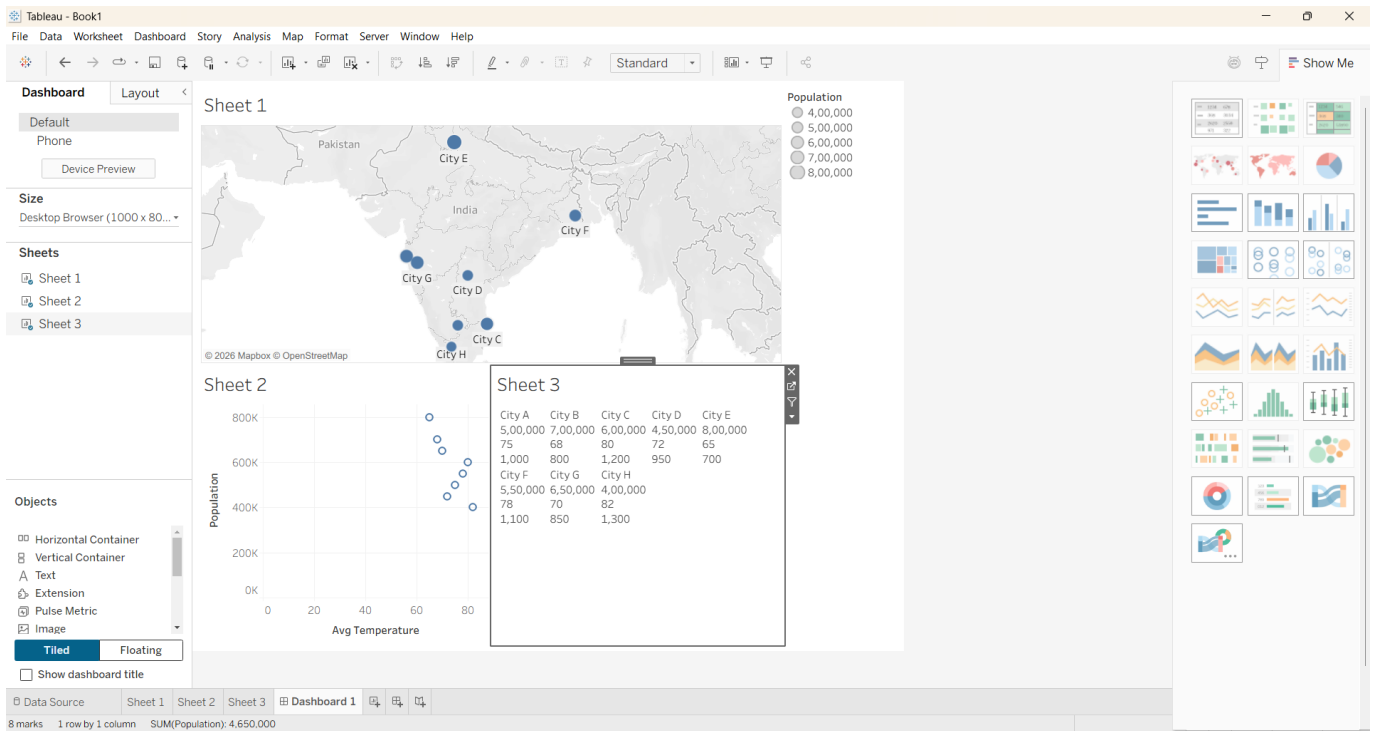
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
2.R.R x Exp9.3.R.R x Exp9.4.R.R x Exp1.0.2.R.R x Exp10.4.R.R x Exp11.1.R.R x Exp11.2.R.R x Exp11.3.R.R x Exp12.1.R x
1 geo_table <- data.frame(
2   City=c("City A", "City B", "City C"),
3   Population=c(500000,700000,600000),
4   Average_Temperature=c(75,68,80),
5   Elevation=c(1000,800,1200)
6 )
7
8 print(gg_table)|

```

```

R - R 4.6.1
+ x=Average_Temperature ,
+ y=Population)+
+ theme_minimal()
> geo_table <- data.frame(
+   City=c("City A", "City B", "City C"),
+   Population=c(500000,700000,600000),
+   Average_Temperature=c(75,68,80),
+   Elevation=c(1000,800,1200)
+ )
> print(gg_table)
  City Population Average_Temperature Elevation
1 City A      5e+05              75         1000
2 City B      7e+05              68           800
3 City C      6e+05              80         1200

```



Result:

The map, scatter plot, and table were successfully created. The interactive dashboard effectively displayed city locations and the relationship between population and average temperature.

Set12:ProductInventoryManagement

Dataset: Product Inventory

ProductID	ProductName	QuantityAvailable	Price(in\$)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

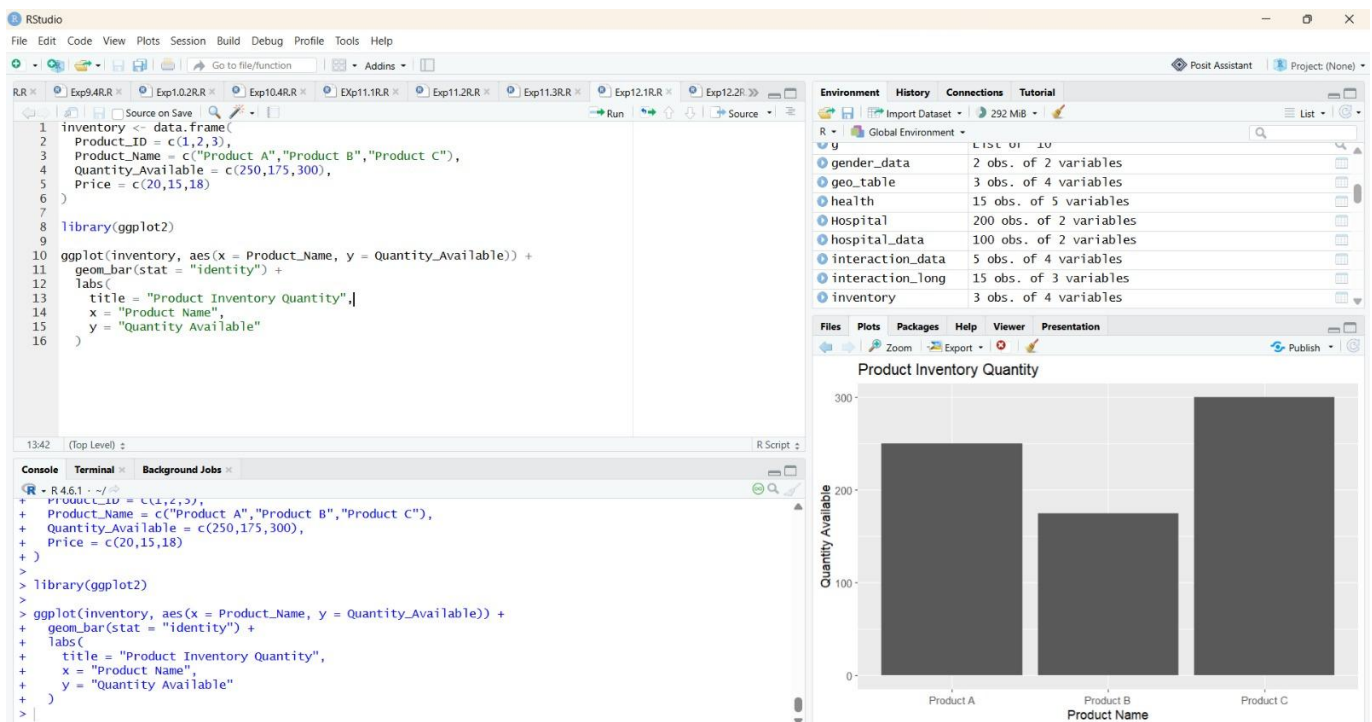
1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different product categories.
3. Build a table to show the inventory data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.

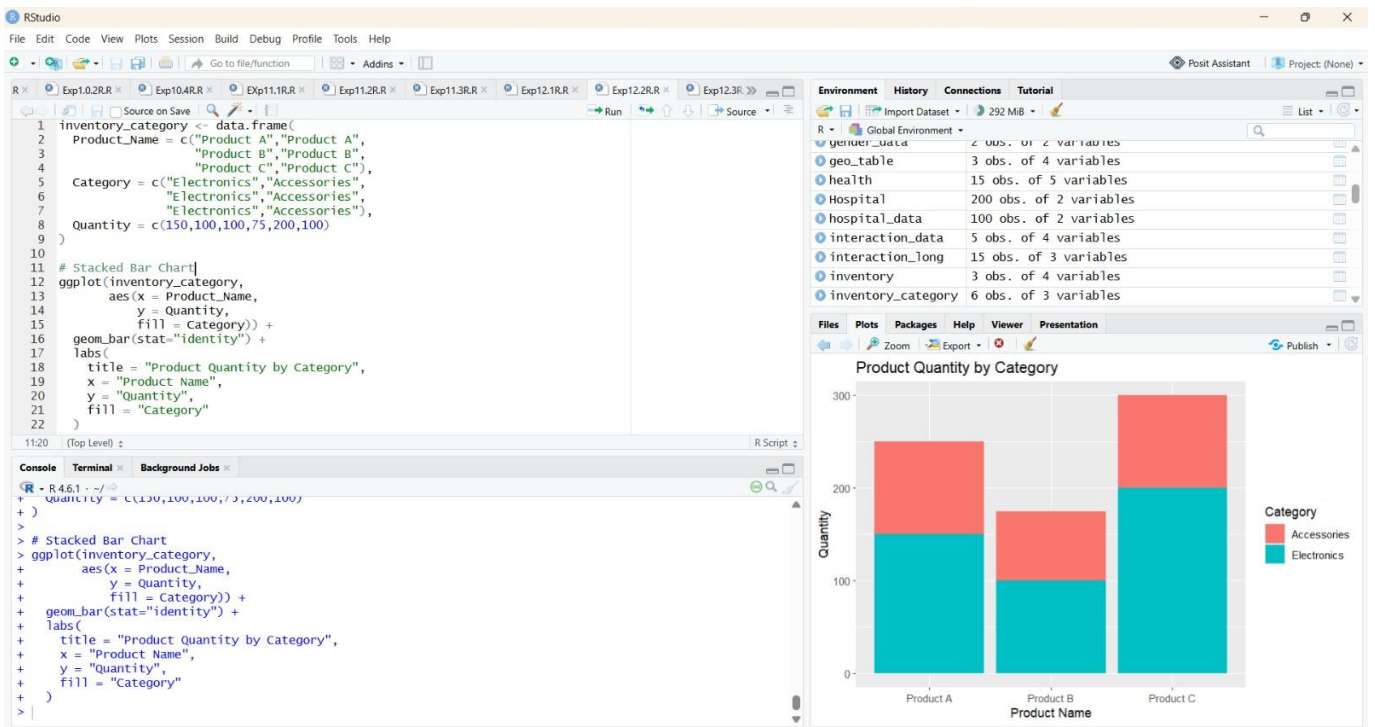
Aim:

To analyze product inventory data using Tableau by creating charts, tables, and an interactive dashboard.

Algorithm:

1. Import the Product Inventory dataset into Tableau.
2. Verify the imported data fields.
3. Create a bar chart showing the quantity available for each product.
4. Generate a stacked bar chart for product quantities by category.
5. Build a table displaying inventory details.
6. Add titles, labels, legends, and formatting.
7. Design an interactive dashboard using the created visualizations.
8. Save the workbook.





RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Exp10.4.R.R Exp11.1.R.R Exp11.2.R.R Exp11.3.R.R Exp12.1.R.R Exp12.2.R.R Exp12.3.R.R Exp16.R.F

Source on Save

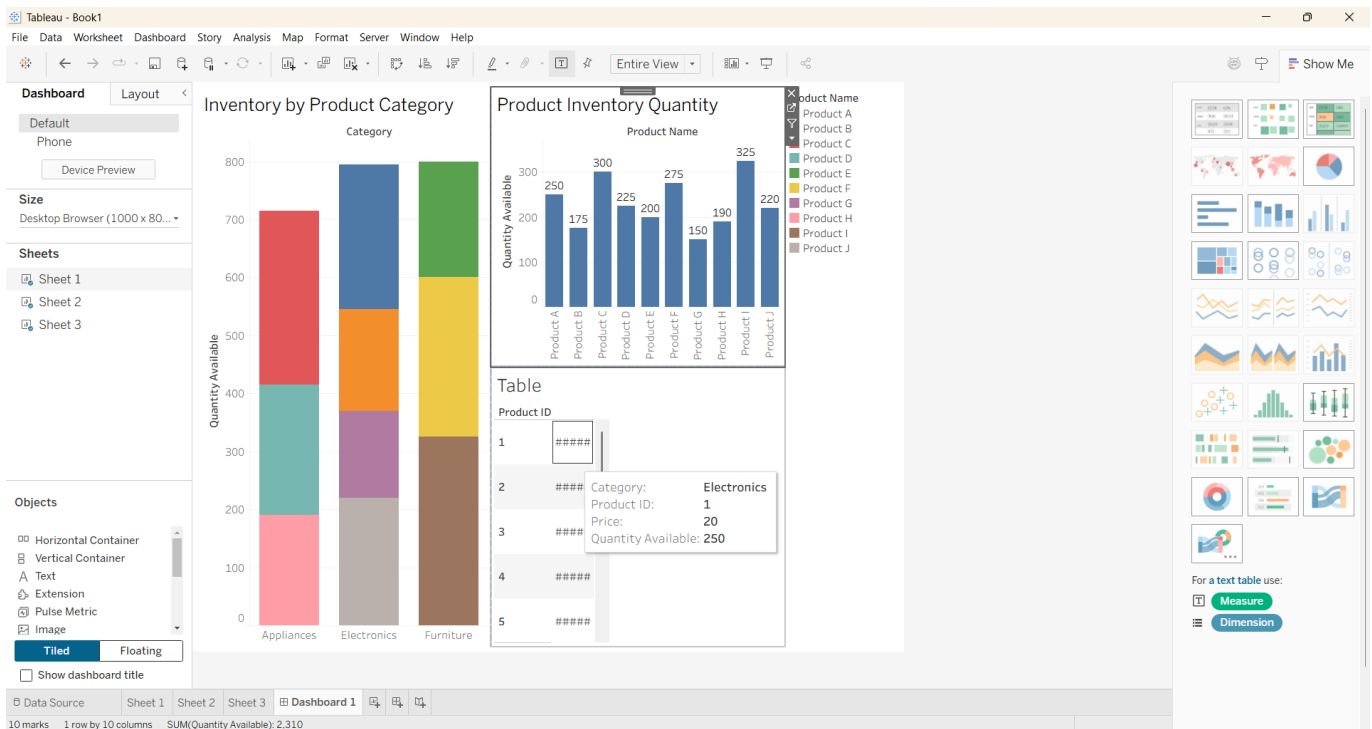
Run

4:43 (Top Level)

R Script

Console Terminal Background Jobs

```
R - R 4.6.1 ~ /  
+ title = Category  
+ )  
+ library(knitr)  
+  
+ kable(inventory,  
+   caption="Product Inventory Details")  
+  
Table: Product Inventory Details  
+ |  
+ | Product_ID | Product_Name | Quantity_Available | Price |  
+ |-----| :-----| :-----| :-----|  
+ | 1 | Product A | 250 | 20 |  
+ | 2 | Product B | 175 | 15 |  
+ | 3 | Product C | 300 | 18 |  
+ |
```



Result:

The dashboard successfully visualized inventory quantities, product categories, and inventory details, enabling easy analysis of inventory status.

Set13:StockAnalysis

Dataset: Stock Prices

Date	StockA	StockB	StockC
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

1. Create a line chart to visualize the stock prices of three companies (StockA, StockB, and StockC) over a specific time period. Label the axes and title the chart.
2. Generate a bar chart showing the daily percentage change in stock prices for StockA. Label the chart elements.
3. Build a table to display the stock price data for each company over the given period. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

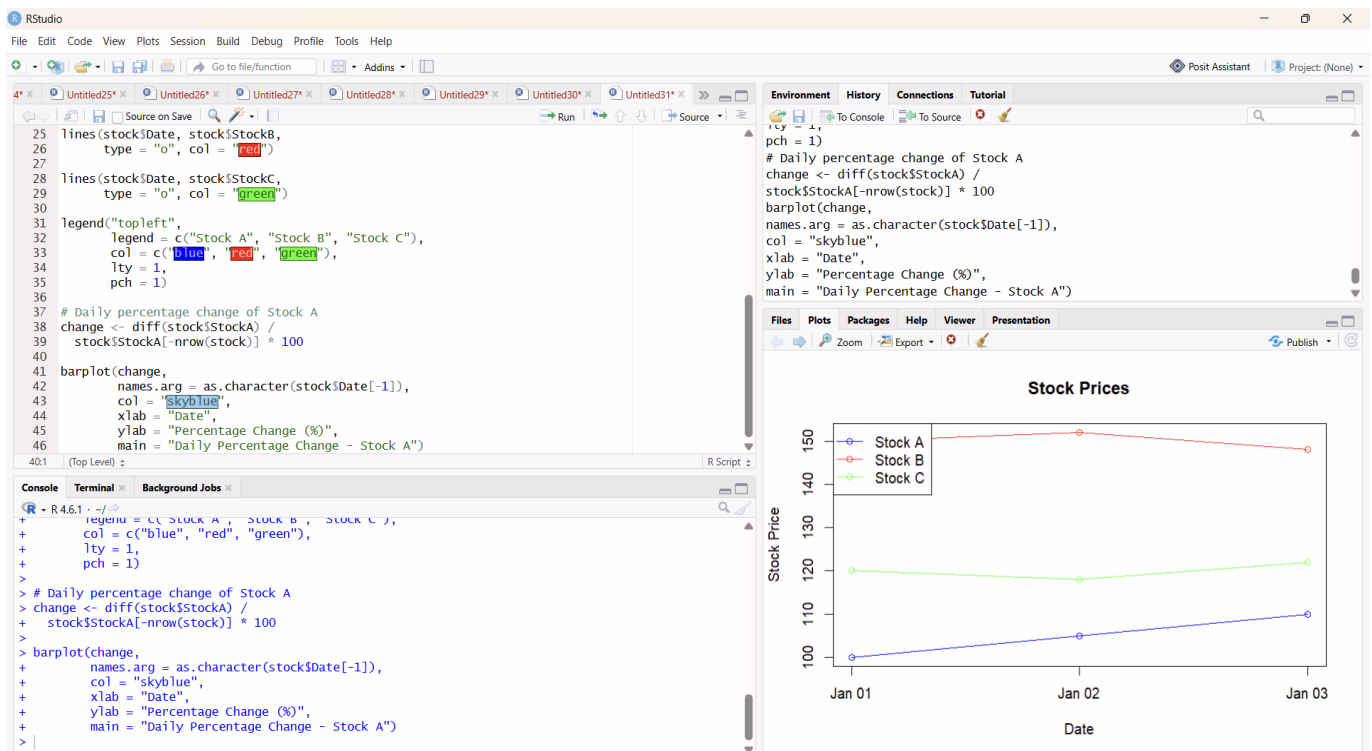
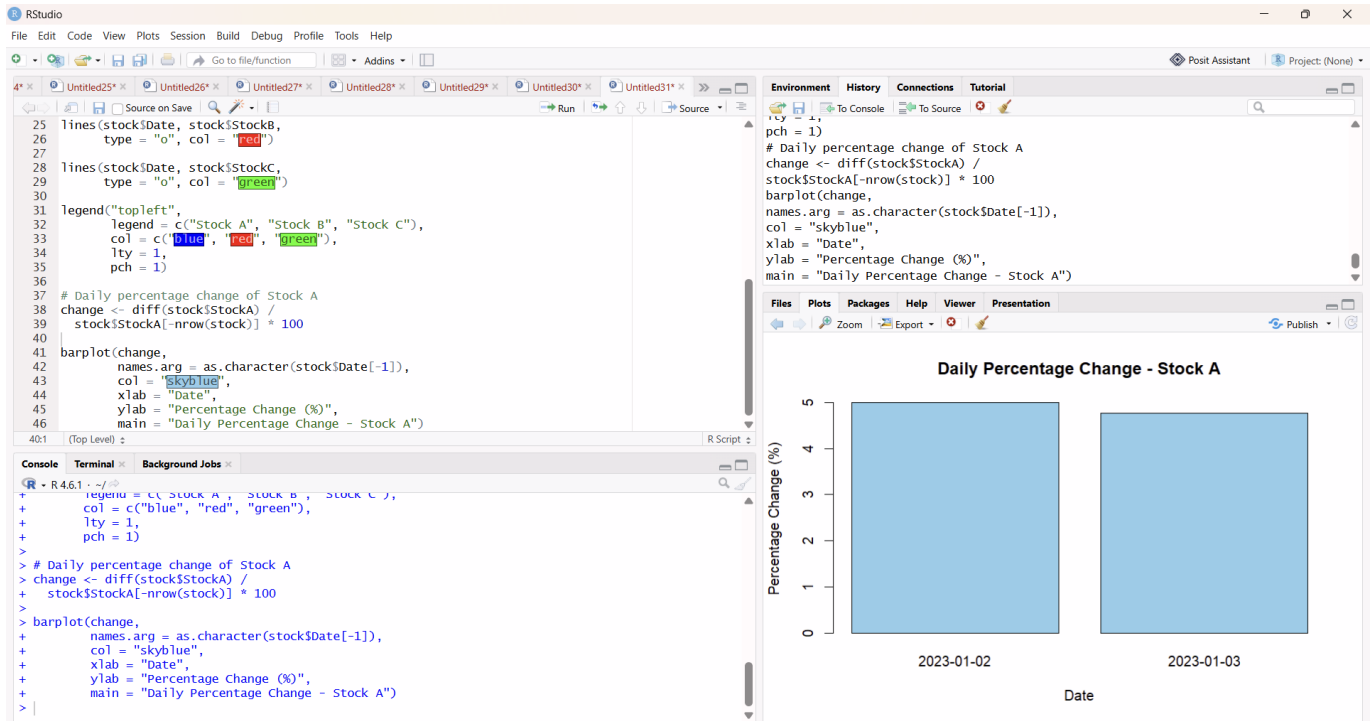
Aim:

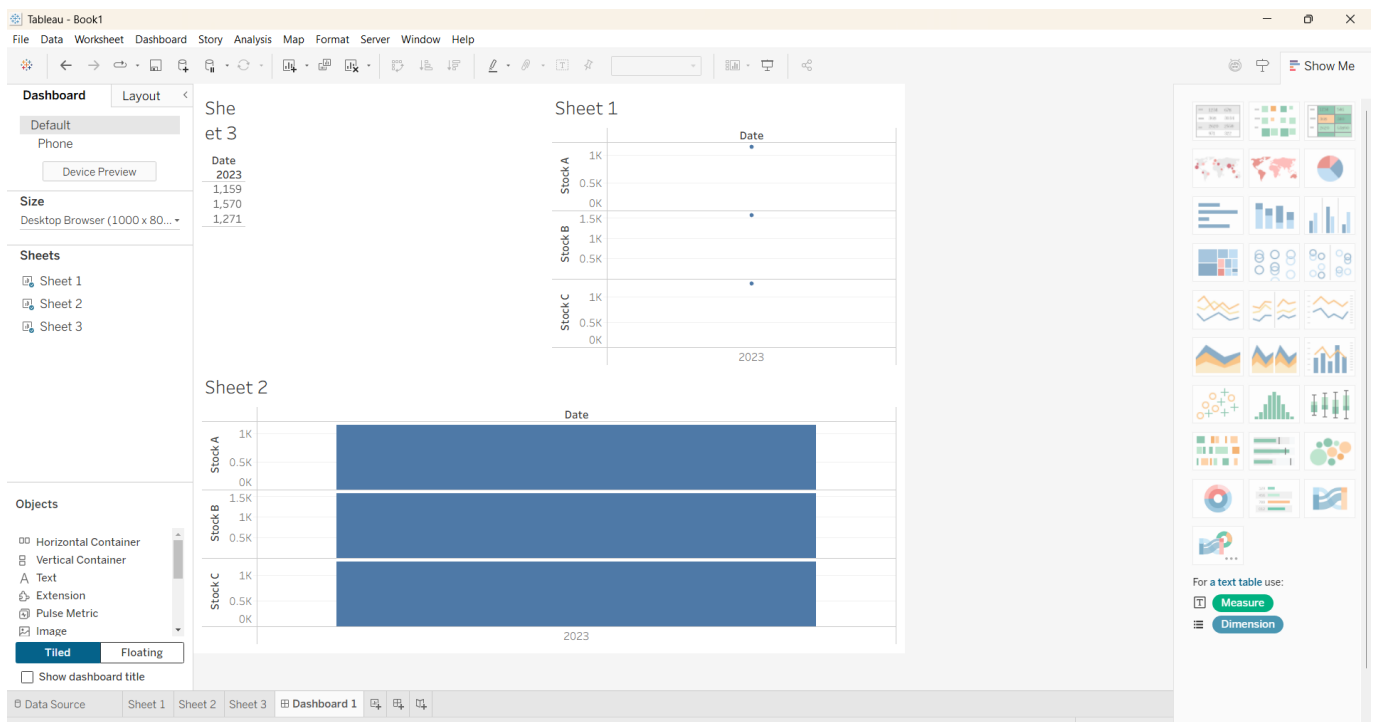
To analyze stock price data using Tableau by creating charts, tables, and an interactive dashboard.

Algorithm:

1. Open Tableau and import the Stock Prices dataset.
2. Verify the imported data.
3. Create a line chart showing stock prices of all three companies.
4. Generate a bar chart displaying the daily percentage change of Stock A.

5. Build a table containing stock price information.
6. Add titles, labels, legends, and formatting.
7. Combine the visualizations into an interactive dashboard.
8. Save the Tableau workbook.





Result:

The stock price trends, daily percentage changes, and stock details were successfully visualized through an interactive Tableau dashboard.

Set14

Dataset:Energy_Consumption_Data.csv

Sector	Region	Month	Temperature (°C)	Units_Consumed (kWh)	Cost (₹)	Renewable_Usage(%)	Peak_Hours
Residential	North	Jan	15	320	2100	22	4
Commercial	South	Jan	24	540	3600	18	6
Industrial	West	Feb	20	880	5900	12	8
Residential	East	Feb	18	350	2300	25	5
Commercial	North	Mar	28	610	4100	20	7
Industrial	South	Mar	30	920	6200	15	9

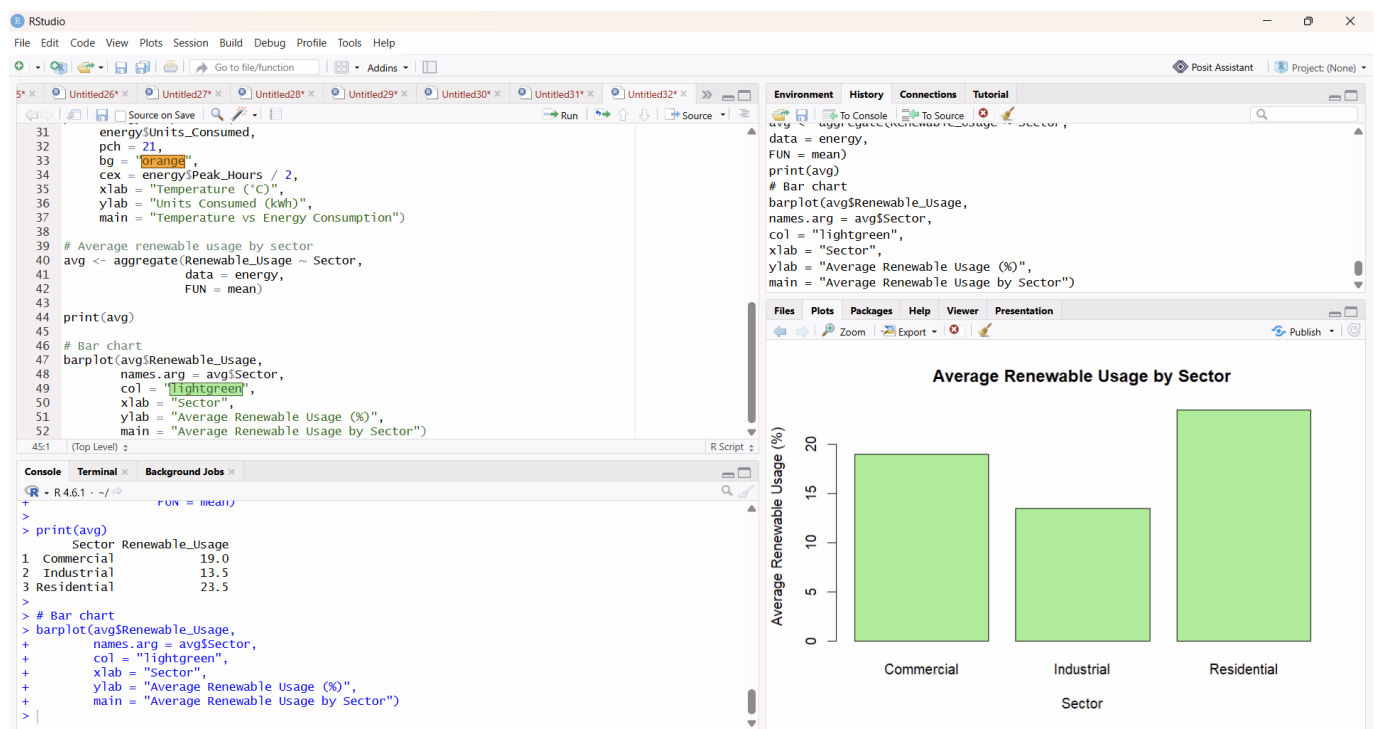
1. Import the dataset in R. Plot a histogram and density plot for **Units_Consumed**. Compare consumption patterns.
2. Using R programming, create a scatter plot between **Temperature** and **Units_Consumed**. Represent **Peak_Hours** using bubble size. Apply transparency.
3. Using R, calculate average **Renewable_Usage** for each **Sector**. Create a bar chart. Interpret renewable energy adoption.
4. Import **Energy_Consumption_Data.csv** into Tableau. Create a line chart for monthly energy usage trends. Create a heatmap (**Sector** vs **Region** vs **Units**). Add filters for **Month** and **Sector**. Design an interactive dashboard.

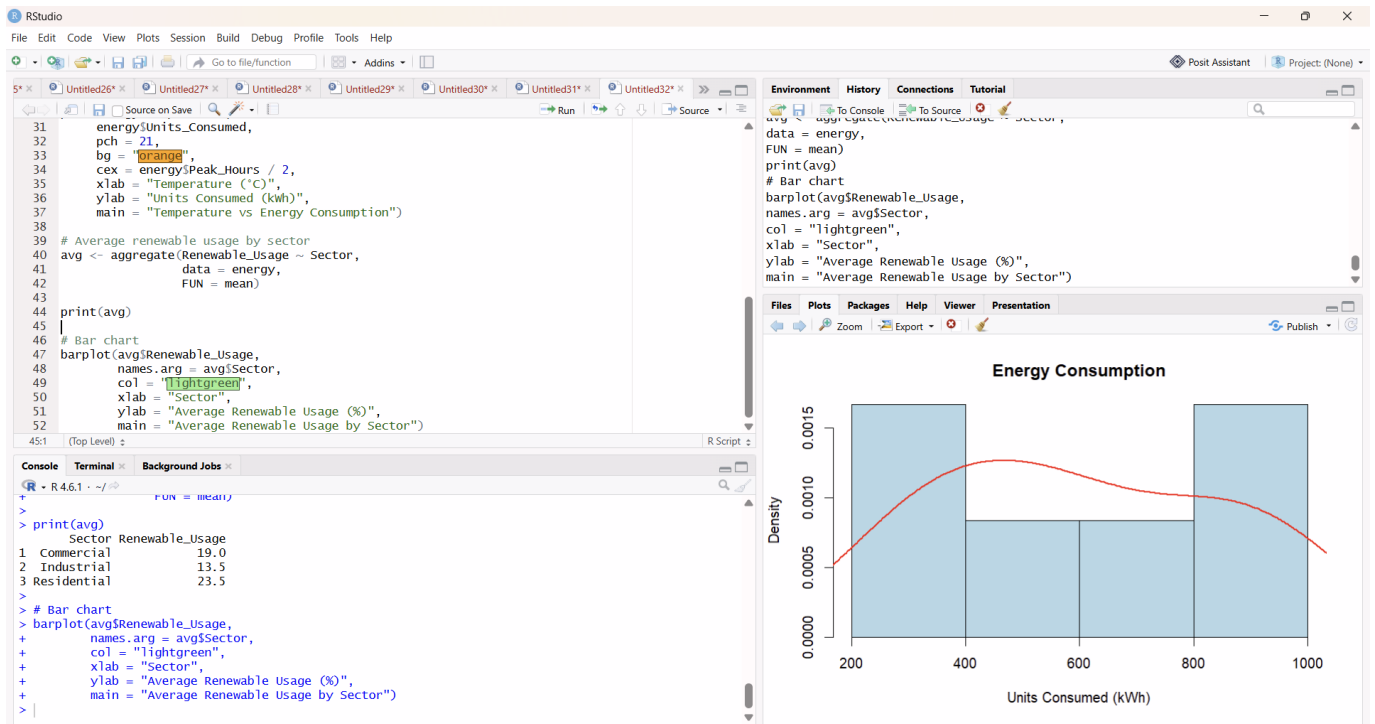
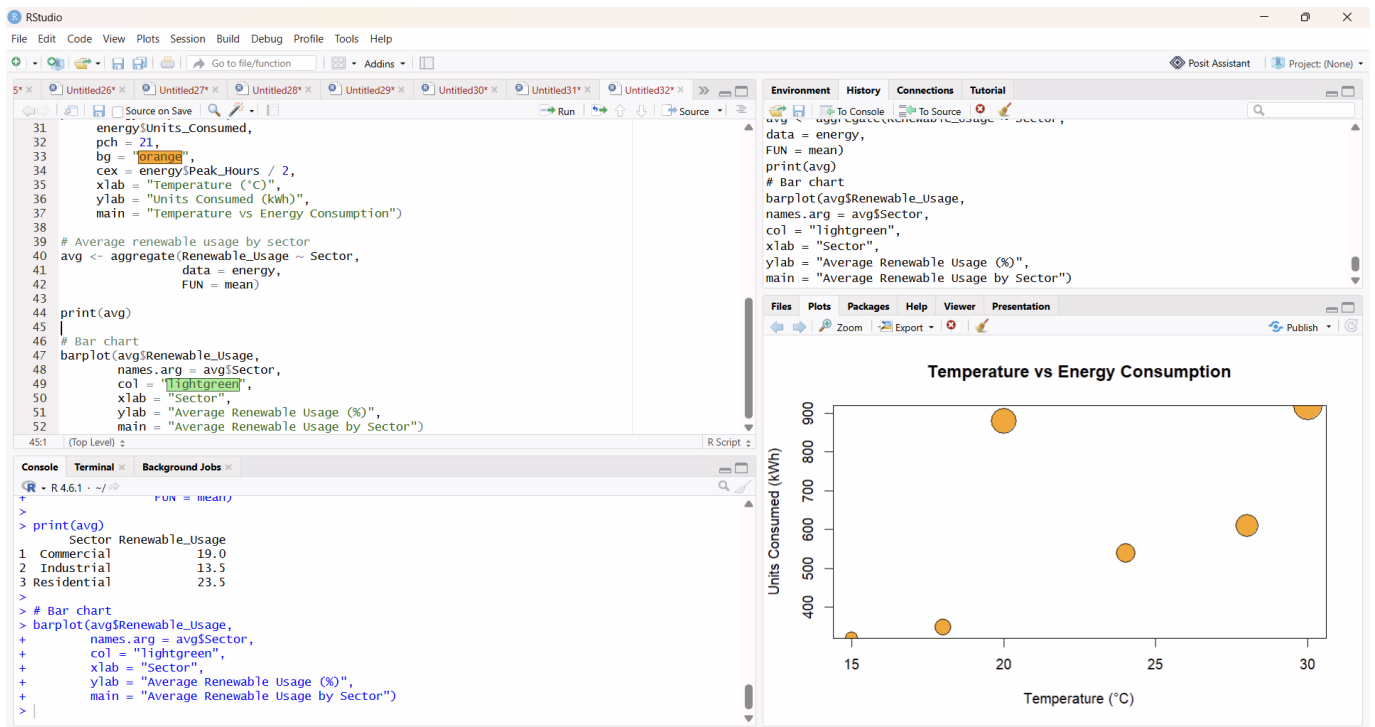
Aim:

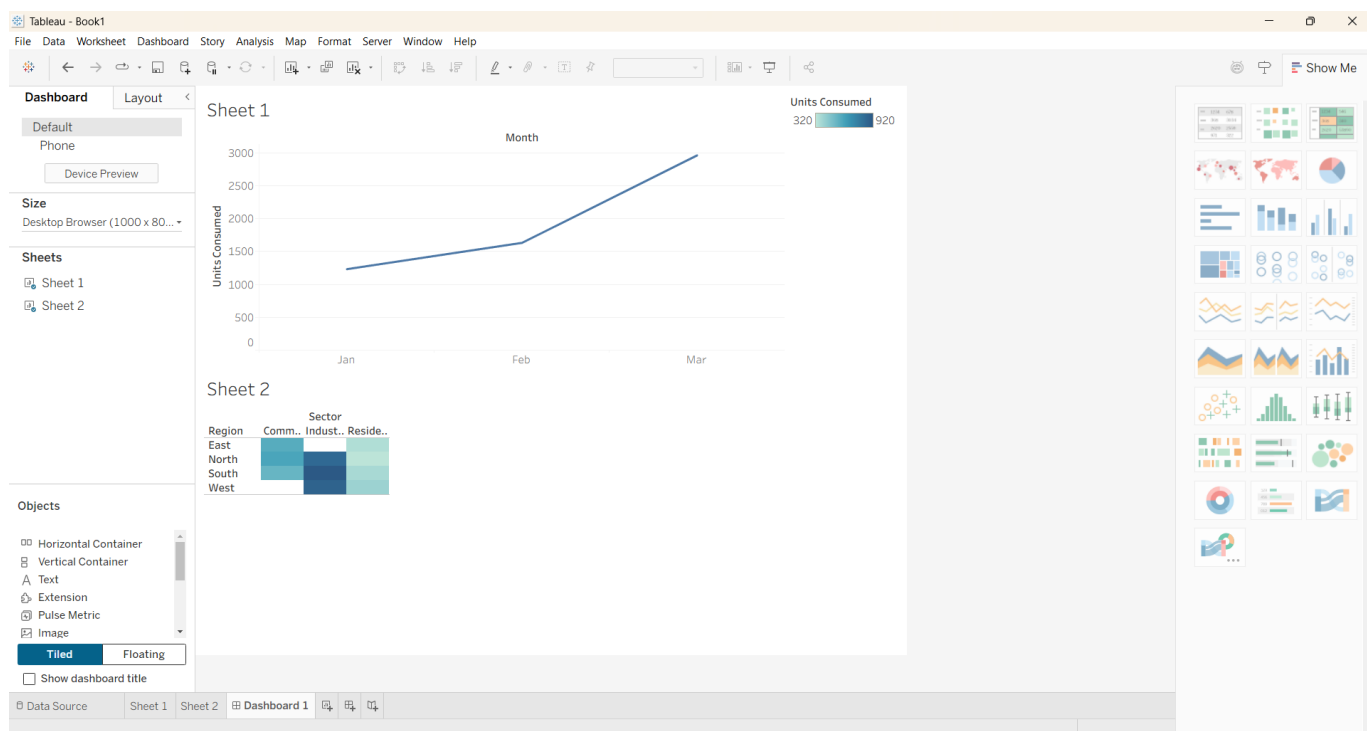
To analyze energy consumption data using R programming and Tableau through statistical visualization and dashboard development.

Algorithm:

1. Import the Energy_Consumption_Data.csv dataset into R.
2. Create a histogram and density plot for Units Consumed.
3. Generate a scatter plot between Temperature and Units Consumed using Peak Hours as bubble size.
4. Calculate the average Renewable Usage for each sector.
5. Create a bar chart representing average Renewable Usage.
6. Import the dataset into Tableau.
7. Create a line chart for monthly energy usage and a heatmap for Sector vs Region vs Units with Month and Sector filters.
8. Design and save an interactive dashboard.







Result:

The required statistical plots and Tableau dashboard were successfully created, providing insights into energy consumption, renewable usage, and monthly energy trends.

SET15–StudentMiniDataAnalysis

Dataset: Student_Mini_Data.csv

Student_ID	Gender	Study_Hours	Attendance	Math_Score	Science_Score	Exam_Date
S01	Male	2.0	78	62	65	2025-01-10
S02	Female	3.5	90	80	85	2025-01-10
S03	Male	1.5	70	55	58	2025-02-12
S04	Female	4.0	95	90	92	2025-02-12
S05	Male	2.8	85	72	74	2025-03-15
S06	Female	3.0	92	82	86	2025-03-15

Questions

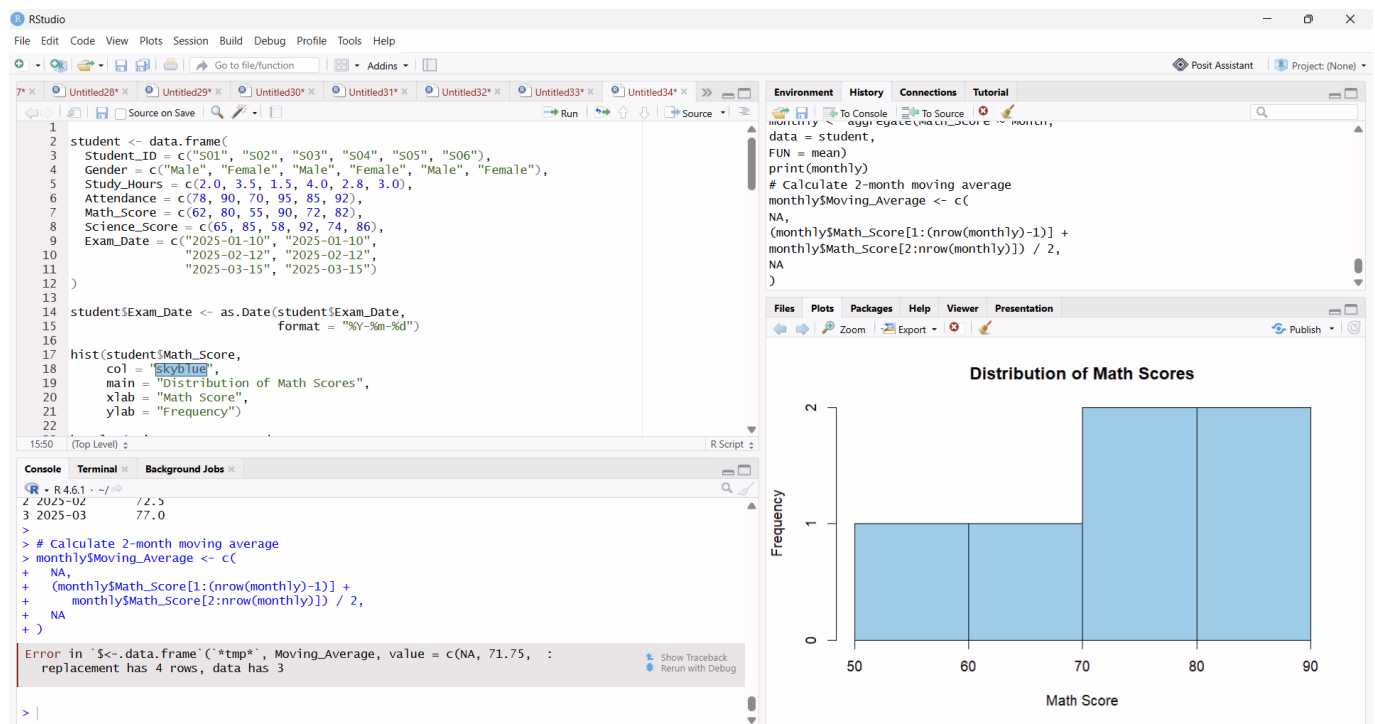
1. Import the dataset in R. Plot a histogram for **Math_Score**. Draw a box plot of **Science_Score** by **Gender**. Add proper title and labels. Interpret the results.
2. Create a scatter plot using R programming between **Study_Hours** and **Math_Score**. Use different colors for **Gender**. Add a regression line. Explain the relationship between study time and performance.
3. Using R, convert **Exam_Date** into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving averages smoothing. Identify any increasing or decreasing pattern.
4. Import **Student_Mini_Data.csv** into Tableau. Create a bar chart showing average Math scores by Gender. Create a line chart for score trends over time. Add a filter for **Exam_Date**. Design a simple interactive dashboard.

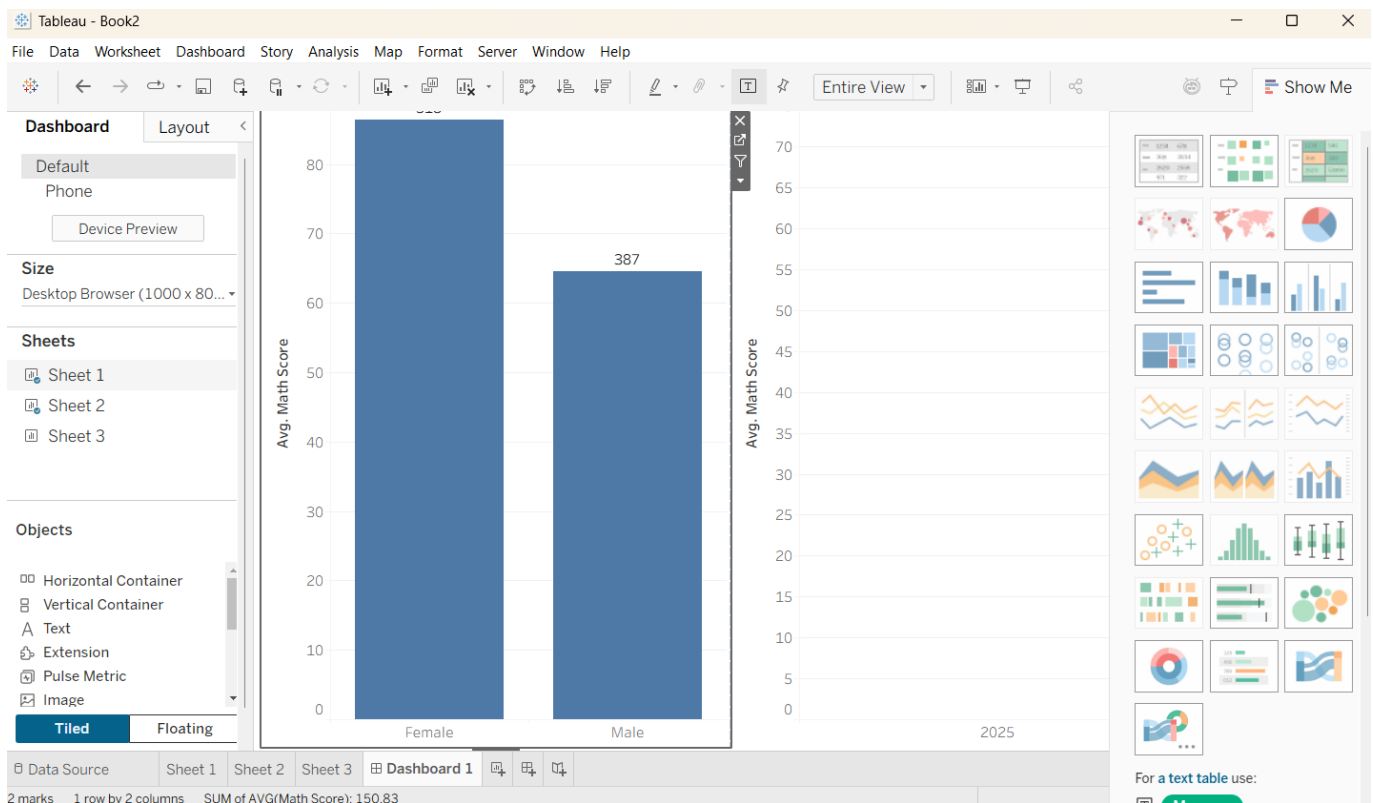
Aim:

To analyze student academic data using R programming and Tableau by creating statistical plots and an interactive dashboard.

Algorithm:

1. Import the Student_Mini_Data.csv dataset into R.
2. Create a histogram for Math Scores.
3. Draw a boxplot of Science Scores grouped by Gender.
4. Create a scatter plot between Study Hours and Math Scores with a regression line.
5. Convert Exam Date into Date format and calculate monthly average Math Scores.
6. Plot a line chart with moving average smoothing.
7. Import the dataset into Tableau and create a bar chart and line chart with an Exam Date filter.
8. Design and save an interactive dashboard.





Result:

The analysis successfully visualized student performance, study habits, and monthly score trends using R and Tableau.

Set16.PatientHealthRiskAnalysis

Dataset

Patient_ID	Age	BMI	BP	Cholesterol
P1	25	22	120	180
P2	40	28	135	210
P3	55	30	145	240
P4	35	26	130	200
P5	60	32	150	260

Questions

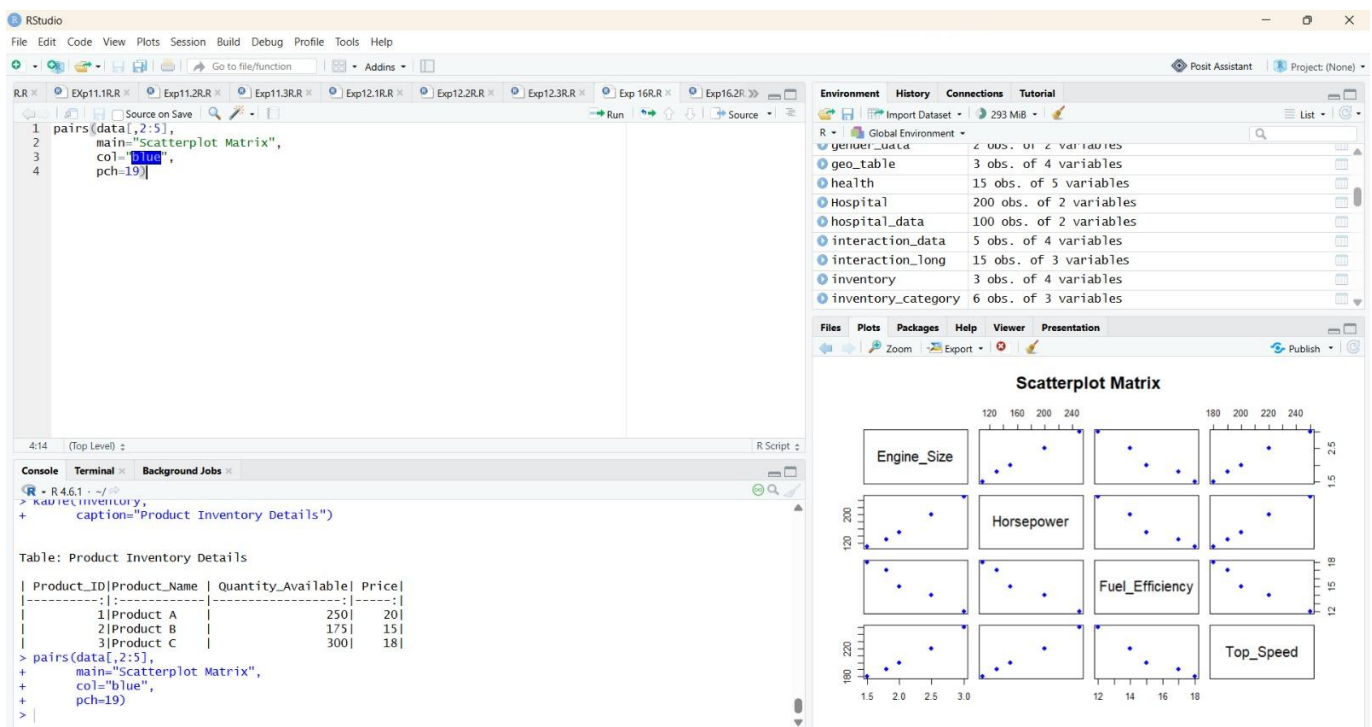
1. Using R, create a scatter plot matrix for Age, BMI, BP, and Cholesterol. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.
2. Construct a Q-Q plot and ECDF for Cholesterol level to assess distribution behaviour. Comment on normality assumption and skewness.
3. Using R, create a bar chart showing the average values of each health indicator (Age, BMI, BP, Cholesterol). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk?
4. Create an interactive Tableau dashboard with a heatmap highlighting high-risk patients and an Age filter for exploring risk patterns. Include a hover tooltip showing all health indicators. Analyze which patients are highest risk, how age affects risk, and how this dashboard can aid healthcare professionals in prioritizing care.

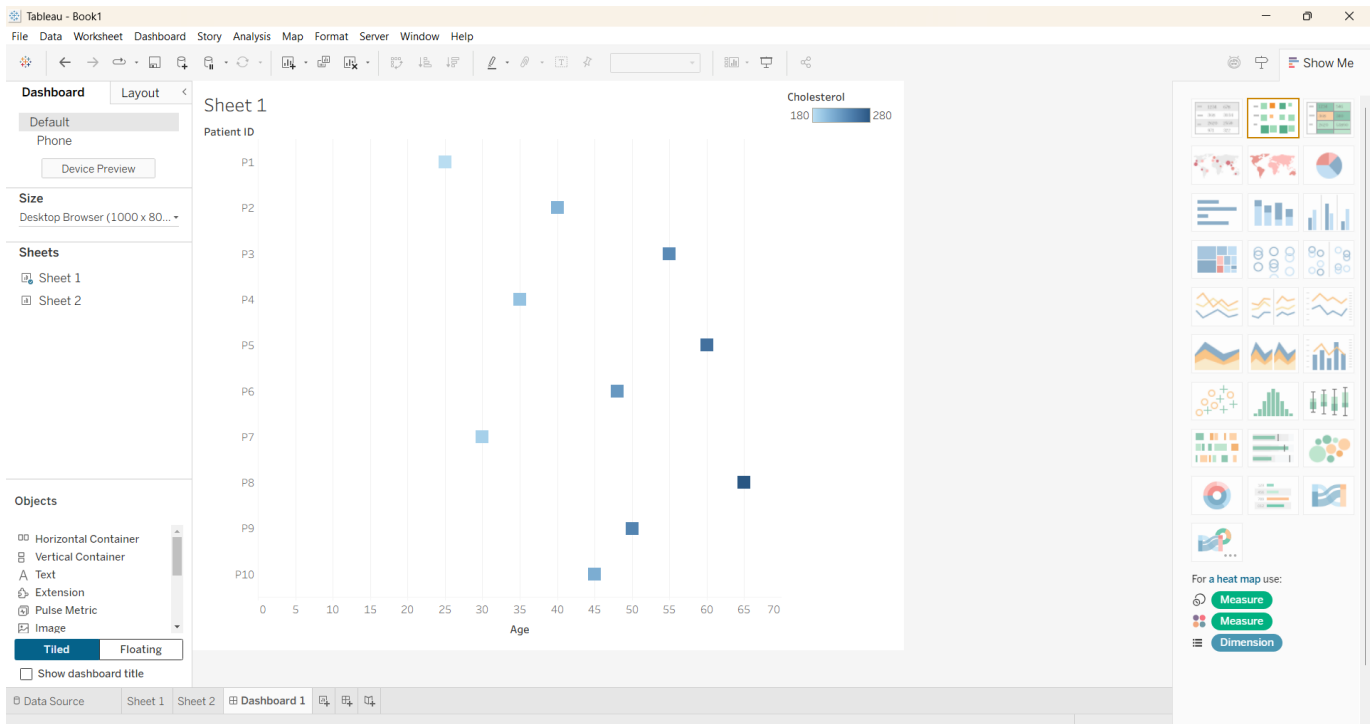
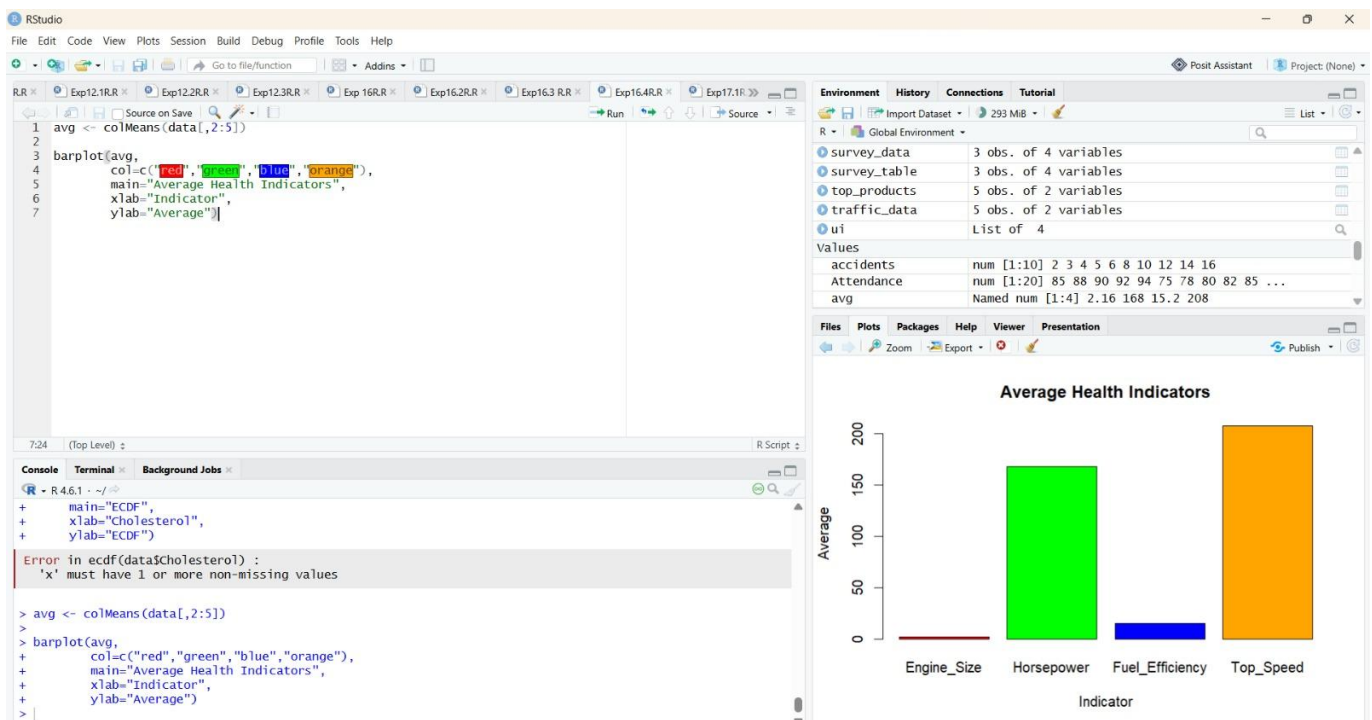
Aim:

To analyze patient health indicators using R programming and Tableau for identifying health risk patterns.

Algorithm:

1. Import the Patient Health dataset into R.
2. Create a scatterplot matrix for Age, BMI, BP, and Cholesterol.
3. Generate a Q-Q plot and ECDF for Cholesterol levels.
4. Calculate the average values of each health indicator.
5. Create a bar chart showing the average health indicators.
6. Import the dataset into Tableau.
7. Create a heatmap highlighting high-risk patients and add an Age filter with tooltips.
8. Save the interactive dashboard.





Result:

The visualizations successfully identified patient health trends and risk patterns, enabling effective analysis through an interactive Tableau dashboard.

SET17–VehiclePerformanceAnalysis

Dataset

Vehicle_ID	Engine_Size(L)	Horsepower	Fuel_Efficiency(km/l)	Top_Speed(km/h)	Safety_Rating
V1	1.5	110	18	180	4
V2	2.0	150	15	200	5
V3	3.0	250	12	250	5
V4	2.5	200	14	220	4
V5	1.8	130	17	190	3

Questions

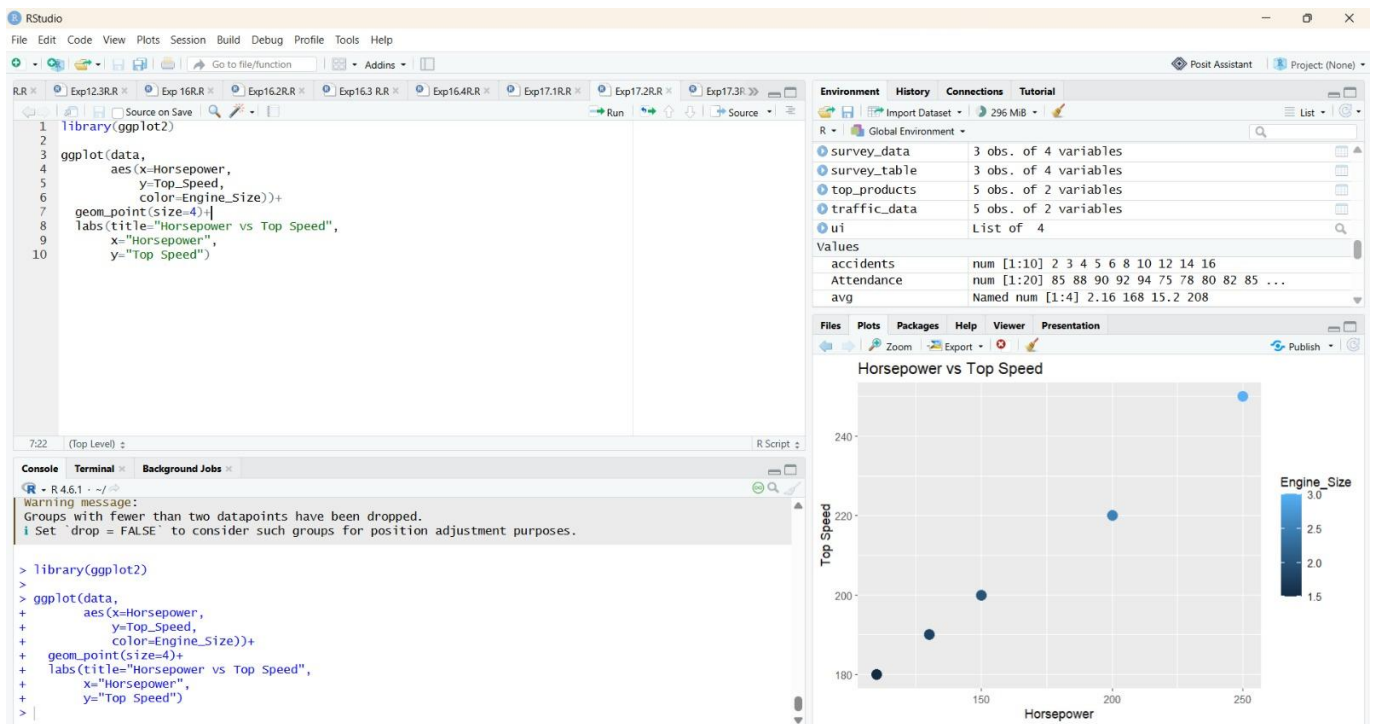
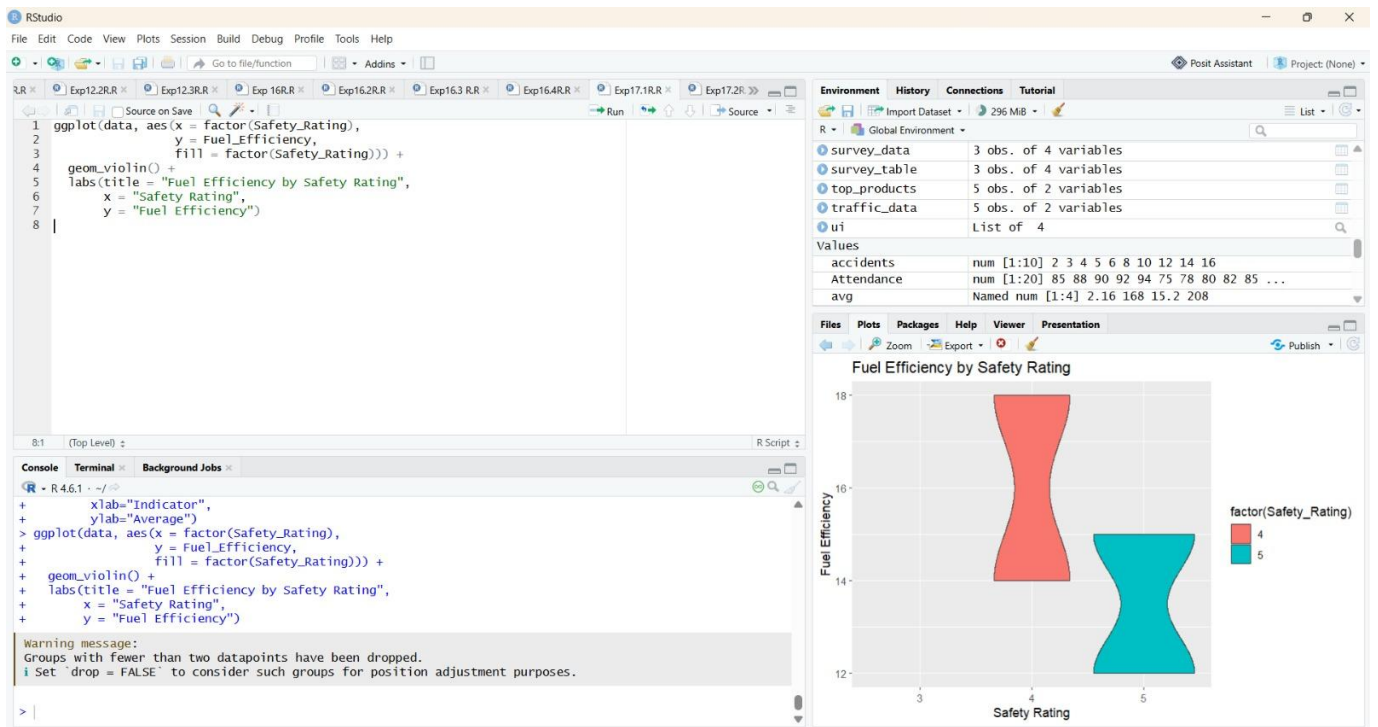
1. Using R, develop a violin plot to compare the distribution of **Fuel_Efficiency** across different **Safety_Rating** categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.
2. Using R, construct a scatter plot to explore the relationship between **Horsepower** and **Top_Speed**, applying color differentiation based on **Engine_Size**. Analyze the strength and direction of correlation and find observable performance trends.
3. Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with **Top_Speed** and their influence on overall vehicle performance.
4. In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for **Engine_Size** and **Safety_Rating**, to highlight vehicles that balance fuel efficiency and high performance. Provide analytical interpretation of the insights derived from the dashboard.

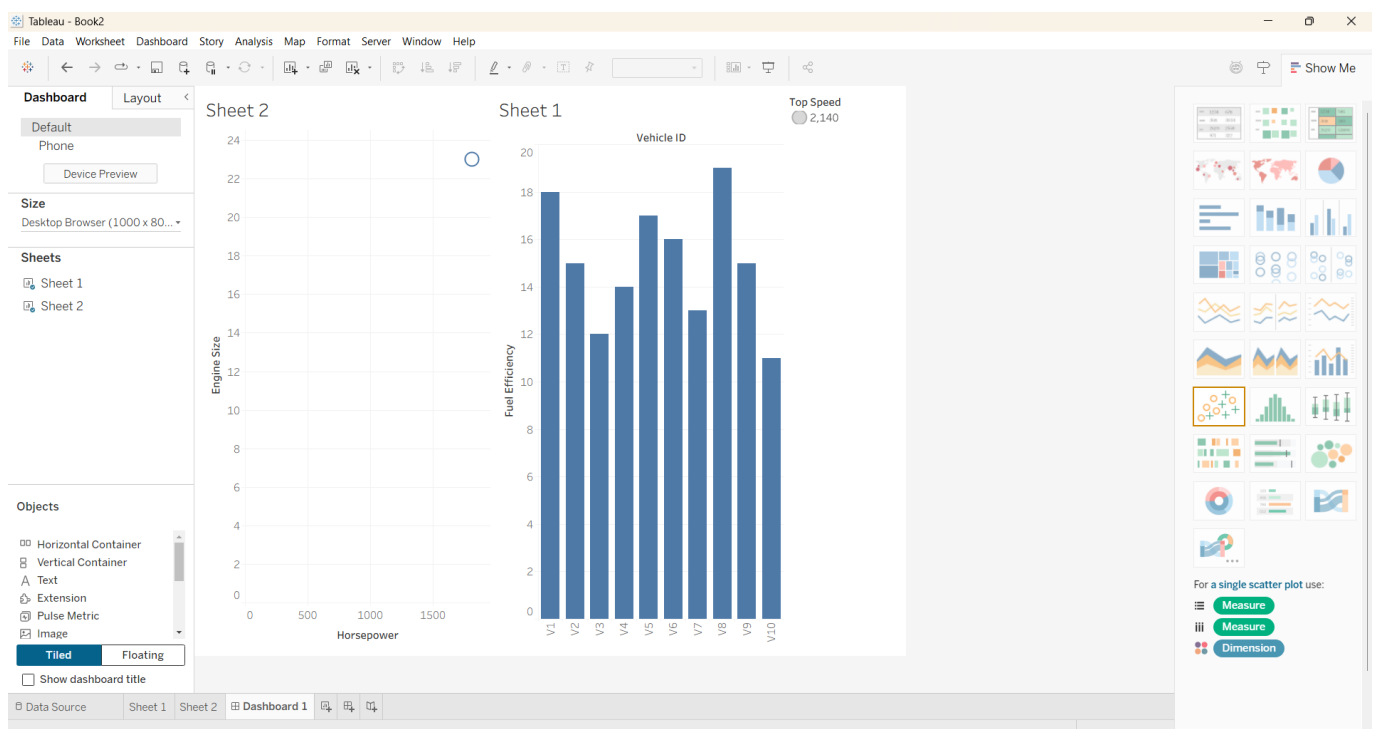
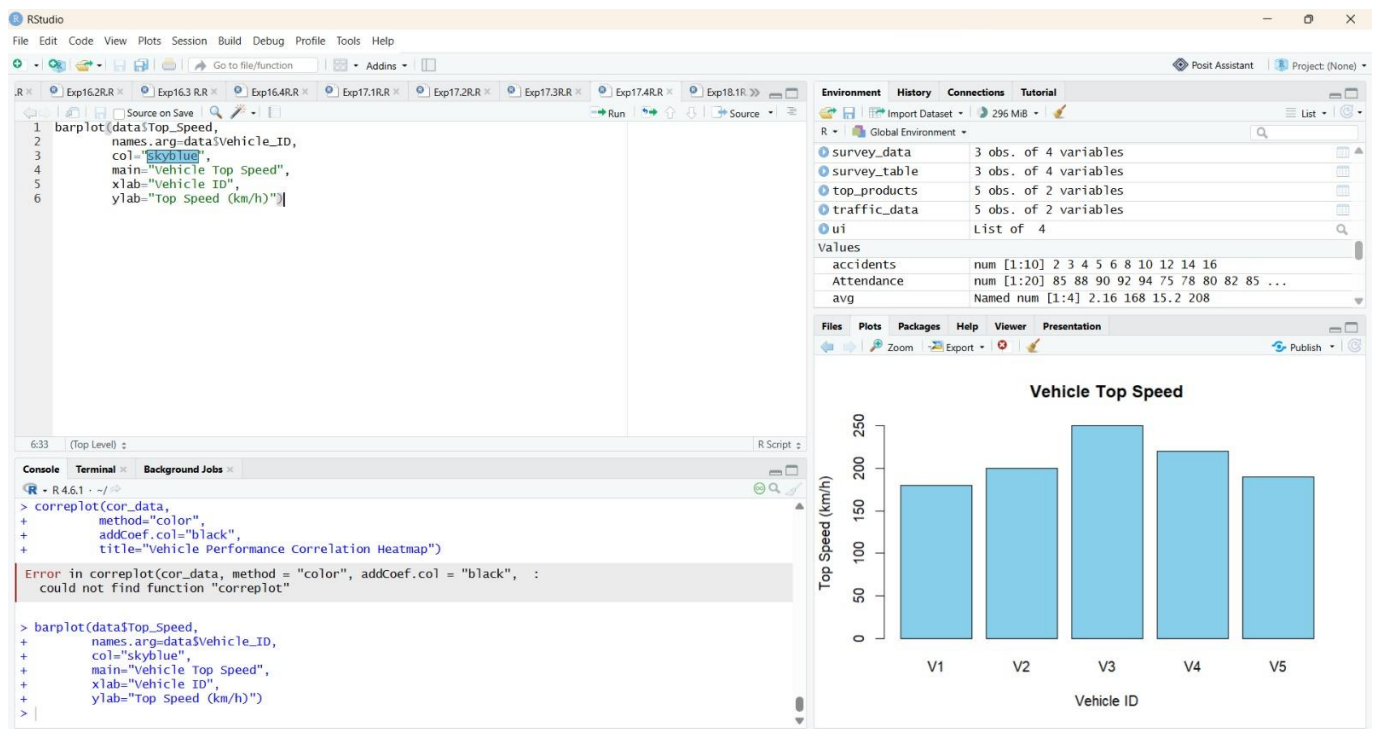
Aim:

To analyze vehicle performance using R programming and Tableau through advanced visualizations and an interactive dashboard.

Algorithm:

1. Import the Vehicle Performance dataset into R.
2. Create a violin plot comparing Fuel Efficiency across Safety Rating categories.
3. Generate a scatter plot between Horsepower and Top Speed using Engine Size as color.
4. Create a correlation heatmap for all numerical variables.
5. Analyze the relationships among performance variables.
6. Import the dataset into Tableau.
7. Create bar charts and scatter plots with Engine Size and Safety Rating filters.
8. Design and save the dashboard.





Result:

The analysis successfully illustrated vehicle performance characteristics, fuel efficiency, and relationships among performance variables using R and Tableau.

SET18–StudentAcademicPerformance

Dataset

Student_ID	Age	Study_Hours	Attendance (%)	Test_Score	Participation_Score
S1	19	12	90	85	8
S2	21	8	70	70	7
S3	20	15	95	92	9
S4	22	10	85	80	8
S5	23	7	60	65	6

Questions

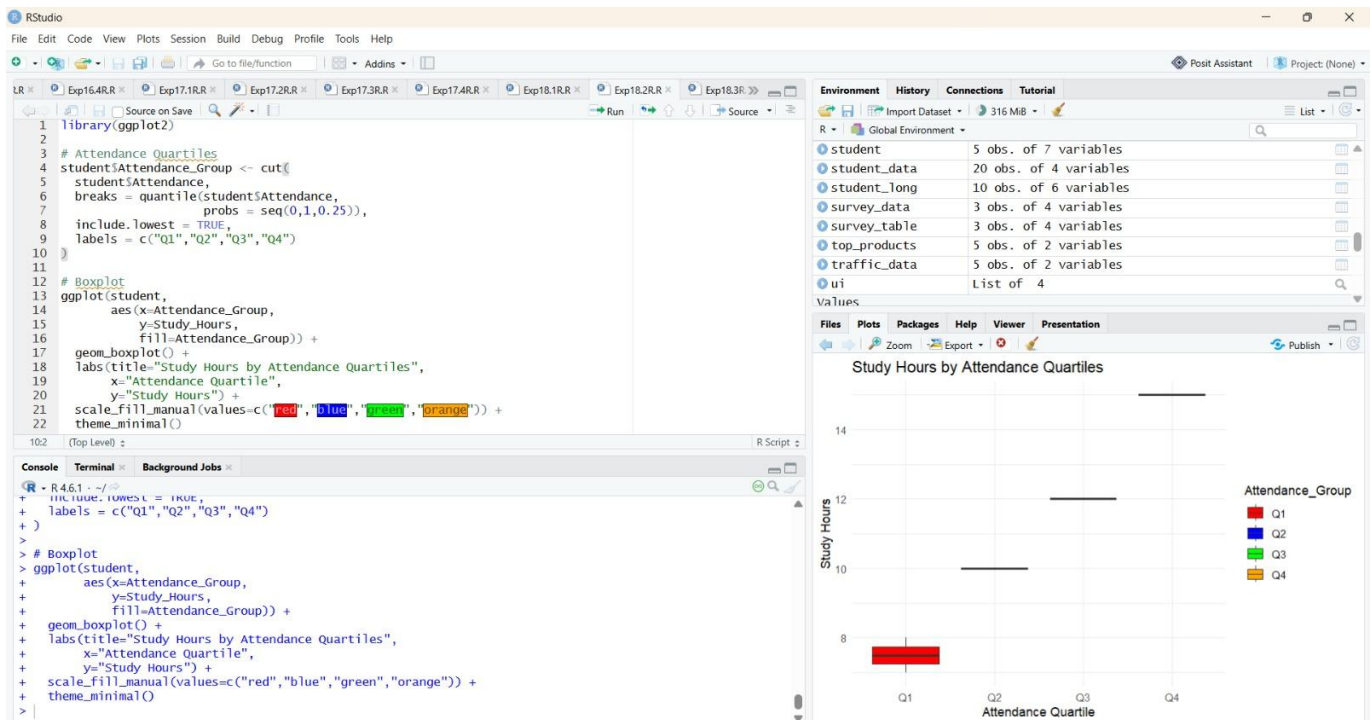
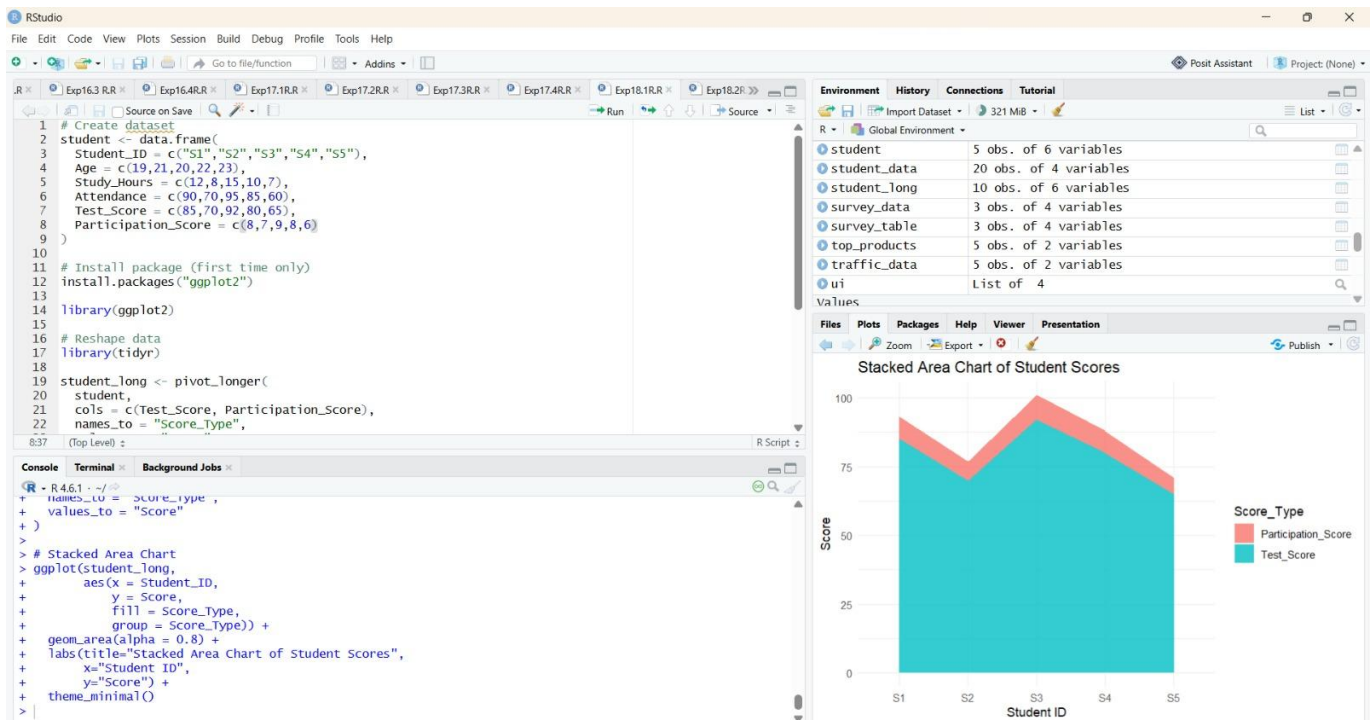
1. Using R, create a stacked area chart showing **Test_Score** and **Participation_Score** across students. Analyze trends.
2. Using R, generate a boxplot of **Study_Hours** grouped by **Attendance** quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.
3. Using R, construct a density plot for **Test_Score** to evaluate score distribution and outliers.
4. In Tableau, design an interactive scatter plot with **Study_Hours** on X-axis, **Test_Score** on Y-axis, colored by **Participation_Score**. Include filters for **Age**.

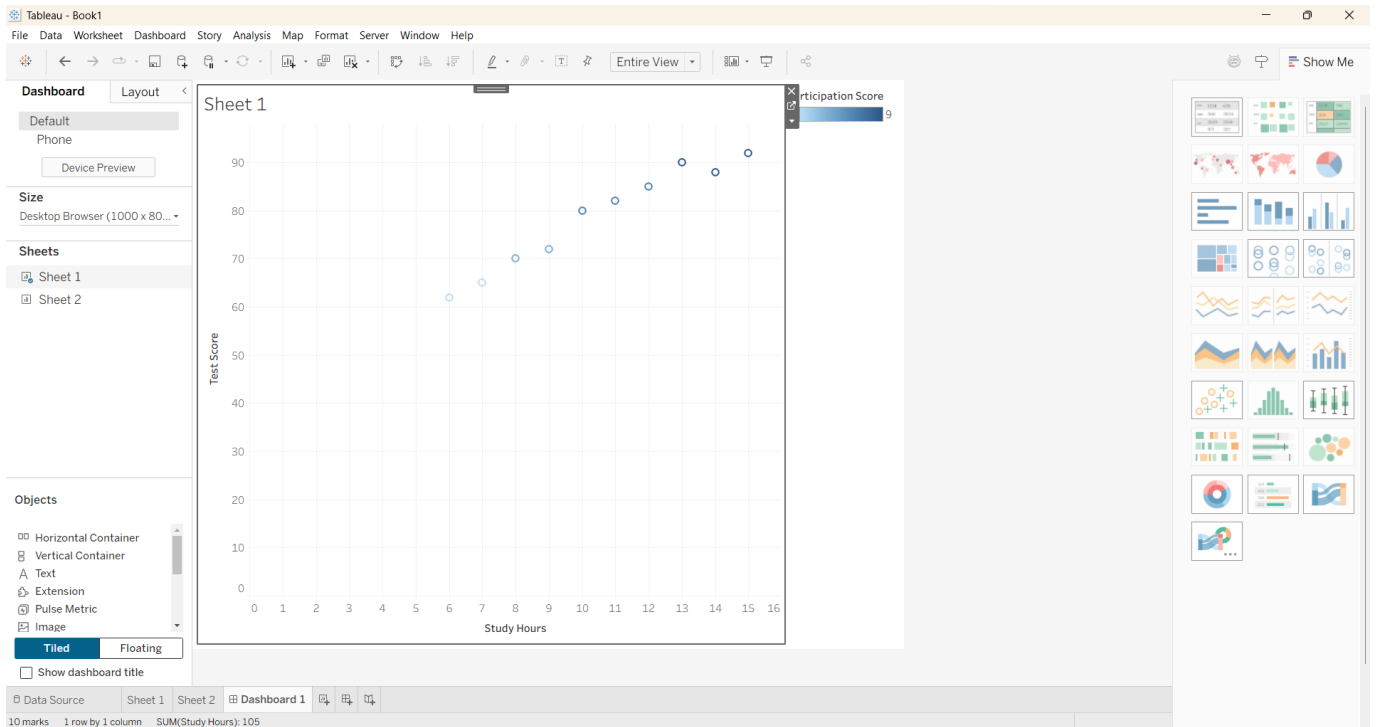
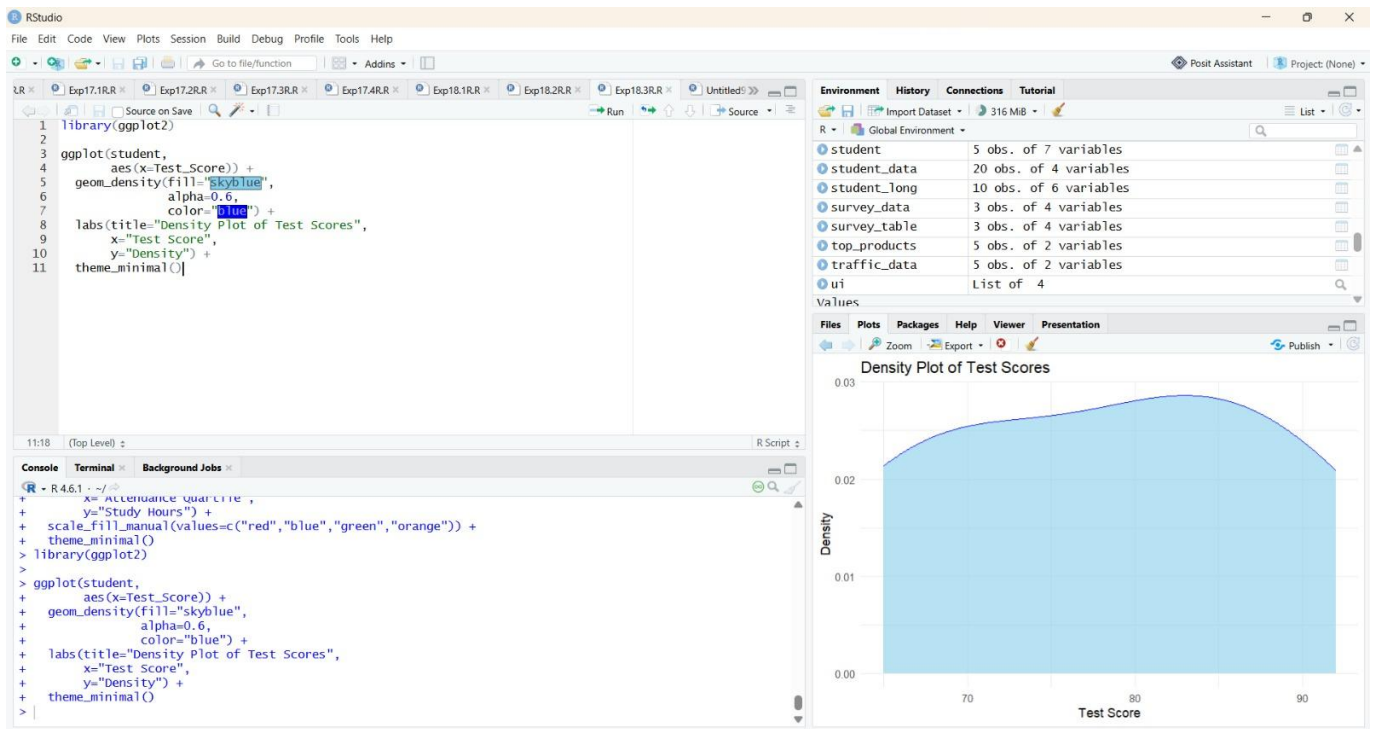
Aim:

To analyze student academic performance using R programming and Tableau by creating statistical visualizations and dashboards.

Algorithm:

1. Import the Student Academic Performance dataset into R.
2. Create a stacked area chart for Test Score and Participation Score.
3. Generate a boxplot of Study Hours grouped by Attendance quartiles.
4. Create a density plot for Test Scores.
5. Add titles, labels, and colors to the visualizations.
6. Import the dataset into Tableau.
7. Create a scatter plot between Study Hours and Test Score colored by Participation Score with an Age filter.
8. Save the dashboard.





Result:

The required visualizations successfully displayed academic performance trends, study patterns, and score distributions through R and Tableau.

SET19–MobileAppUsageAnalysis

Dataset: Mobile_App_Usage.csv

User_ID	Gender	Age	Screen_Time (hrs)	App_Usage_Count	Data_Used(GB)	Satisfaction	Usage_Date
U01	Male	20	4.5	18	2.4	3	2025-01-08
U02	Female	22	6.0	25	3.8	5	2025-01-08
U03	Male	19	3.2	12	1.6	3	2025-02-11
U04	Female	21	7.1	30	4.5	5	2025-02-11
U05	Male	23	2.8	10	1.2	2	2025-03-14
U06	Female	20	5.4	22	3.1	4	2025-03-14

Questions

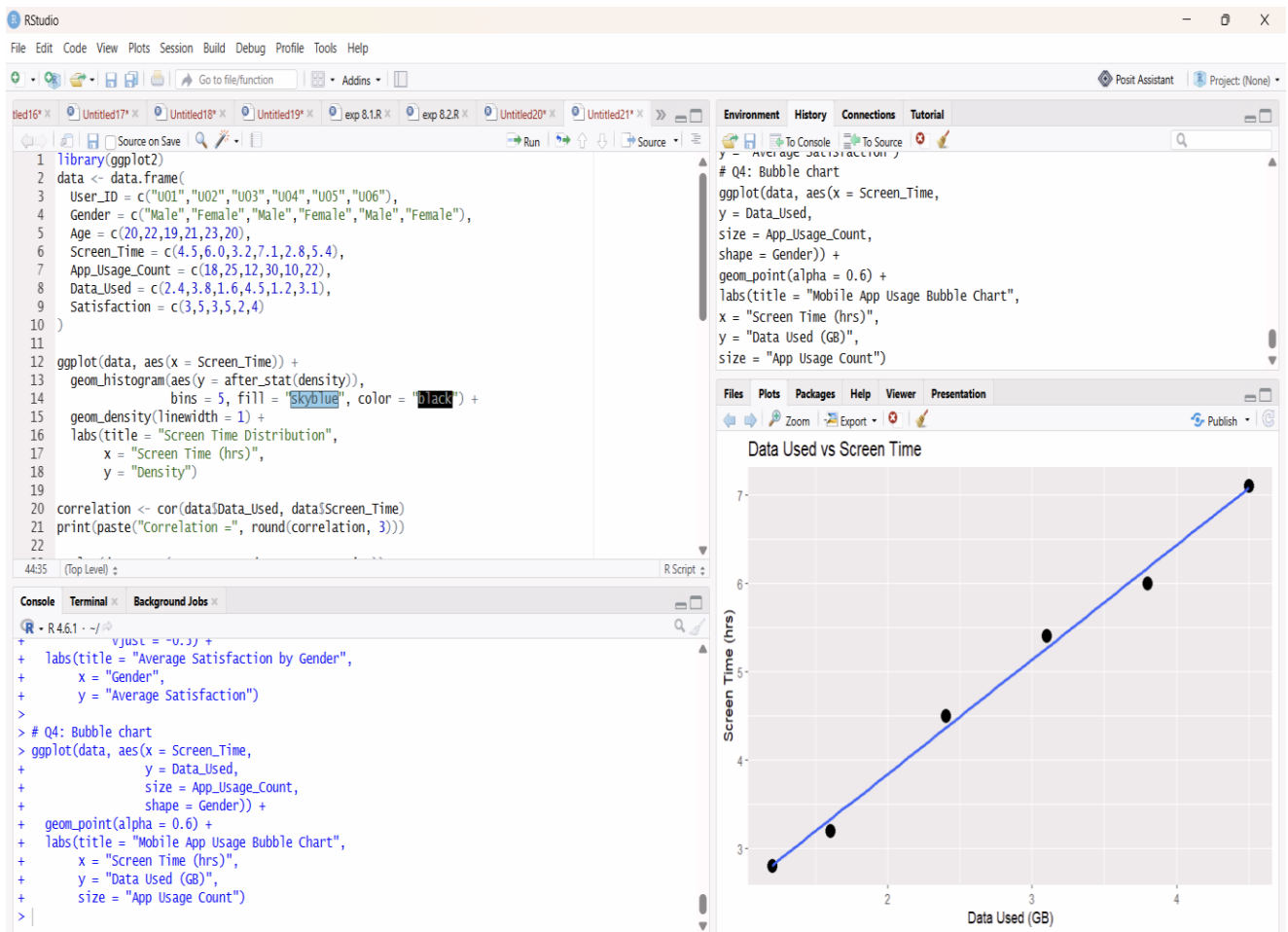
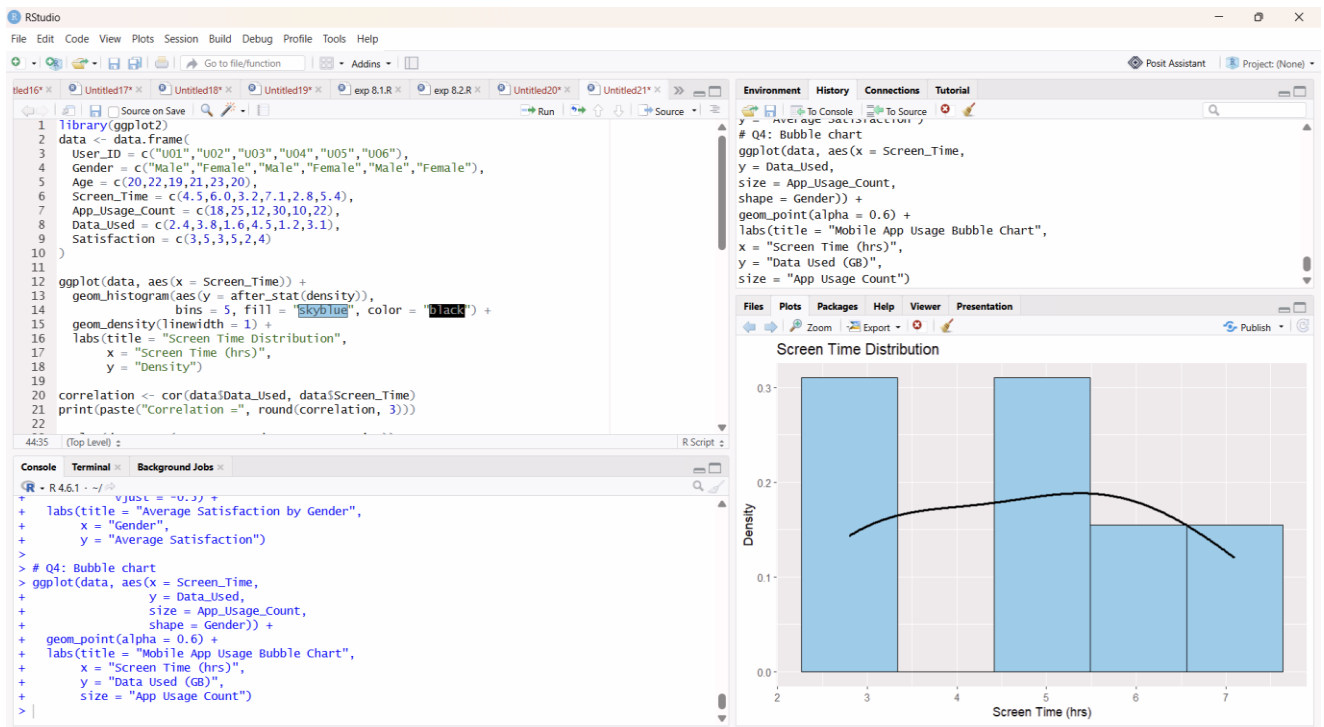
1. Import the dataset in R. Plot a histogram and density plot for **Screen_Time**. Add suitable colors and labels. Interpret the screen-time distribution.
2. Create a scatter plot using R programming between **Data_Used** and **Screen_Time**. Calculate the correlation between them. Add a trend line. Explain the relationship.
3. Using R, find the average **Satisfaction** score for each **Gender**. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.
4. Import **Mobile_App_Usage.csv** into Tableau. Create a bubble chart (**Screen_Time vs Data_Used**). Include filters for **Gender** and **Usage_Date**. Design an interactive dashboard and summarize the key insights.

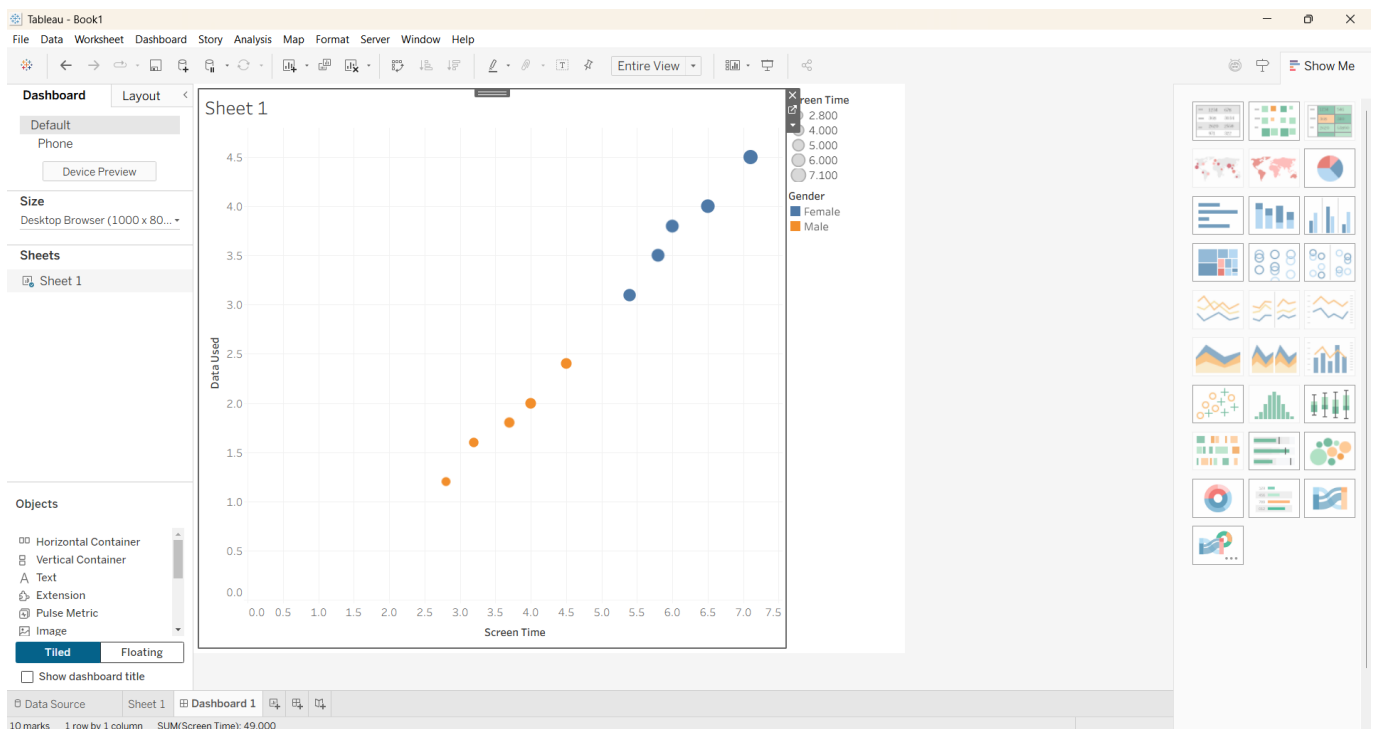
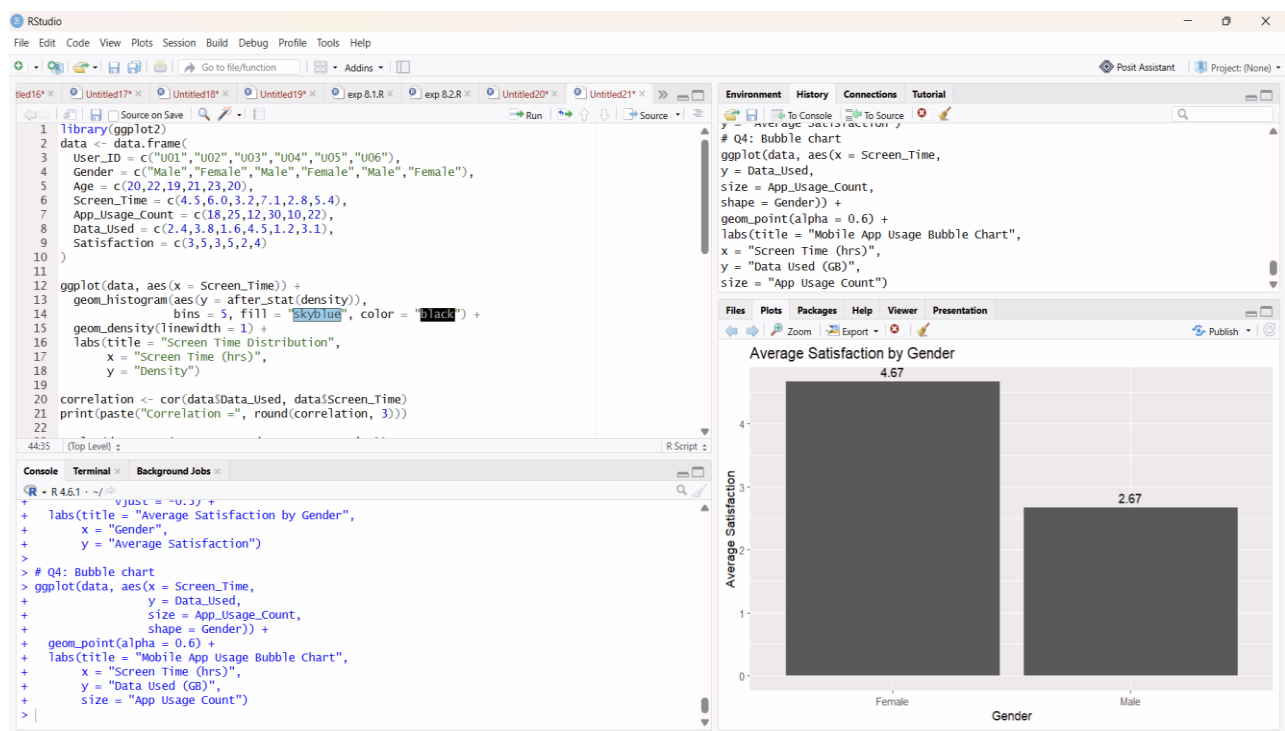
Aim:

To analyze mobile application usage data using R programming and Tableau through statistical visualizations and an interactive dashboard.

Algorithm:

1. Import the Mobile_App_Usage.csv dataset into R.
2. Create a histogram and density plot for Screen Time.
3. Generate a scatter plot between Data Used and Screen Time with a trend line.
4. Calculate the correlation between Screen Time and Data Used.
5. Compute the average Satisfaction score by Gender.
6. Create a bar chart with value labels.
7. Import the dataset into Tableau and create a bubble chart with Gender and Usage Date filters.
8. Save the interactive dashboard.





Result:

The visualizations successfully analyzed user screen time, data usage, satisfaction levels, and overall mobile application usage patterns.

SET20.HospitalAppointmentAnalysis

Dataset:HospitalAppointments

PatientID	Age	WaitingTime(Days)	AppointmentStatus
1	25	2	Attended
2	40	5	Missed
3	35	1	Attended
4	50	7	Missed
5	29	3	Attended

Questions

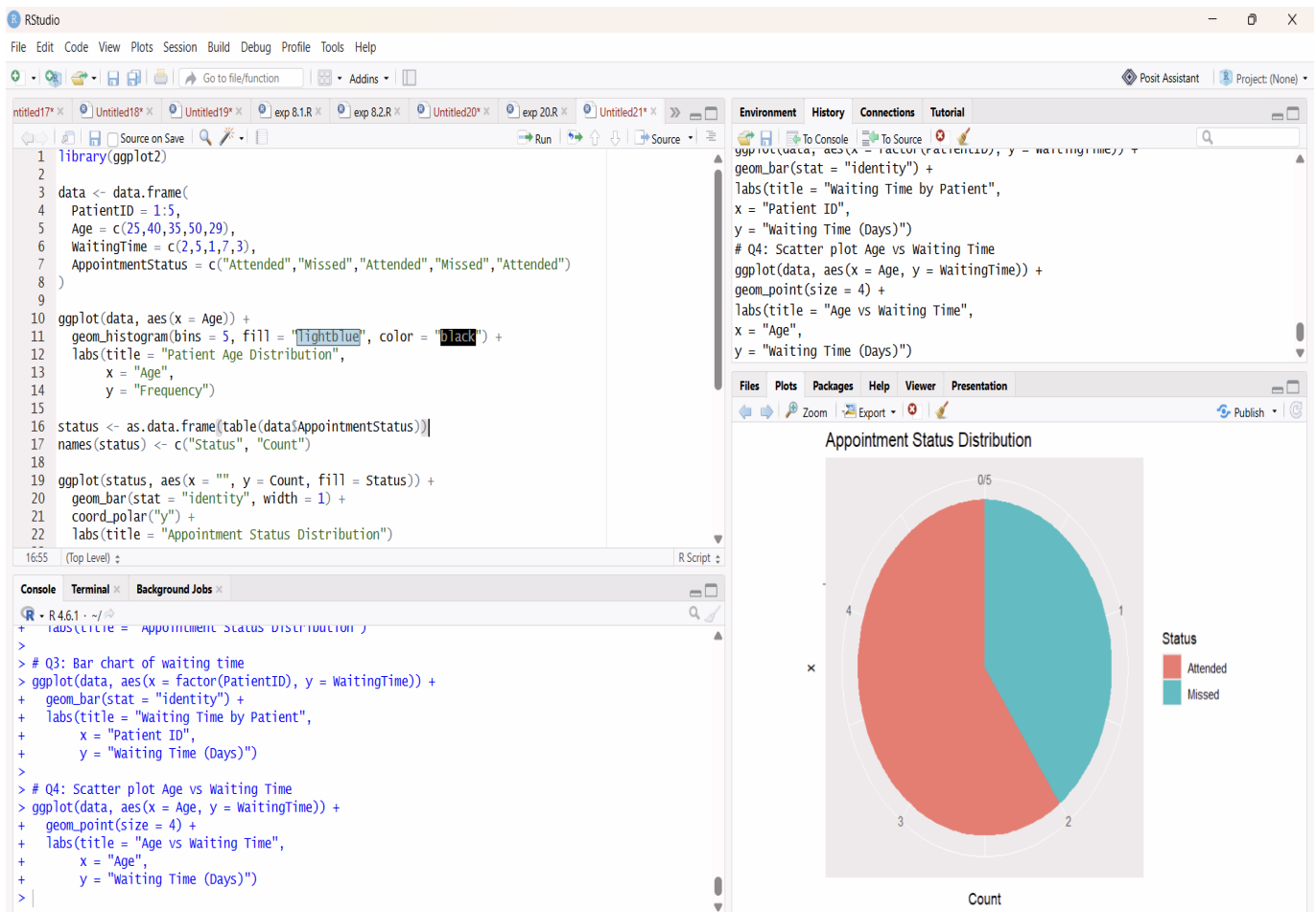
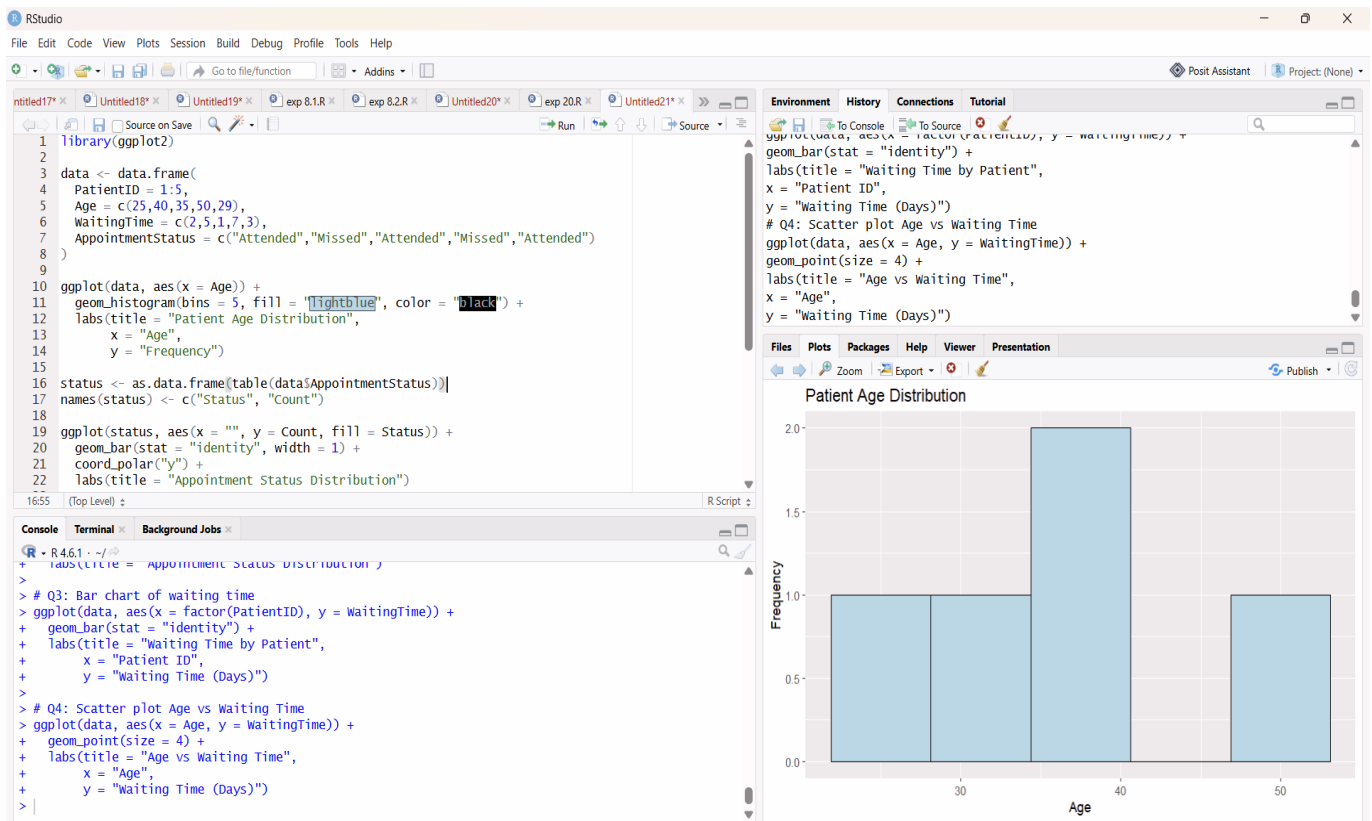
1. Create a histogram showing patient ages.
2. Generate a pie chart for appointment status distribution.
3. Build a bar chart of waiting times by patient.
4. Create a scatter plot of age vs waiting time.

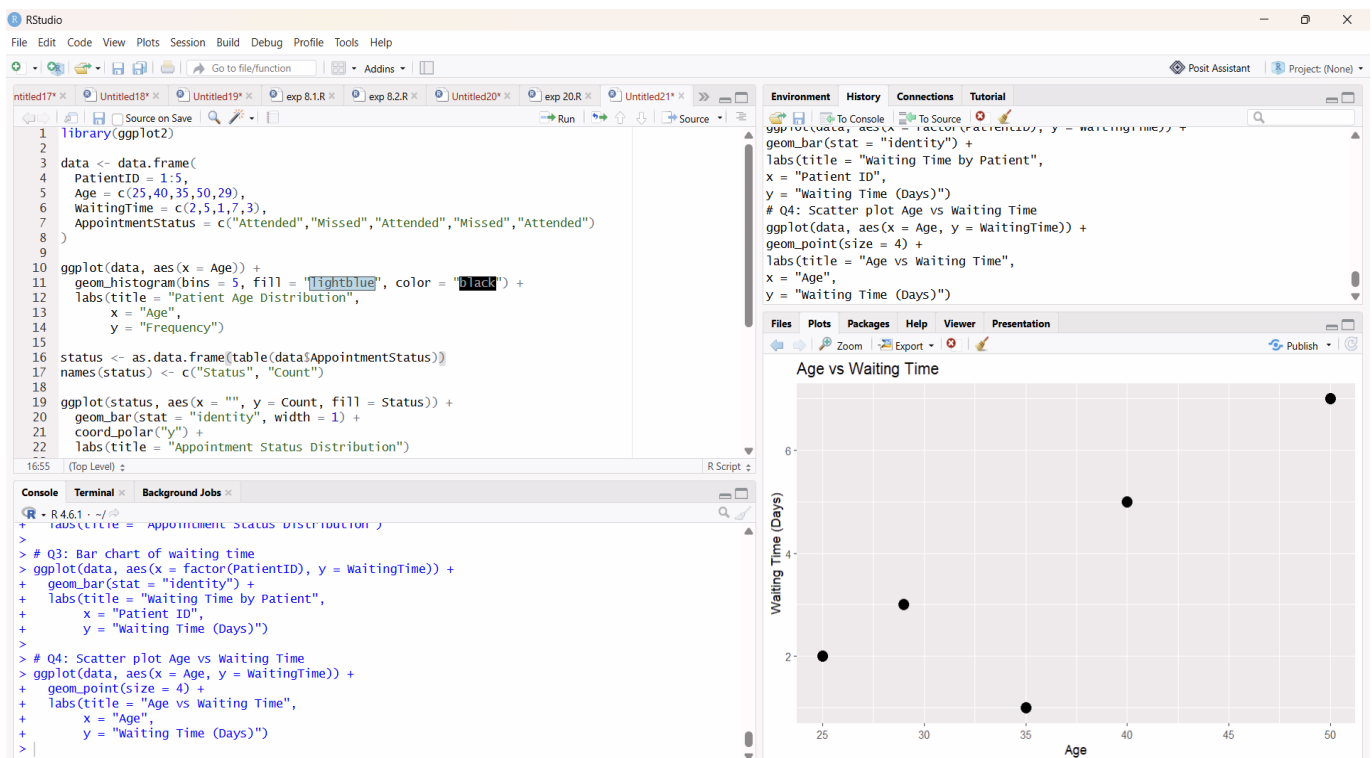
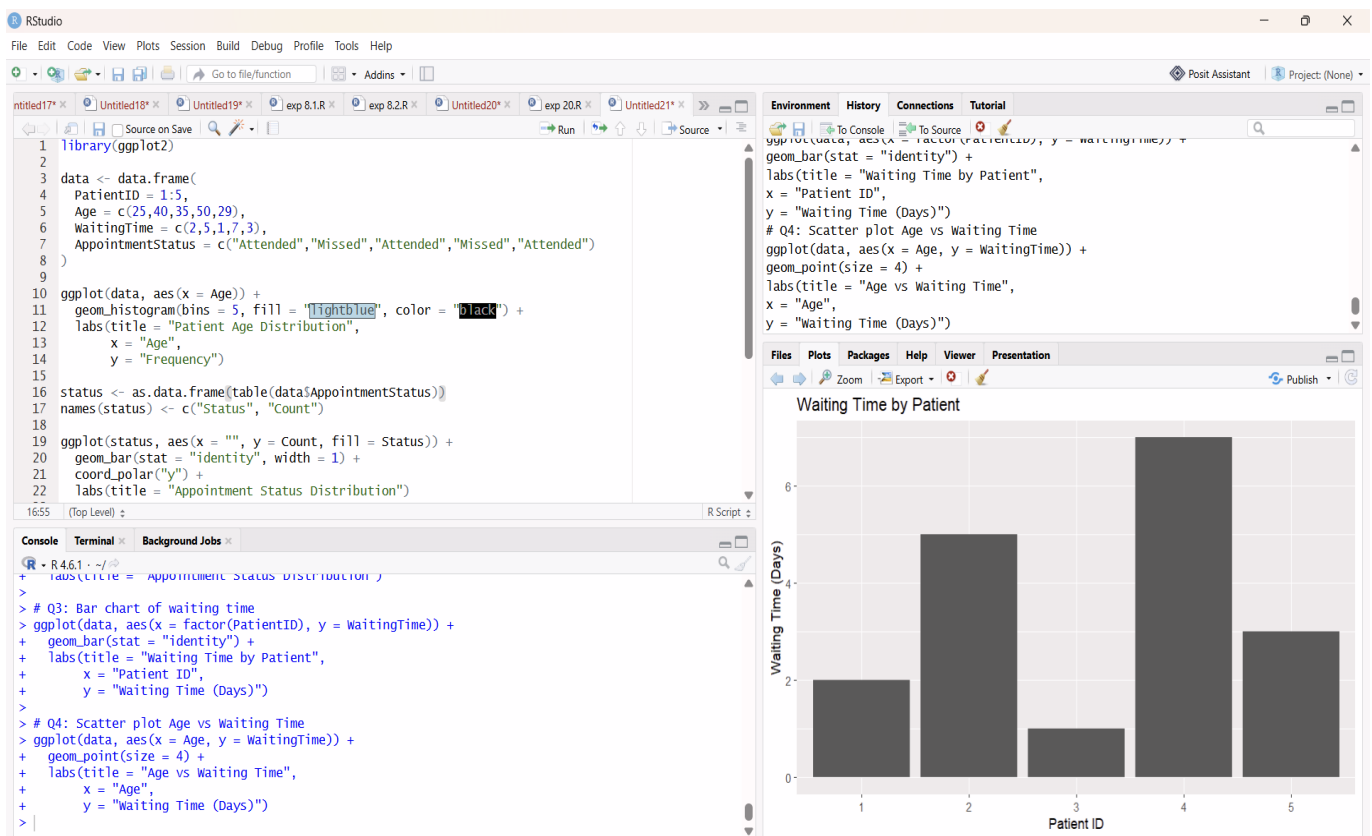
Aim:

To analyze hospital appointment data using Tableau by creating visualizations for patient age, appointment status, waiting time, and their relationship.

Algorithm:

1. Open Tableau and import the Hospital Appointments dataset.
2. Verify the imported data fields.
3. Create a histogram showing patient age distribution.
4. Generate a pie chart for appointment status.
5. Build a bar chart displaying waiting time by patient.
6. Create a scatter plot between Age and Waiting Time.
7. Add titles, labels, legends, and formatting.
8. Save the Tableau workbook.





Result:

The required visualizations were successfully created, providing a clear understanding of patient age distribution, appointment status, waiting times, and the relationship between age and waiting time.

SET21.MovieRatingsAnalysis

Dataset:MovieRatings

Movie ID	Genre	Rating	Duration(Min)
1	Action	4.5	120
2	Comedy	3.8	90
3	Drama	4.2	140
4	Action	4.7	130
5	Comedy	3.5	95

Questions

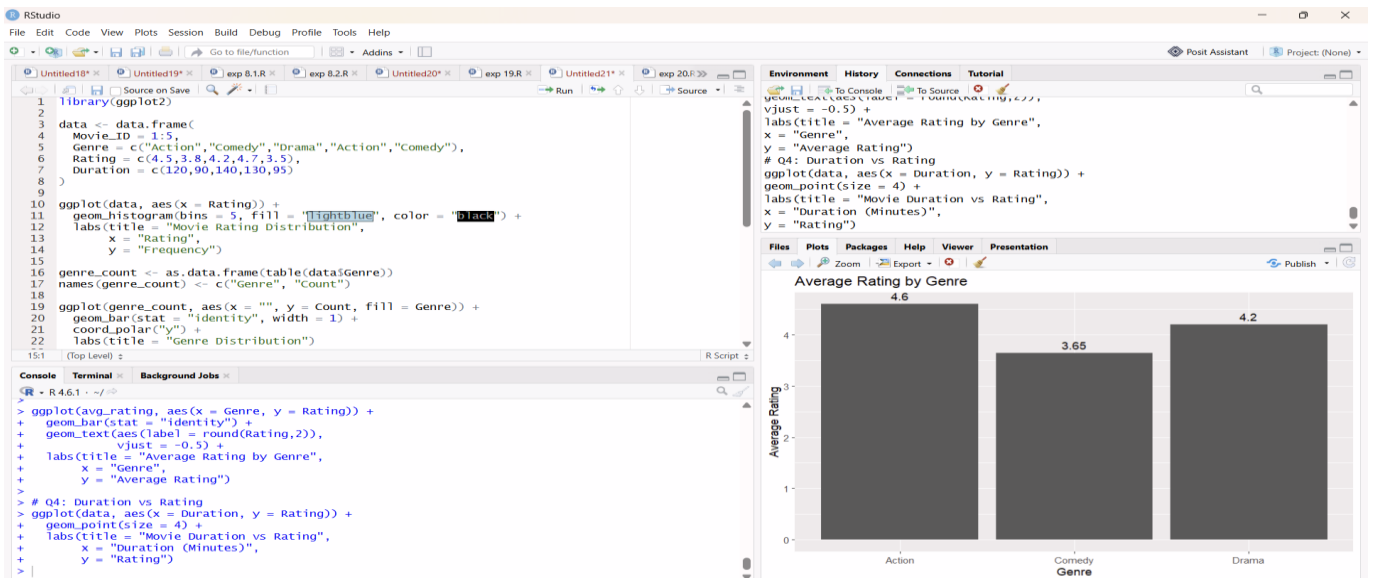
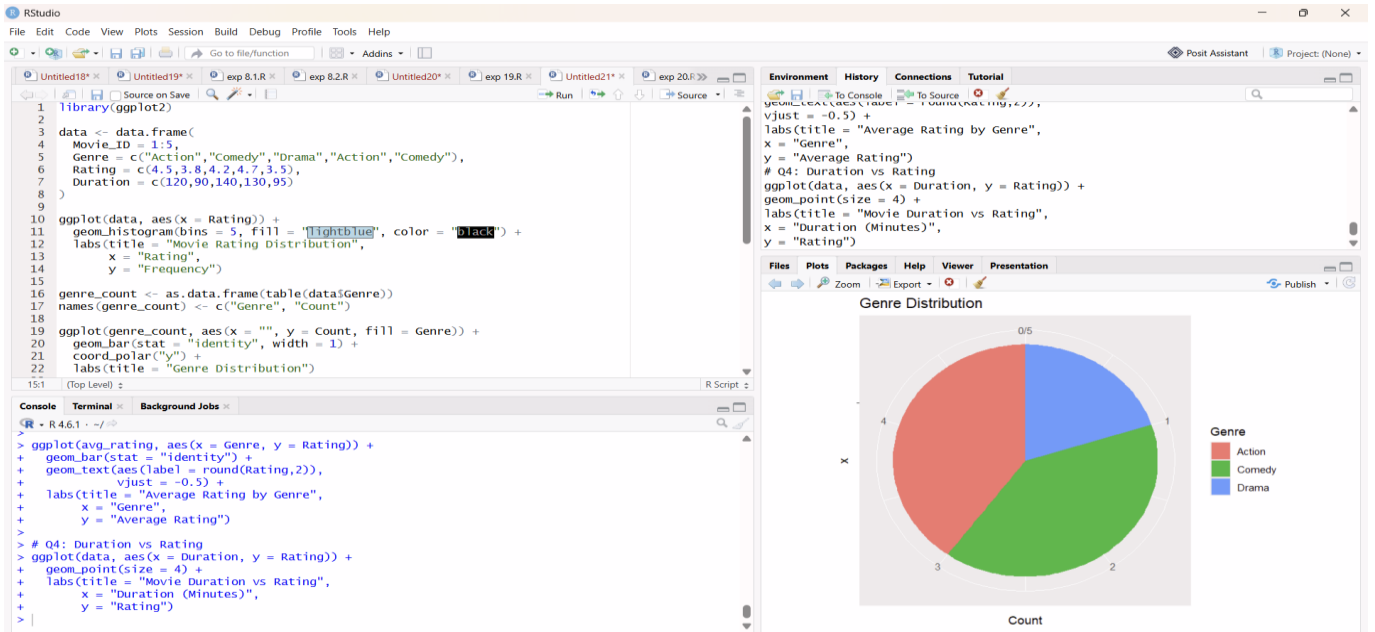
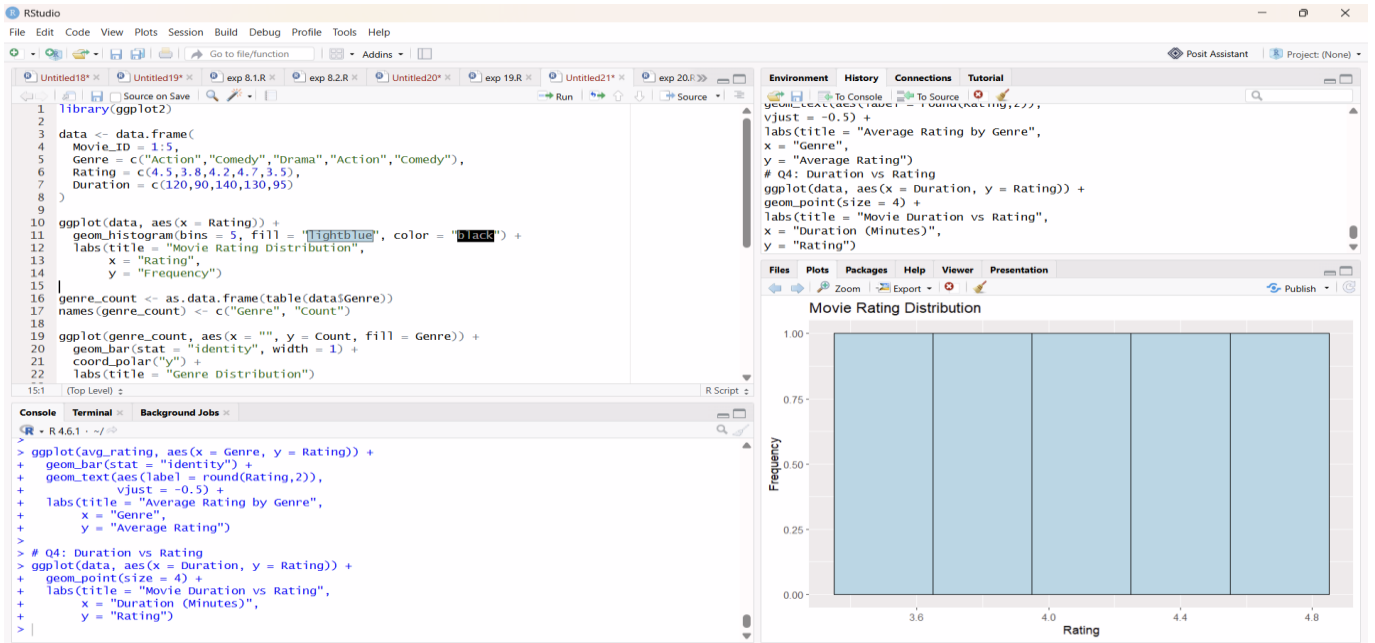
1. Create a histogram of movie ratings.
2. Generate a pie chart showing genre distribution.
3. Build a bar chart of average ratings by genre.
4. Create a scatter plot of duration vs rating.

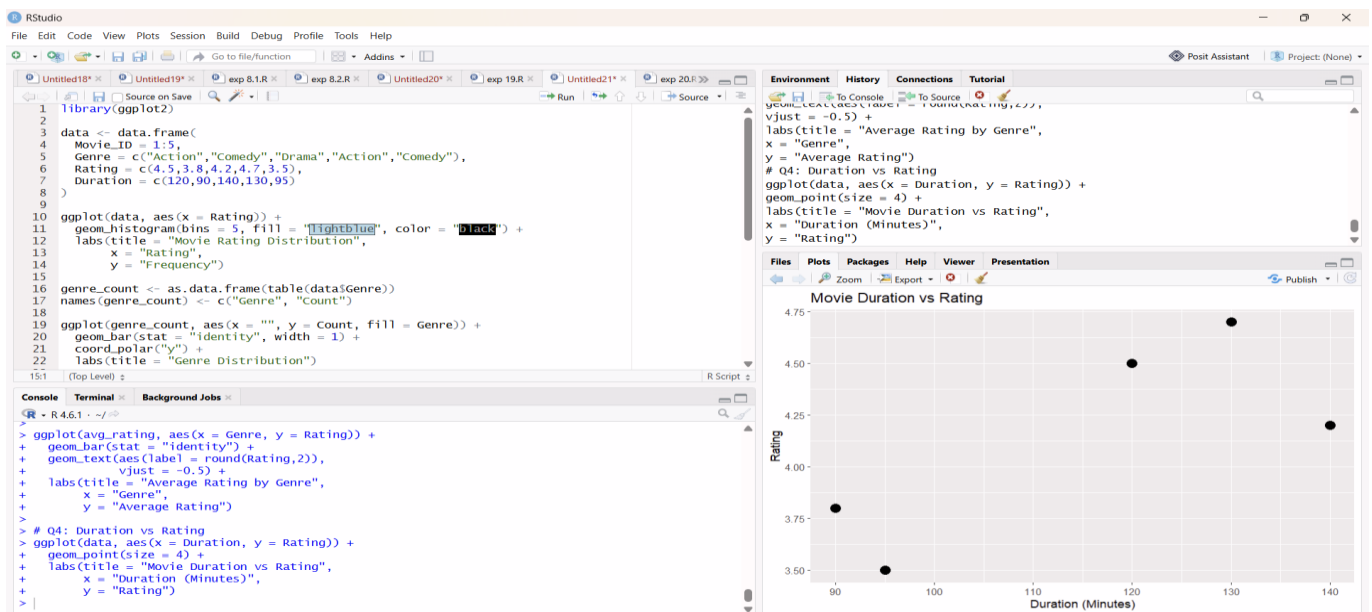
Aim:

To analyze movie ratings data using Tableau by creating visualizations for ratings, genres, and movie duration.

Algorithm:

1. Open Tableau and import the Movie Ratings dataset.
2. Verify the imported data fields.
3. Create a histogram for movie ratings.
4. Generate a pie chart showing the distribution of movie genres.
5. Build a bar chart displaying average ratings by genre.
6. Create a scatter plot between movie duration and ratings.
7. Add appropriate titles, labels, legends, and formatting.
8. Save the Tableau workbook.





Result:

The required visualizations were successfully created, providing insights into movie ratings, genre distribution, average ratings, and the relationship between movie duration and ratings.

SET22.LibraryBorrowingRecords

Dataset:LibraryUsage

UserID	BooksBorrowed	Days Kept	FineAmount
1	2	10	0
2	5	25	15
3	3	14	0
4	6	30	20
5	1	7	0

Questions

1. Create a histogram of books borrowed.
2. Generate a pie chart for users with and without fines.
3. Build a bar chart of fine amounts by user.
4. Create a scatter plot of days kept vs fine amount.

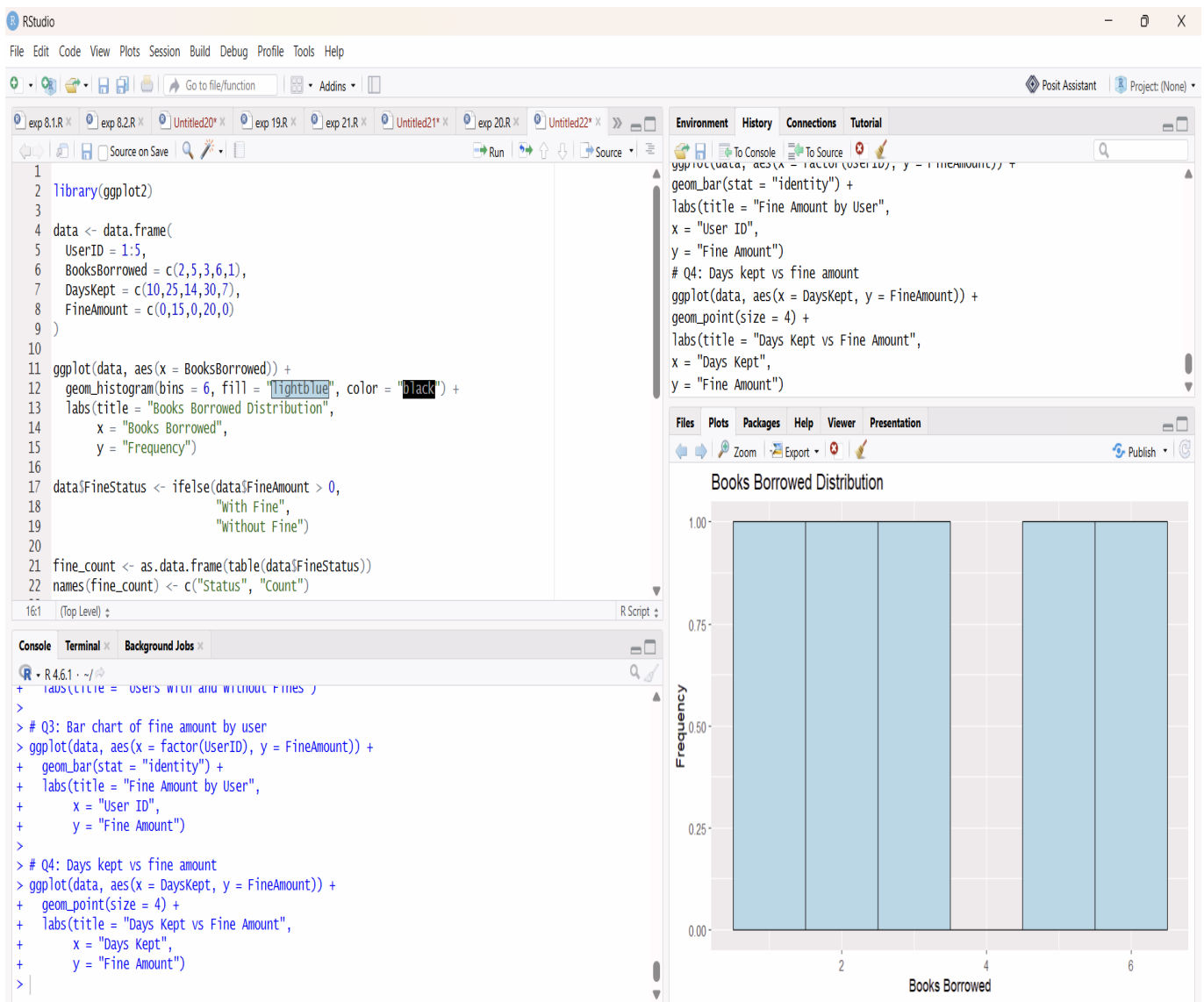
Aim:

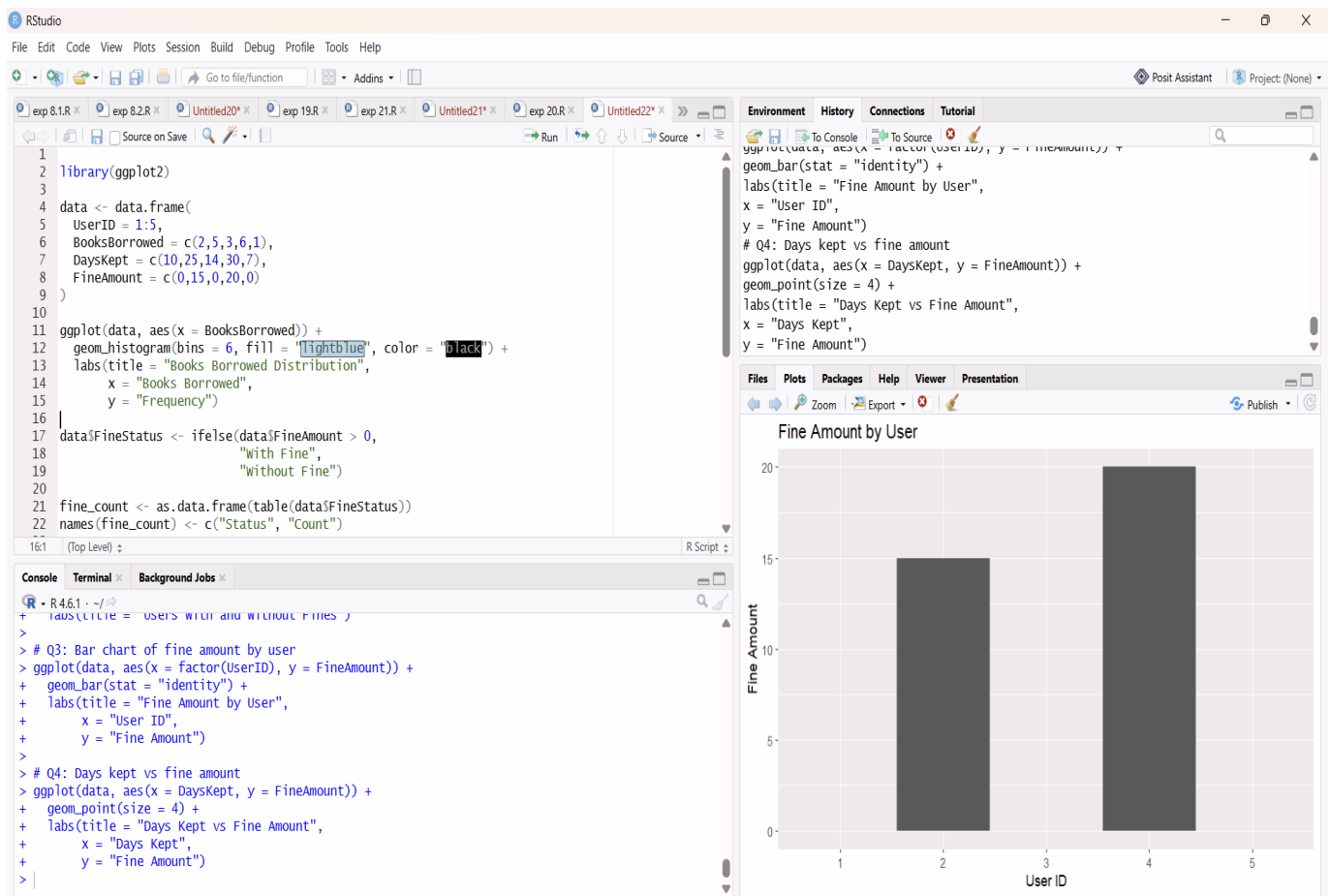
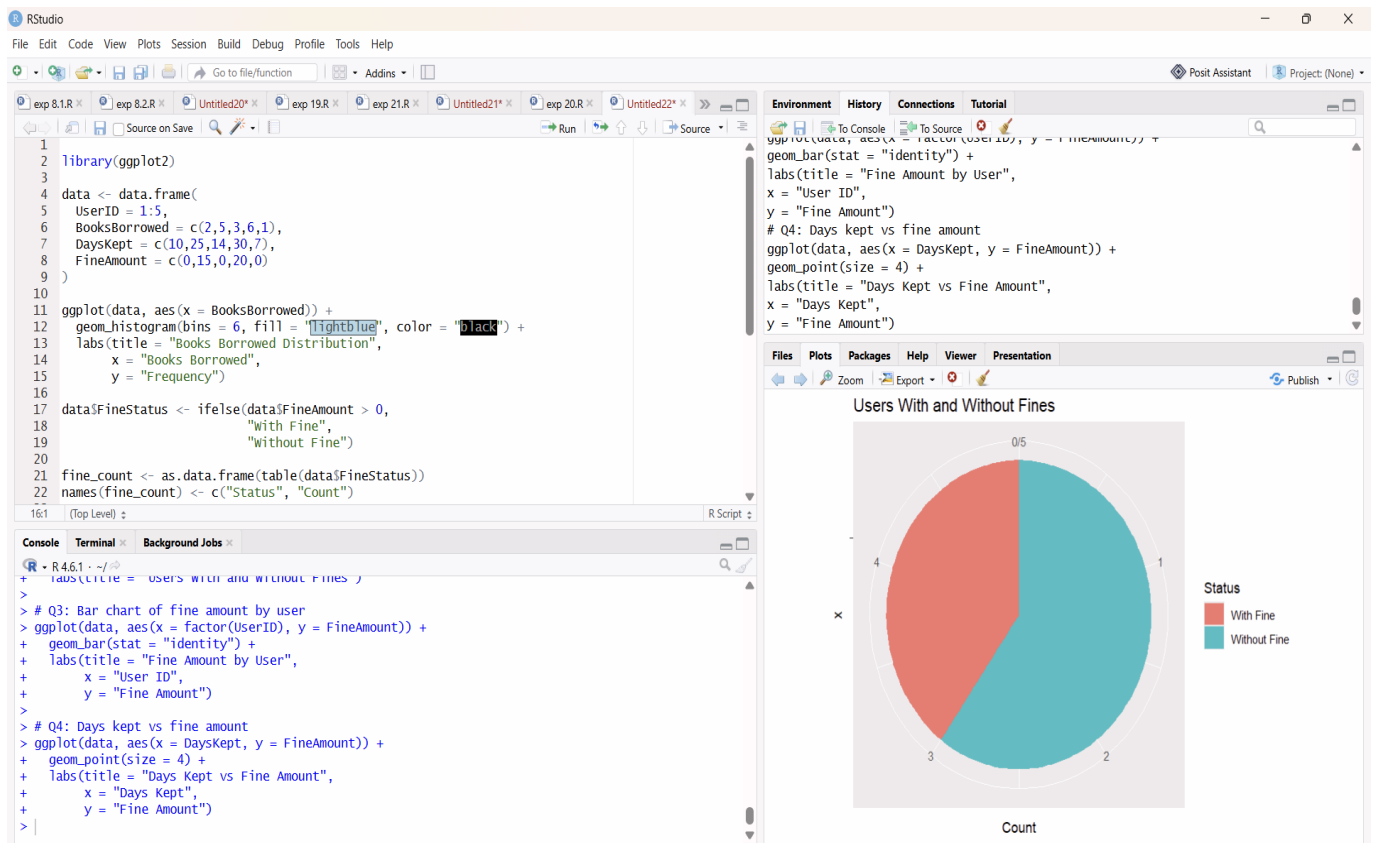
To analyze library borrowing records using Tableau by visualizing borrowing patterns and fine details.

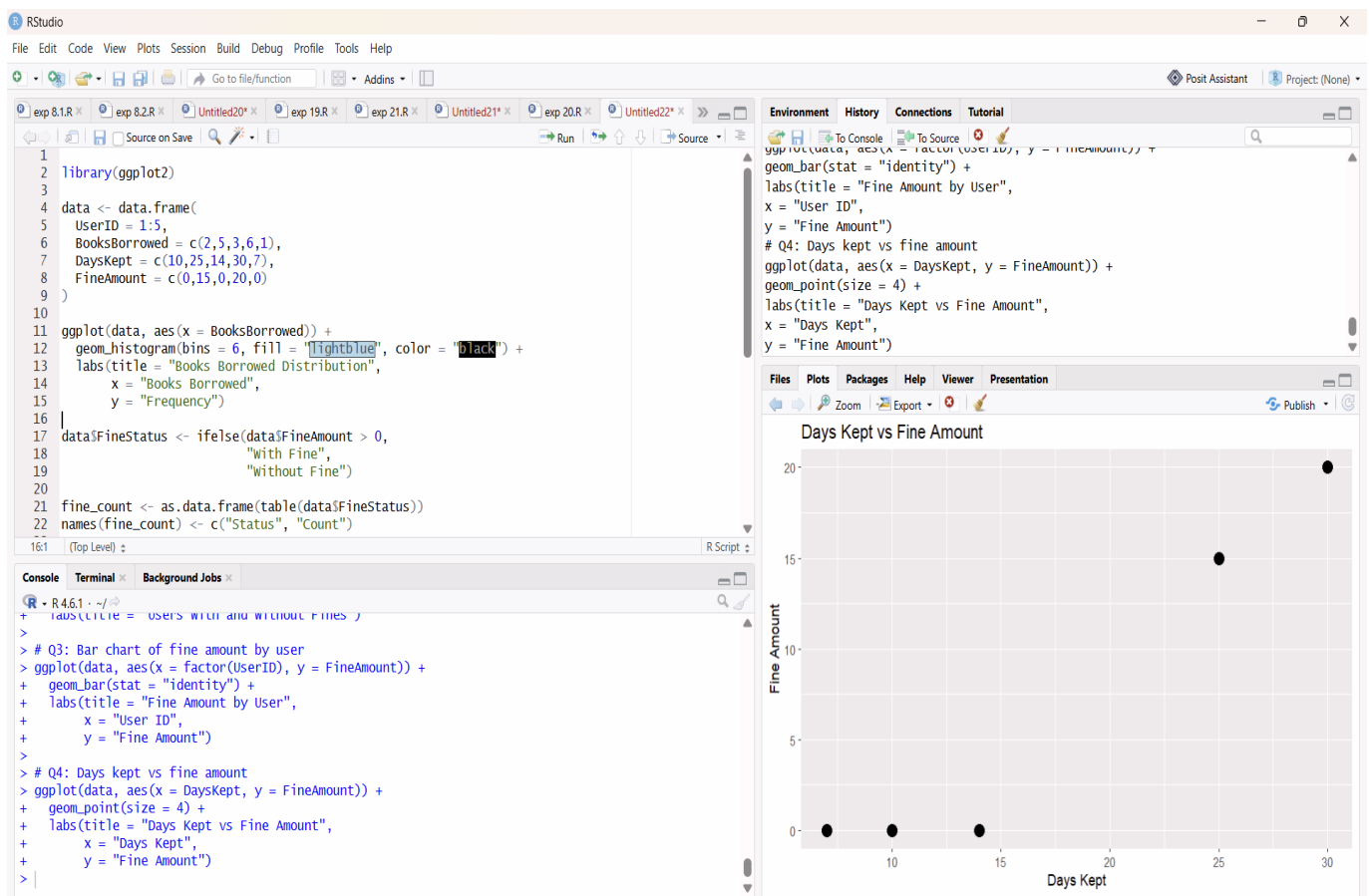
Algorithm:

1. Open Tableau and import the Library Usage dataset.
2. Verify the imported data fields.
3. Create a histogram showing the distribution of books borrowed.
4. Generate a pie chart representing users with and without fines.
5. Build a bar chart displaying fine amounts by user.
6. Create a scatter plot between days kept and fine amount.

7. Add suitable titles, labels, and legends.
8. Save the Tableau workbook.







Result:

The visualizations successfully represented borrowing behavior, fine distribution, and the relationship between book retention period and fines.

SET23.AirlinePassengerAnalysis

Dataset:AirlinePassengers

PassengerID	Age	FlightHours	Satisfaction
1	28	2	High
2	45	8	Medium
3	33	5	High
4	52	10	Low
5	39	6	Medium

Questions

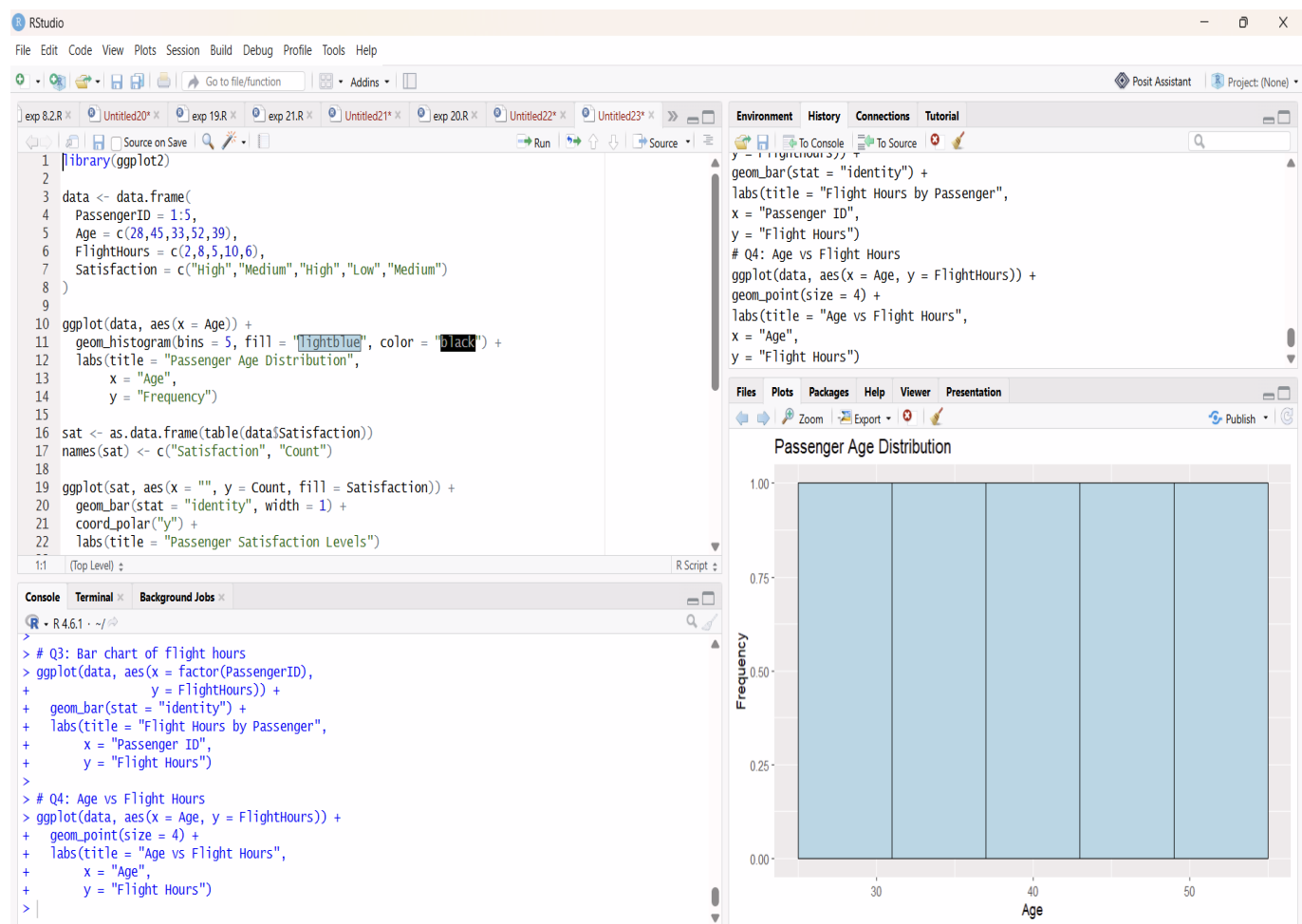
1. Create a histogram of passenger ages.
2. Generate a pie chart of satisfaction levels.
3. Build a bar chart of flight hours by passenger.
4. Create a scatter plot of age vs flight hours.

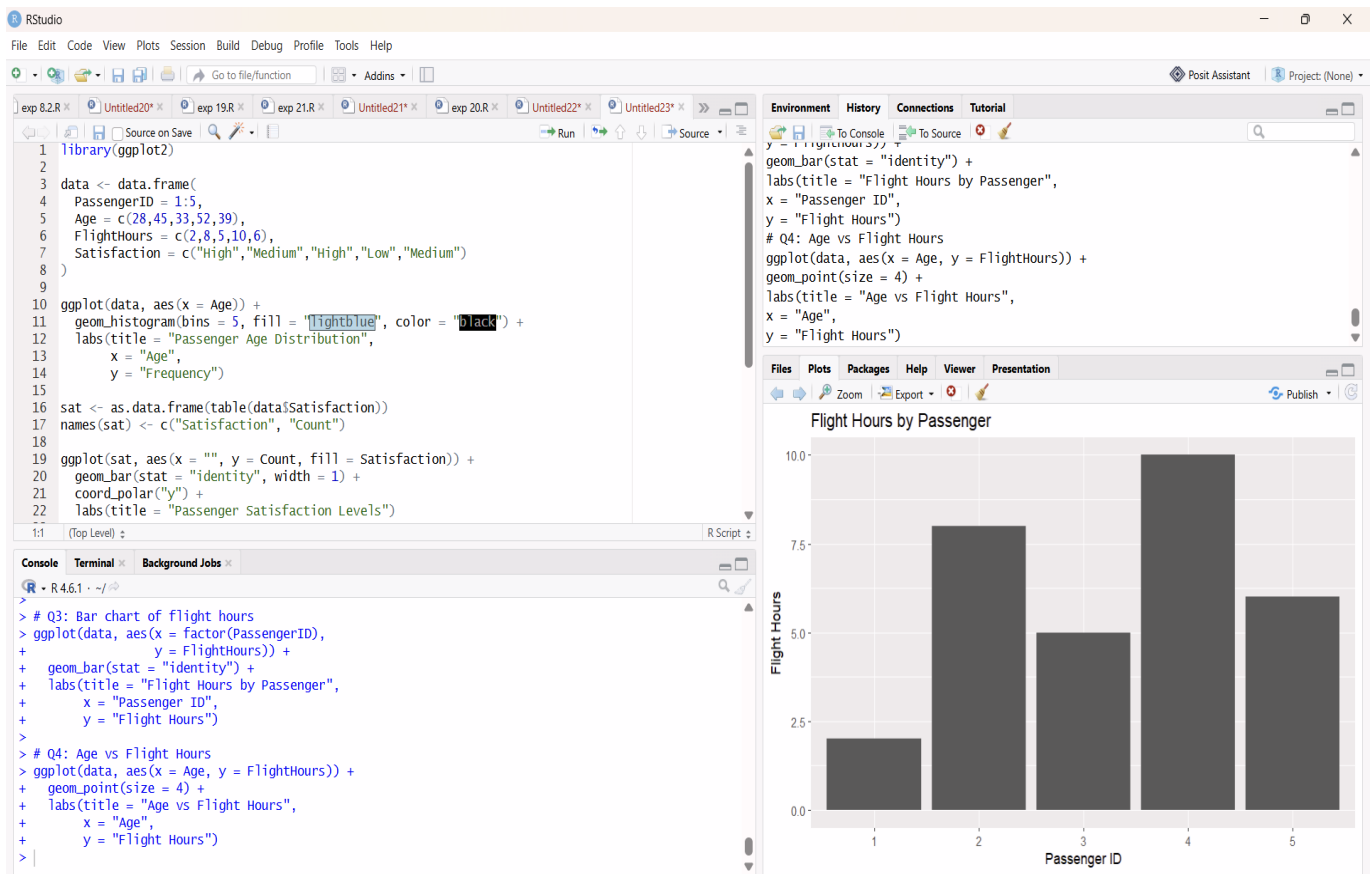
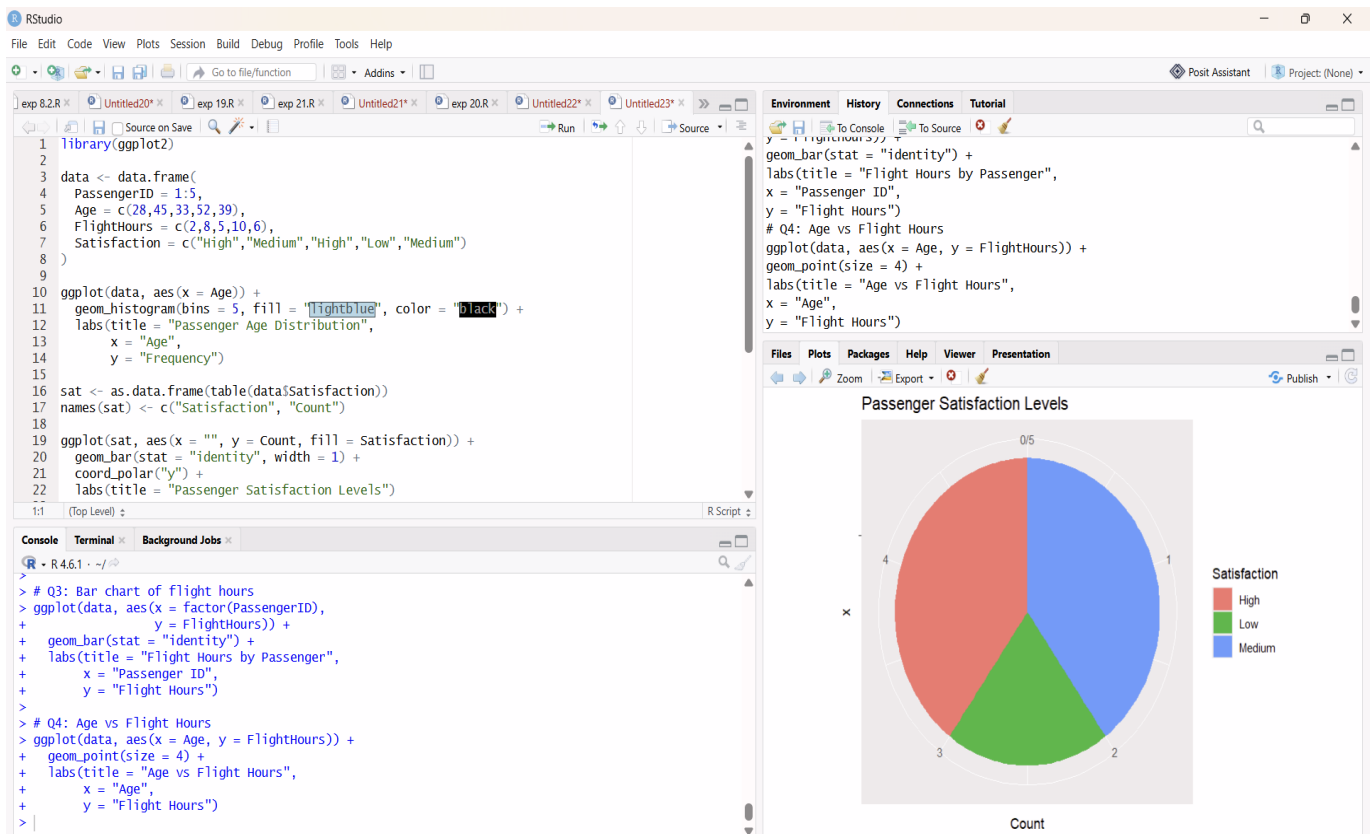
Aim:

To analyze airline passenger data using Tableau by creating visualizations for age, satisfaction, and flight hours.

Algorithm:

1. Import the Airline Passengers dataset into Tableau.
2. Verify the imported data.
3. Create a histogram of passenger ages.
4. Generate a pie chart showing satisfaction levels.
5. Build a bar chart displaying flight hours by passenger.
6. Create a scatter plot between age and flight hours.
7. Add titles, labels, legends, and formatting.
8. Save the workbook.





Result:

The required visualizations were successfully created, providing insights into passenger demographics, satisfaction levels, and flight hour patterns.

SET24.HotelBookingAnalysis

Dataset:HotelBookings

BookingID	StayNights	Guests	Room Type
1	2	2	Standard
2	5	4	Deluxe
3	3	1	Standard
4	7	3	Suite
5	4	2	Deluxe

Questions

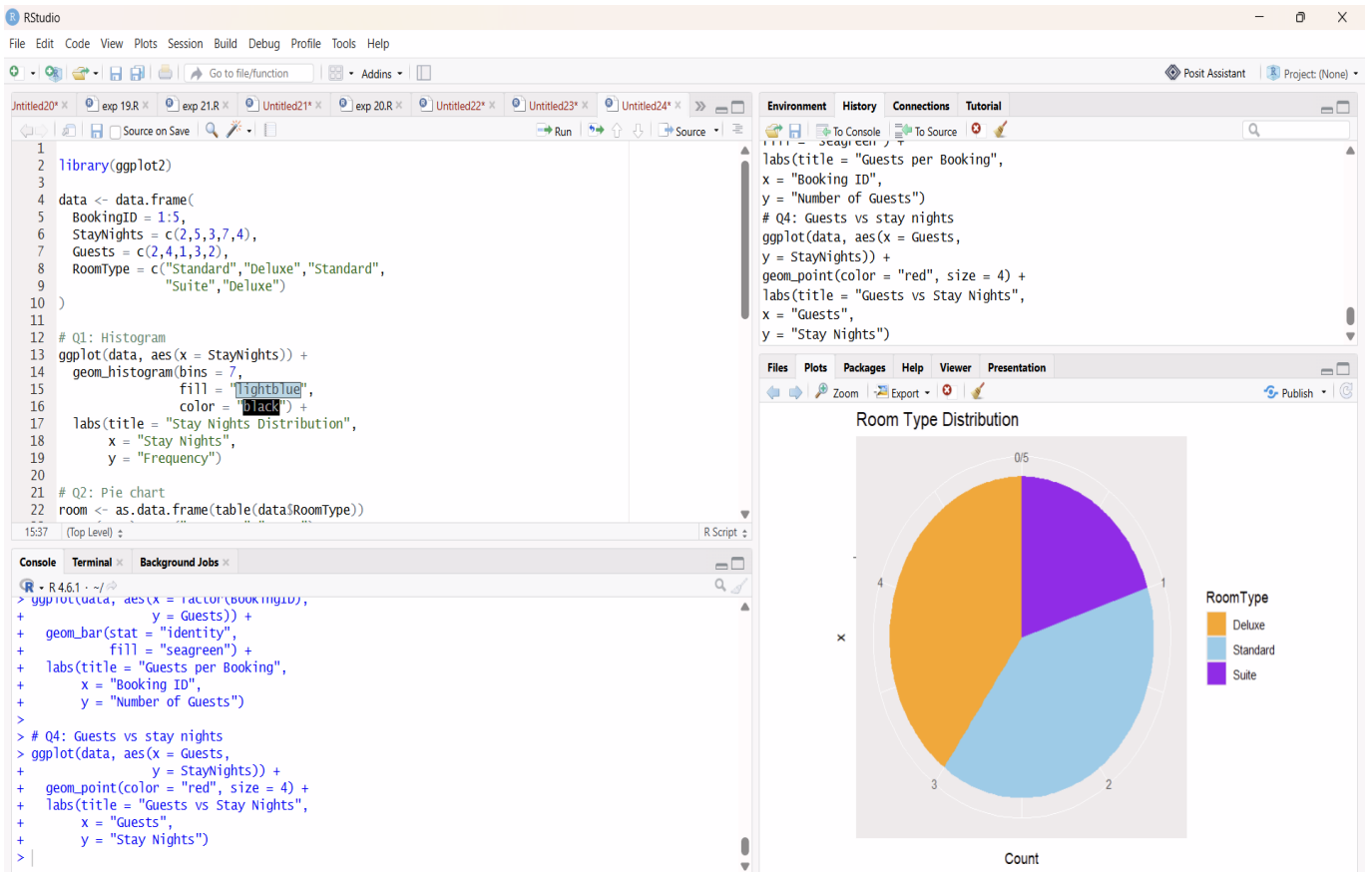
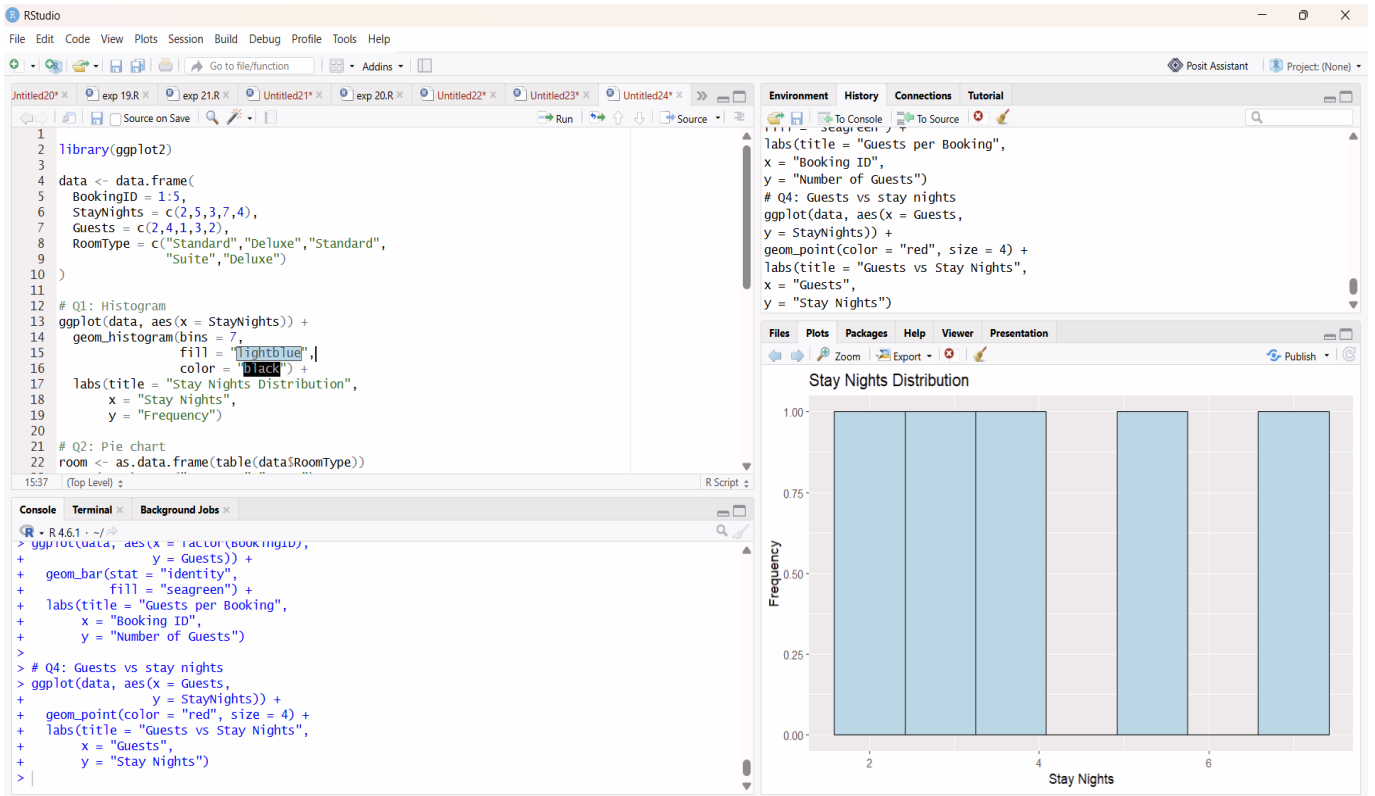
1. Createahistogramofstaynights.
2. Generateapiechartofroomtypes.
3. Buildabarchartofguestsperbooking.
4. Createascatterplotofguestsvsstaynights.

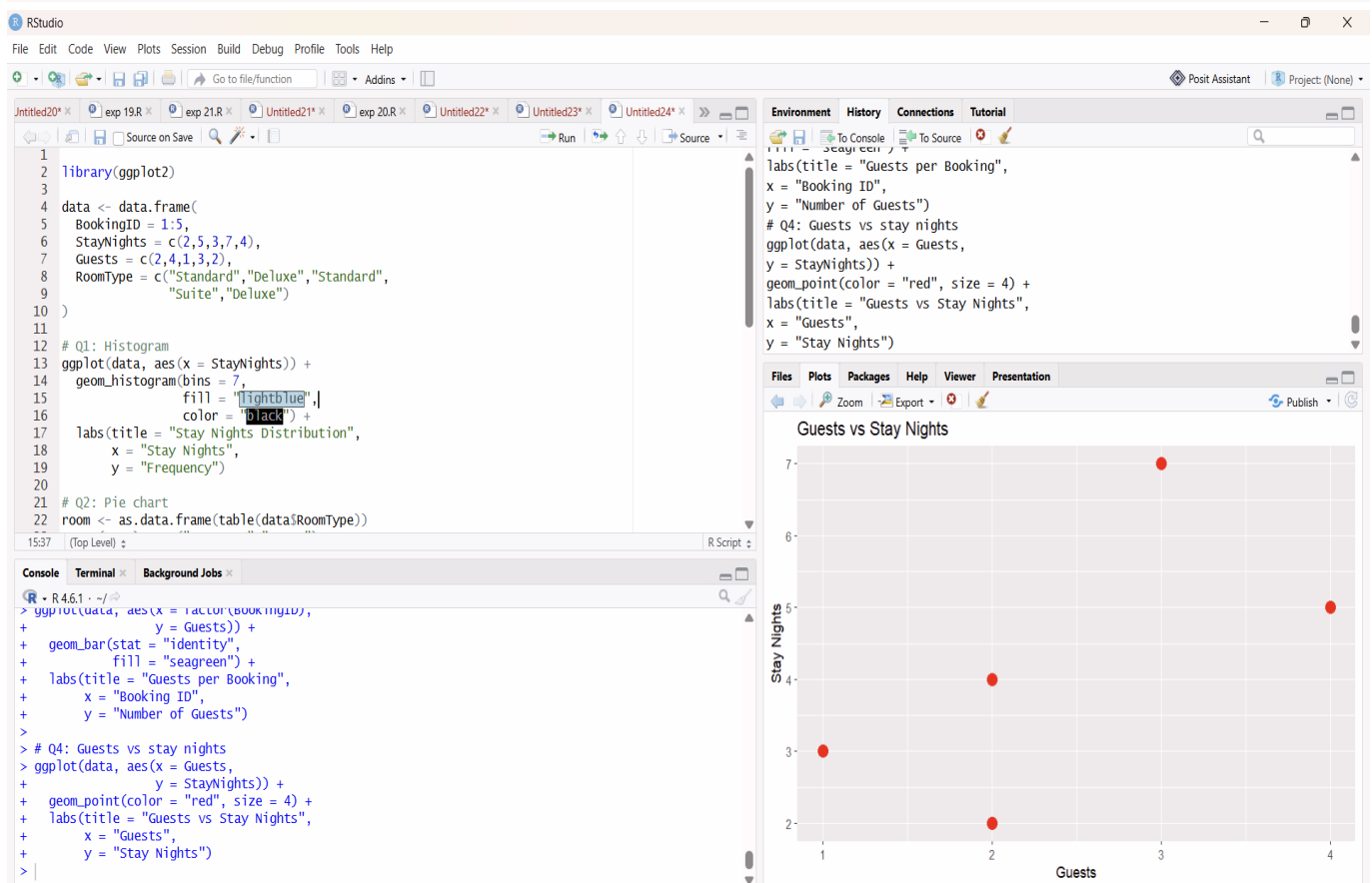
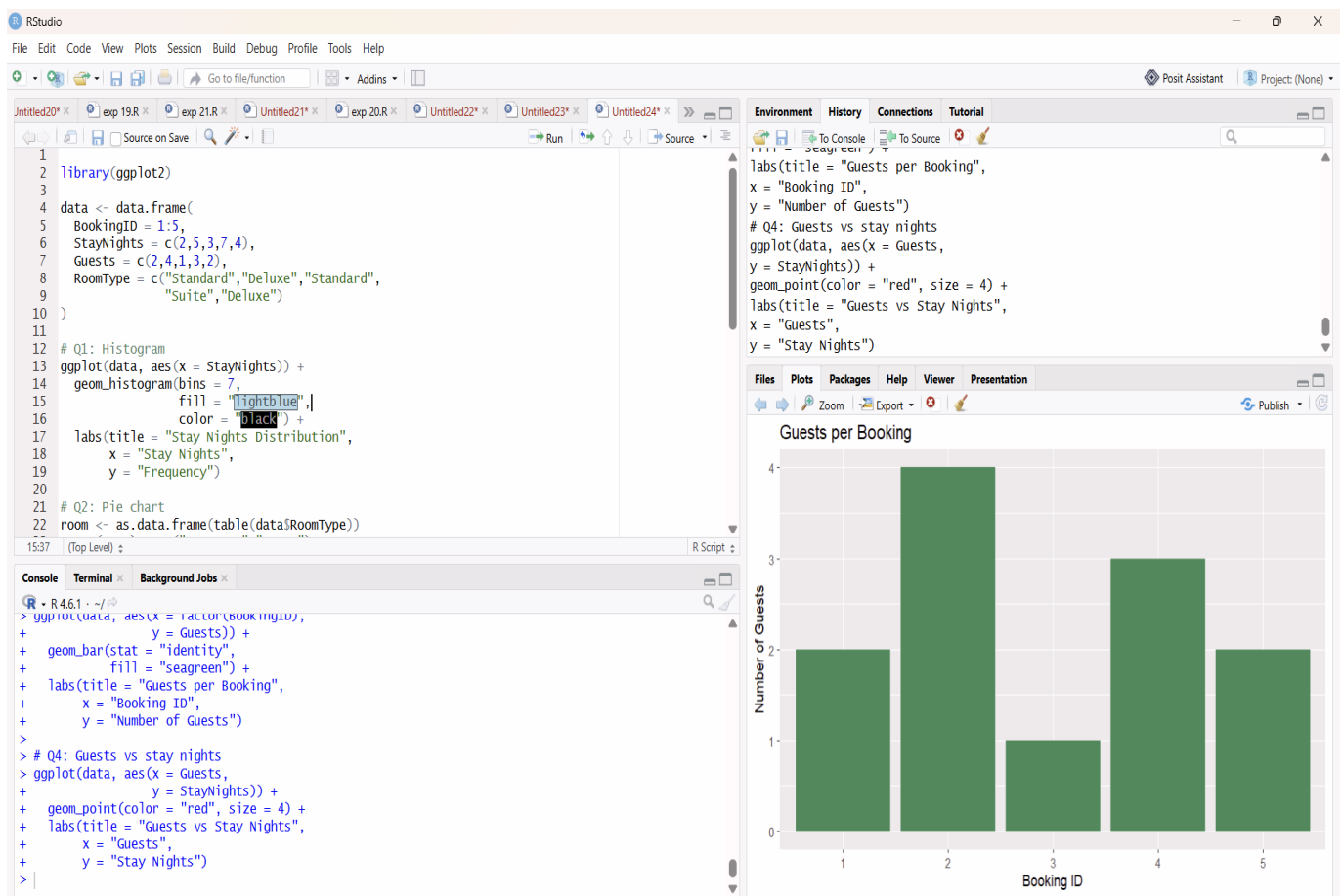
Aim:

To analyze hotel booking data using Tableau by visualizing stay duration, room types, and guest information.

Algorithm:

1. Open Tableau and import the Hotel Bookings dataset.
2. Verify the imported fields.
3. Create a histogram showing stay nights.
4. Generate a pie chart representing room type distribution.
5. Build a bar chart displaying the number of guests per booking.
6. Create a scatter plot between guests and stay nights.
7. Add titles, labels, and legends.
8. Save the Tableau workbook.





Result:

The visualizations successfully displayed booking patterns, room type distribution, guest information, and the relationship between guests and stay duration.

SET25.FitnessTrackerAnalysis

Dataset:DailyFitnessActivity

UserID	Steps	CaloriesBurned	ActiveMinutes
1	7000	250	40
2	10000	400	60
3	8500	320	50
4	12000	500	75
5	6500	220	35

Questions

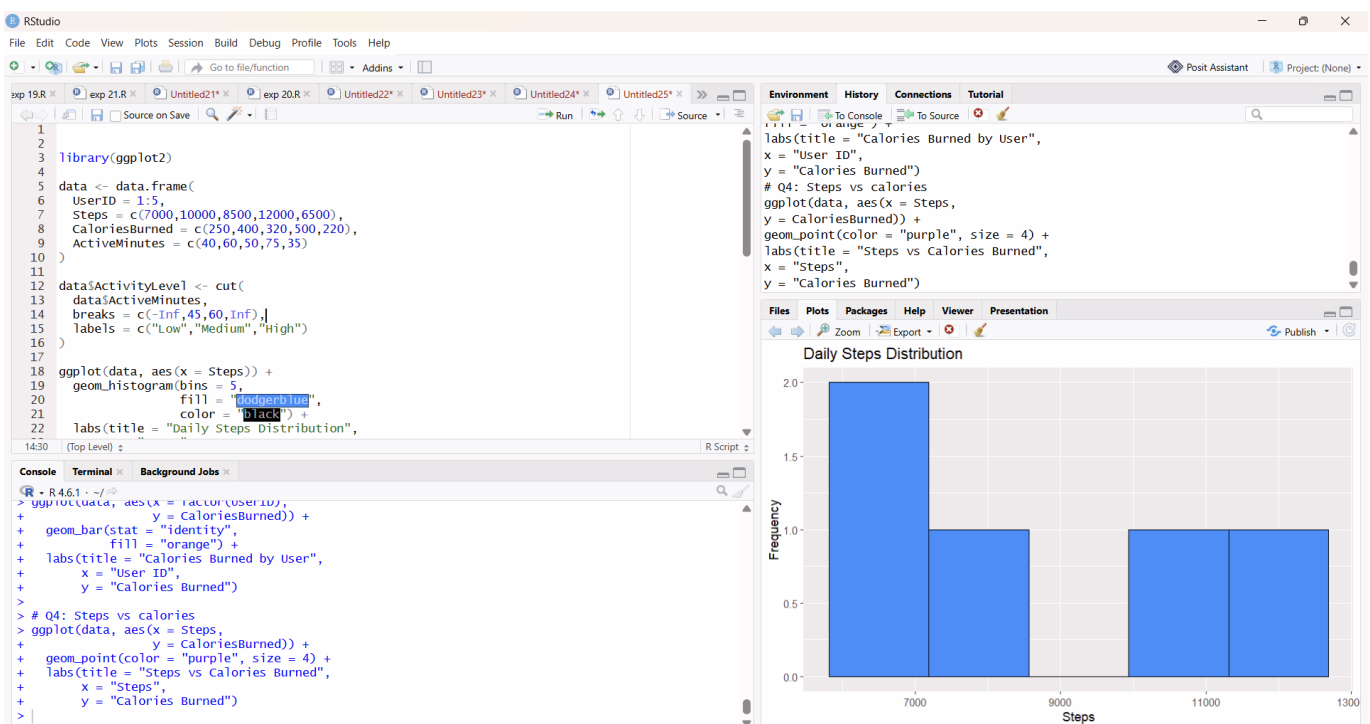
1. Create a histogram of daily steps.
2. Generate a pie chart of activity level categories.
3. Build a bar chart of calories burned by user.
4. Create a scatter plot of steps vs calories burned.

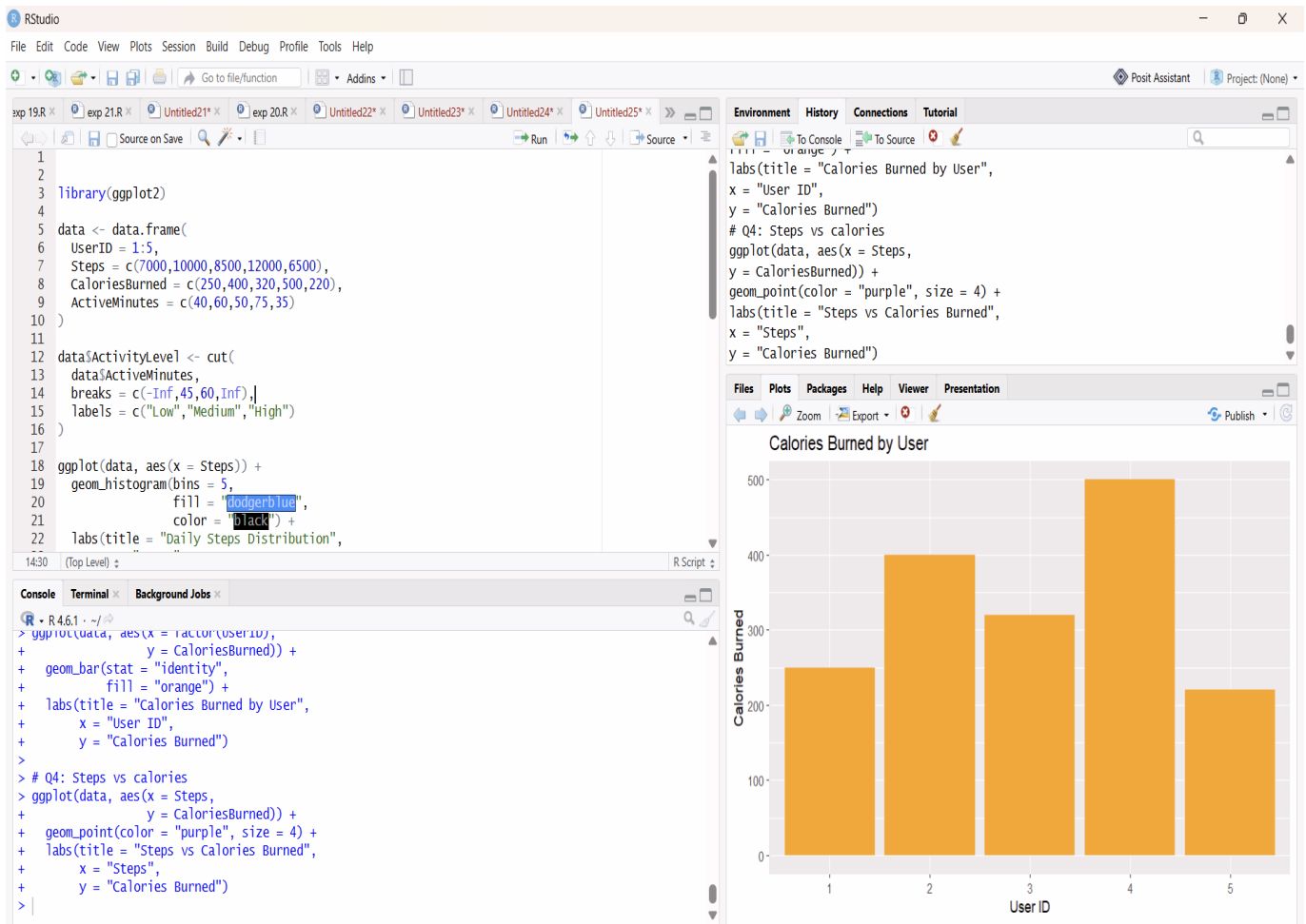
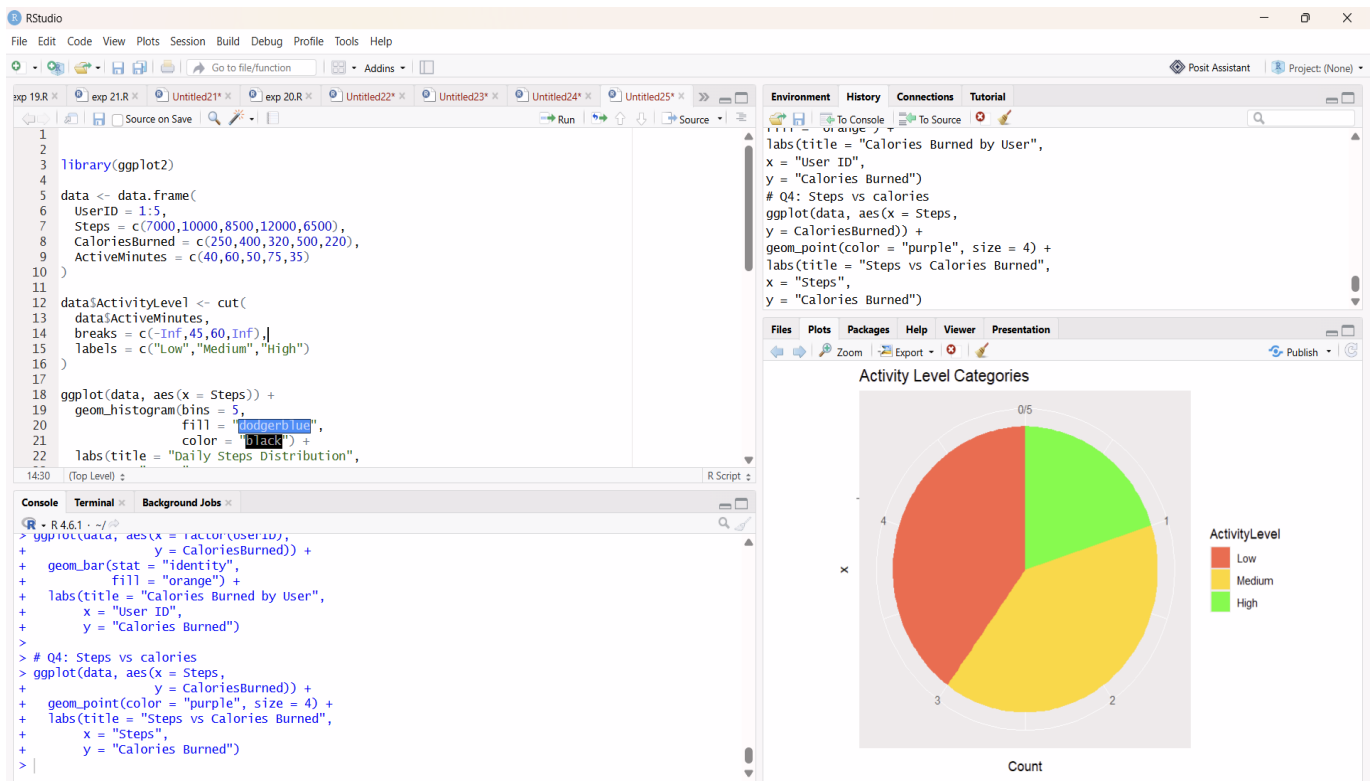
Aim:

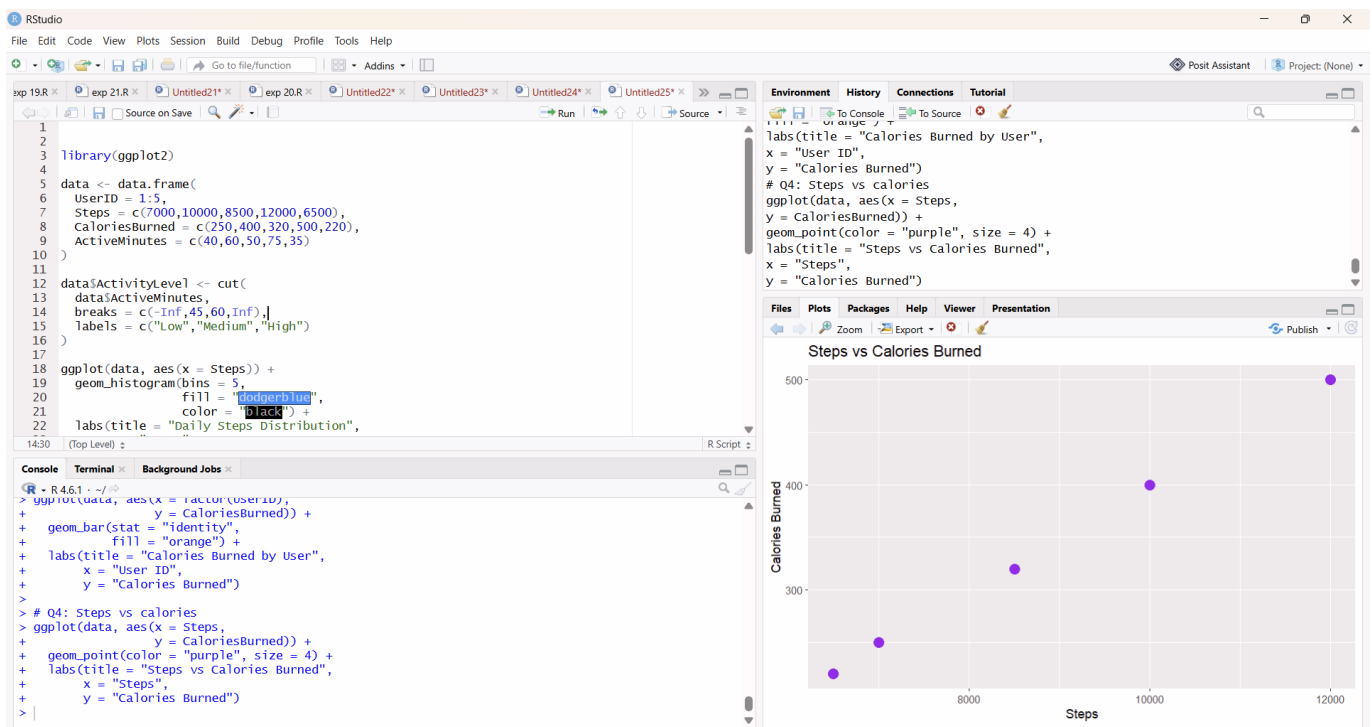
To analyze daily fitness activity using Tableau by creating charts for steps, calories burned, and activity levels.

Algorithm:

1. Import the Daily Fitness Activity dataset into Tableau.
2. Verify the imported data.
3. Create a histogram showing daily steps.
4. Generate a pie chart representing activity level categories.
5. Build a bar chart displaying calories burned by each user.
6. Create a scatter plot between steps and calories burned.
7. Add appropriate titles, labels, legends, and formatting.
8. Save the Tableau workbook.







Result:

The required visualizations were successfully created, helping analyze daily physical activity, calorie expenditure, and the relationship between steps and calories burned.

SET26.RestaurantOrdersAnalysis

Dataset:Restaurant Orders

Order ID	ItemsOrdered	BillAmount	DiningType
1	2	25	Dine-In
2	5	60	Takeaway
3	3	35	Dine-In
4	4	50	Delivery
5	2	20	Takeaway

Questions

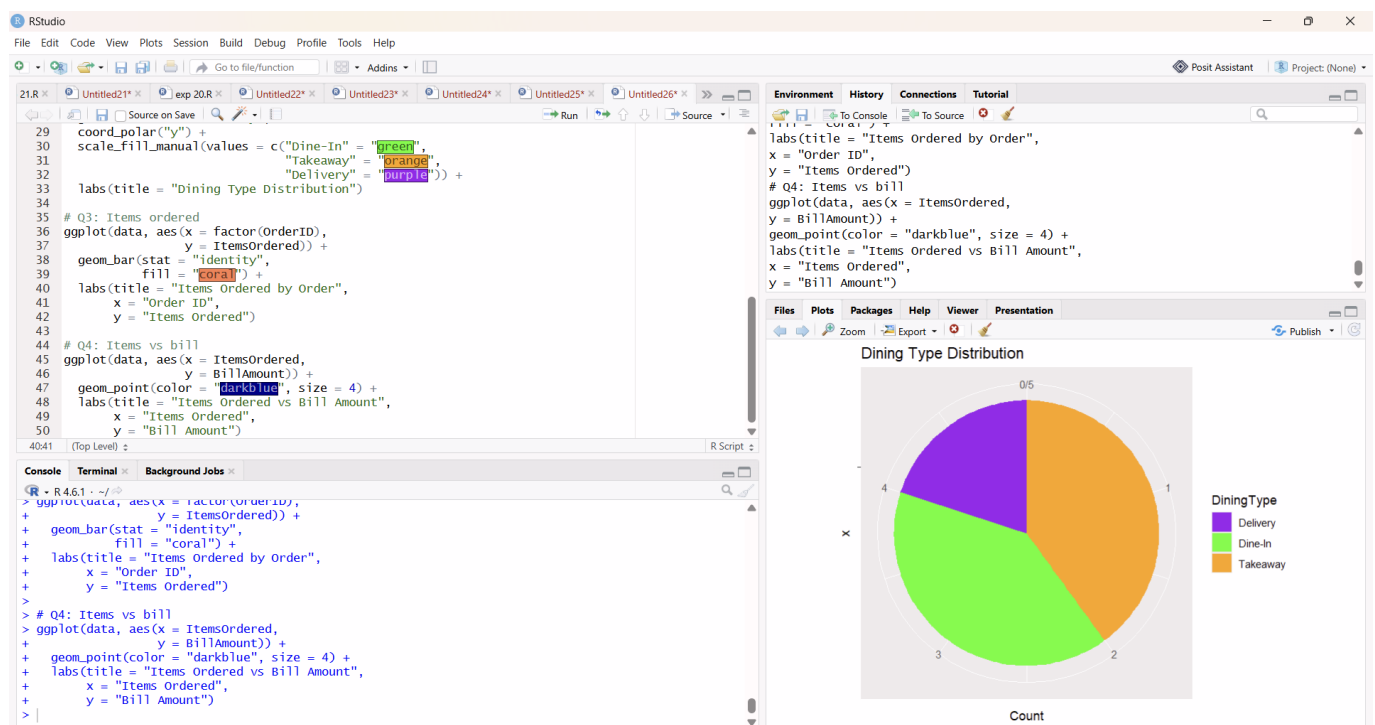
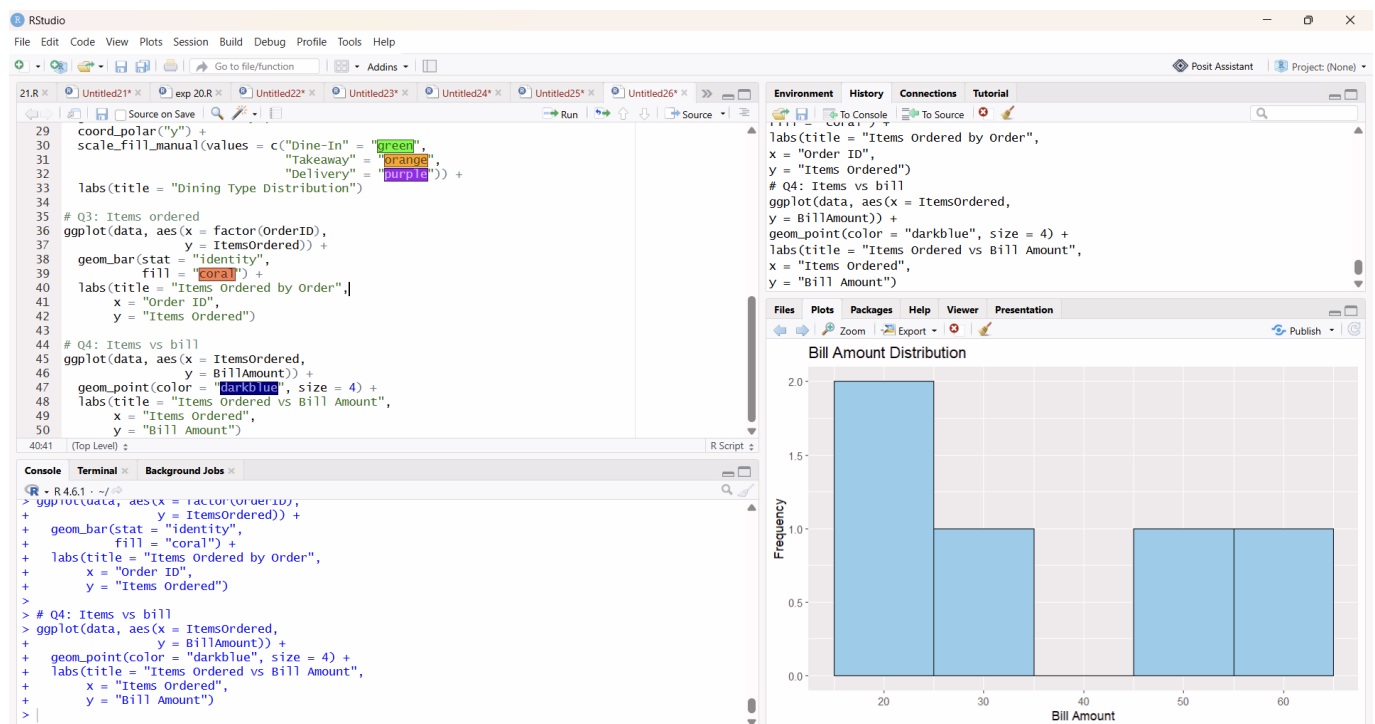
1. Create a histogram of bill amounts.
2. Generate a pie chart of dining types.
3. Build a bar chart of items ordered by order.
4. Create a scatter plot of items ordered vs bill amount.

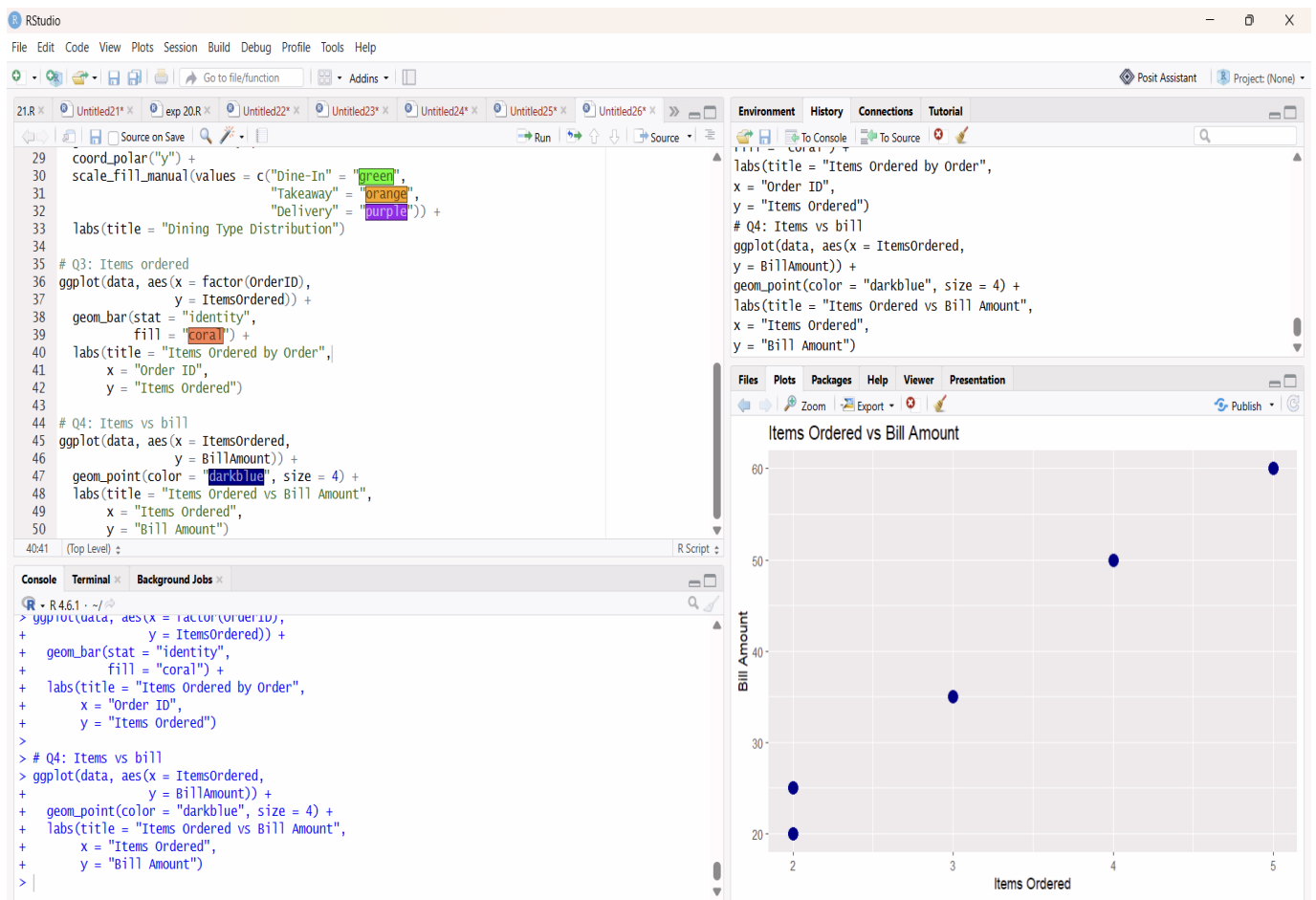
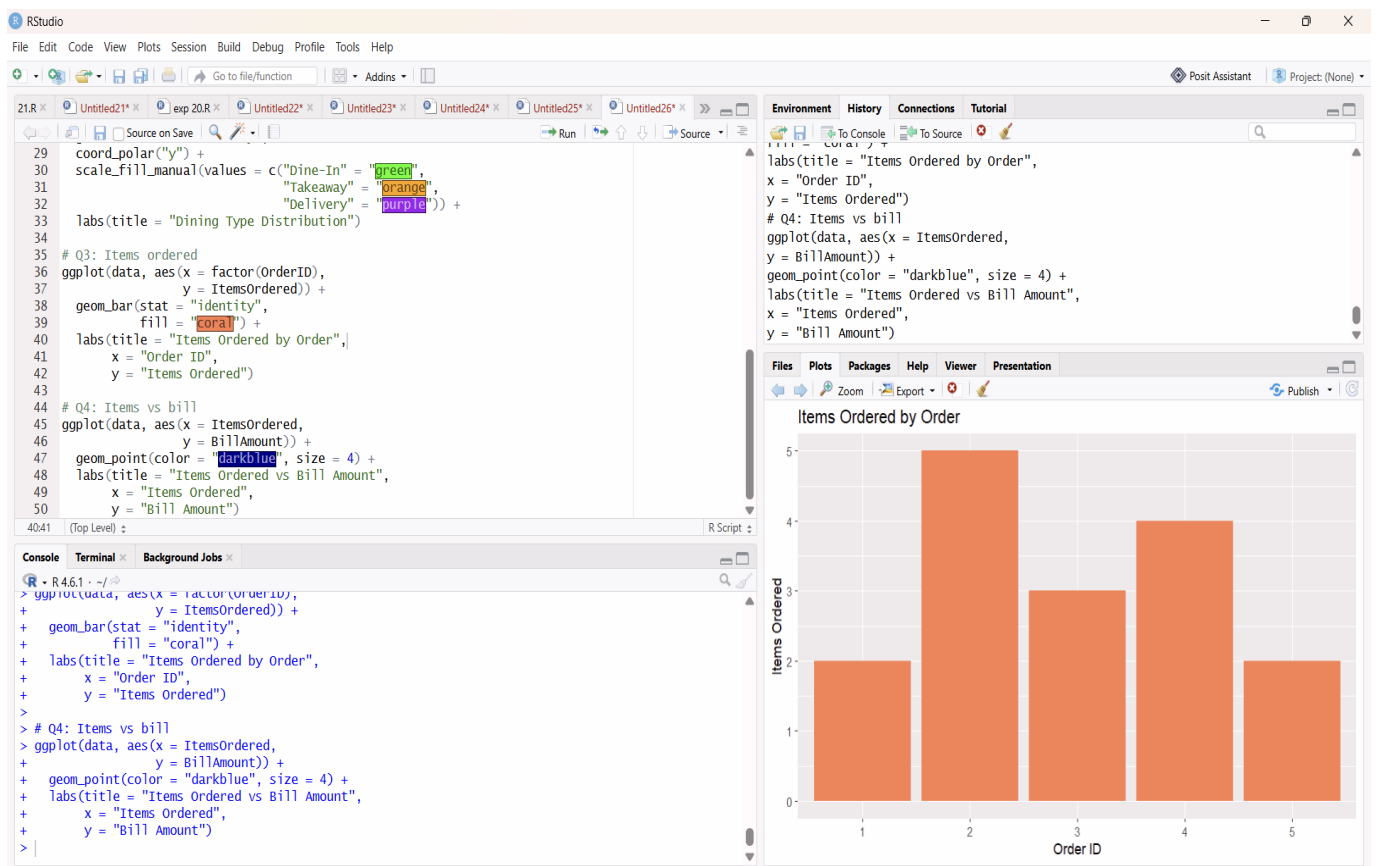
Aim:

To analyze restaurant order data using Tableau by creating visualizations for billing, dining type, and ordered items.

Algorithm:

1. Open Tableau and import the Restaurant Orders dataset.
2. Verify the imported data fields.
3. Create a histogram of bill amounts.
4. Generate a pie chart showing dining type distribution.
5. Build a bar chart displaying the number of items ordered.
6. Create a scatter plot between items ordered and bill amount.
7. Add titles, labels, and legends.
8. Save the Tableau workbook.





Result:

The visualizations successfully represented restaurant order patterns, bill distribution, dining preferences, and the relationship between ordered items and bill amount.

SET27.EnergyPlantMonitoring

Dataset:PowerPlantOperations

Plant ID	Output(MW)	Temperature	Status
1	120	65	Active
2	150	70	Active
3	100	60	Maintenance
4	170	75	Active
5	110	62	Maintenance

Questions

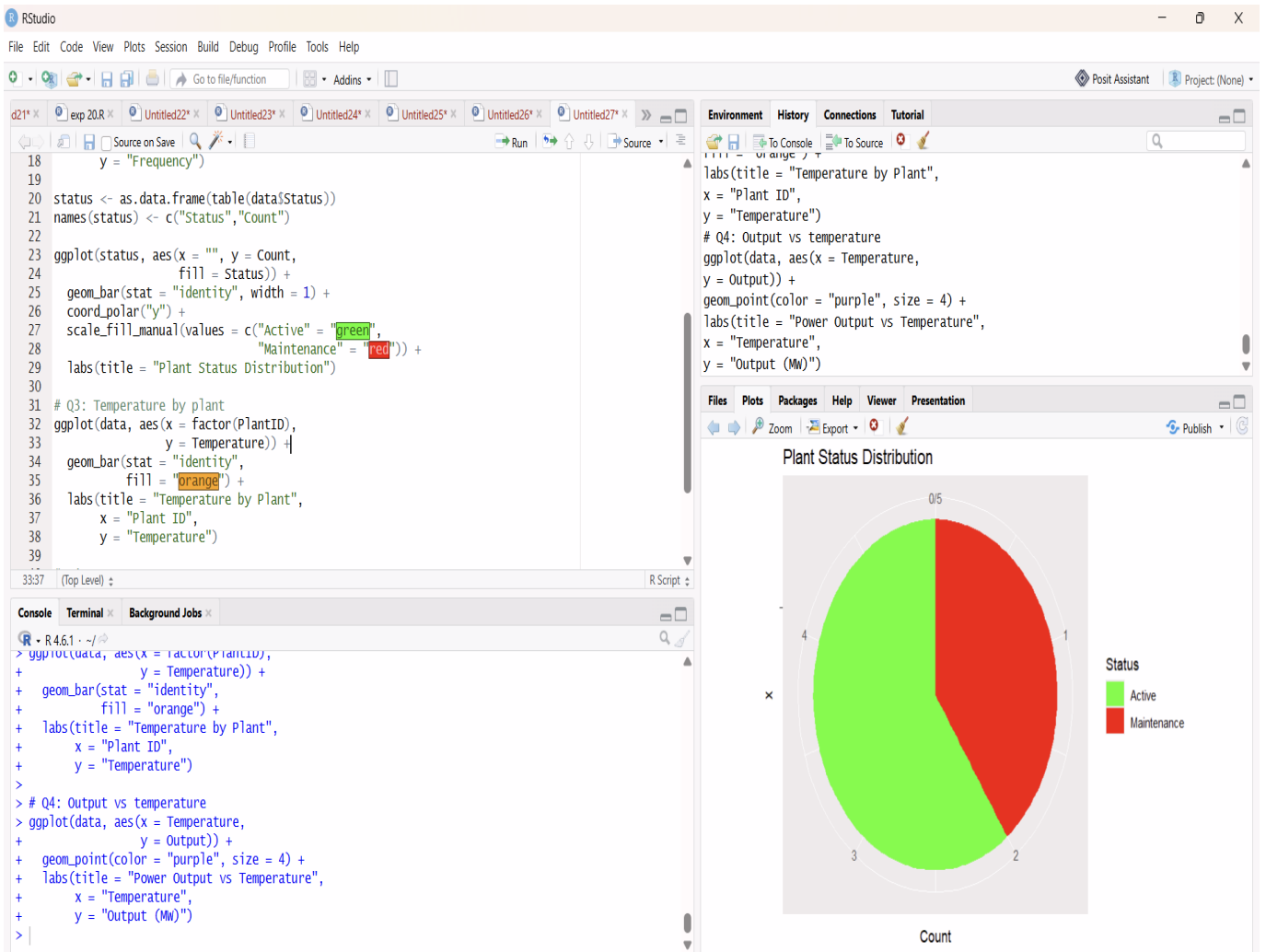
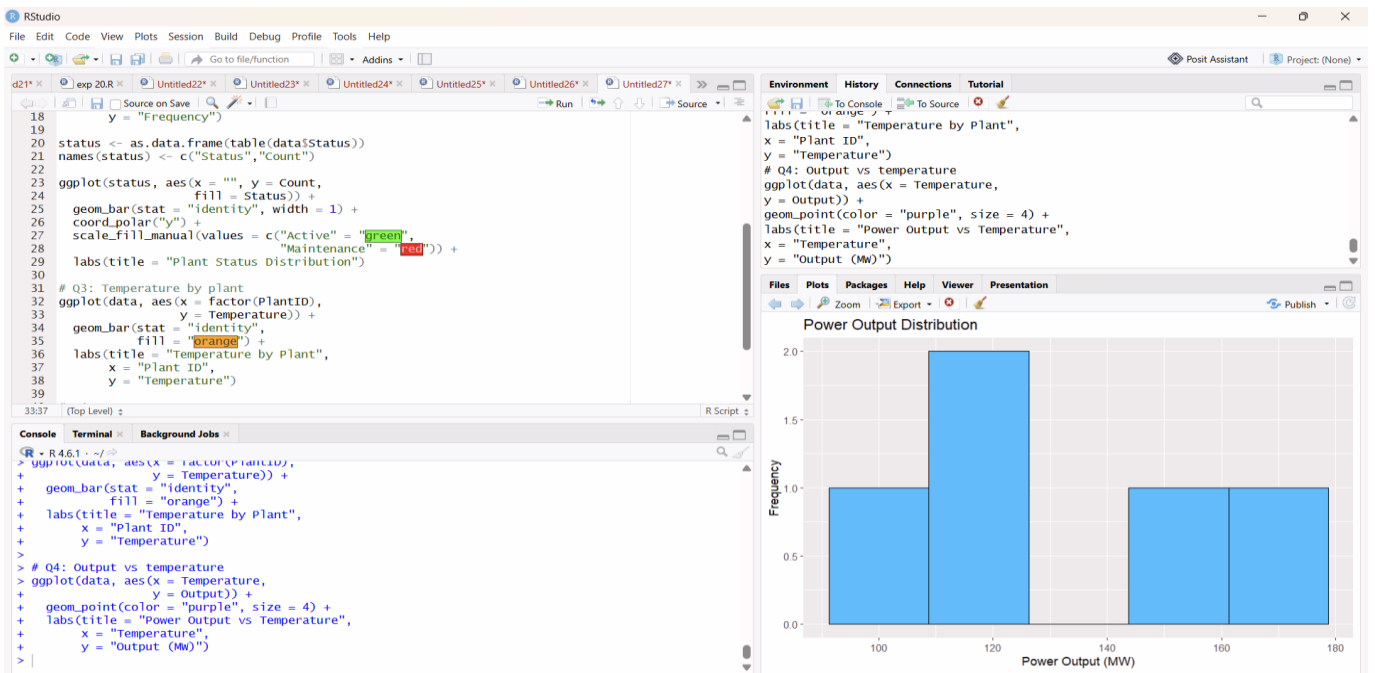
1. Createahistogramofpoweroutput.
2. Generateapiechartofplantstatus.
3. Buildabarchartoftemperaturesbyplant.
4. Createascatterplotofoutputvstemperature.

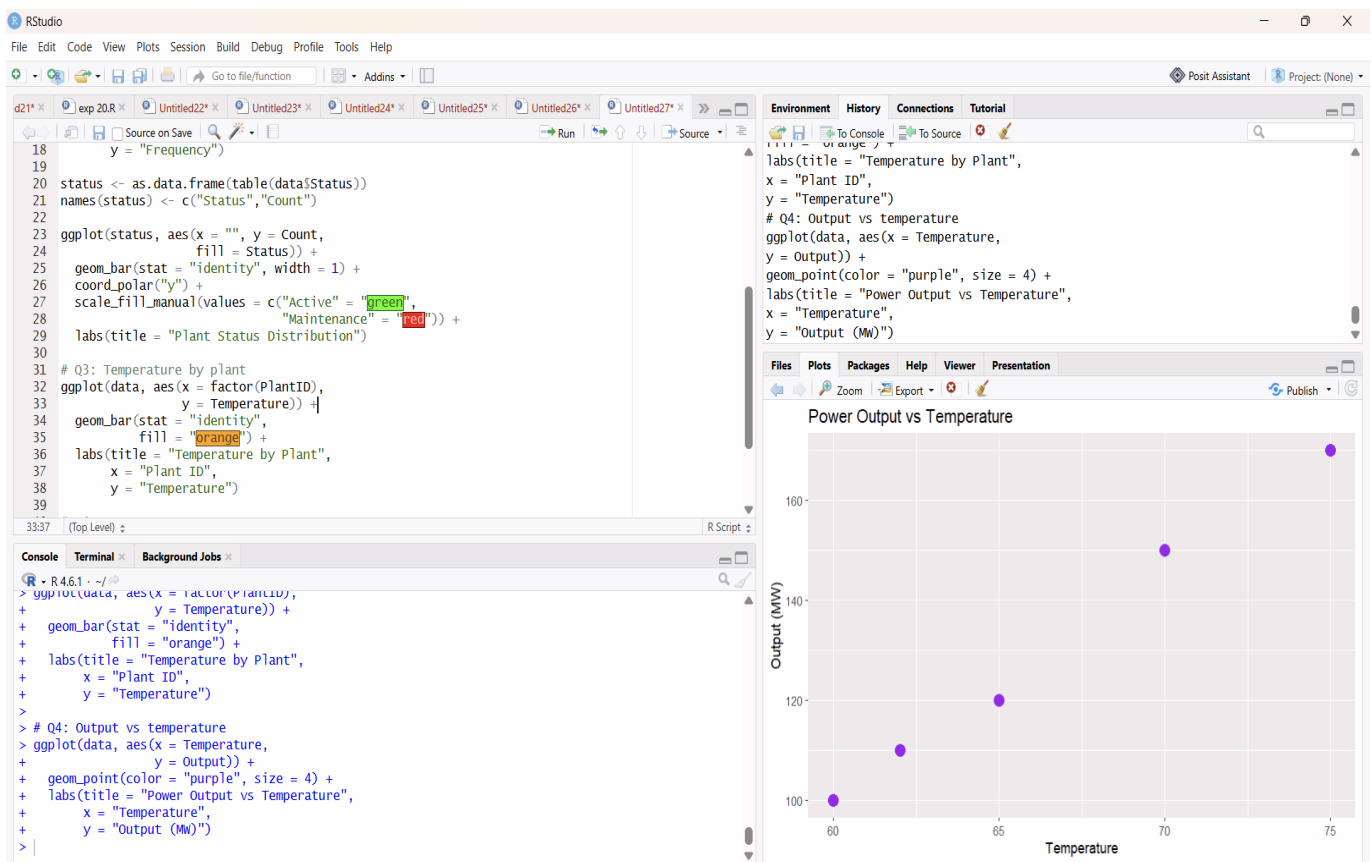
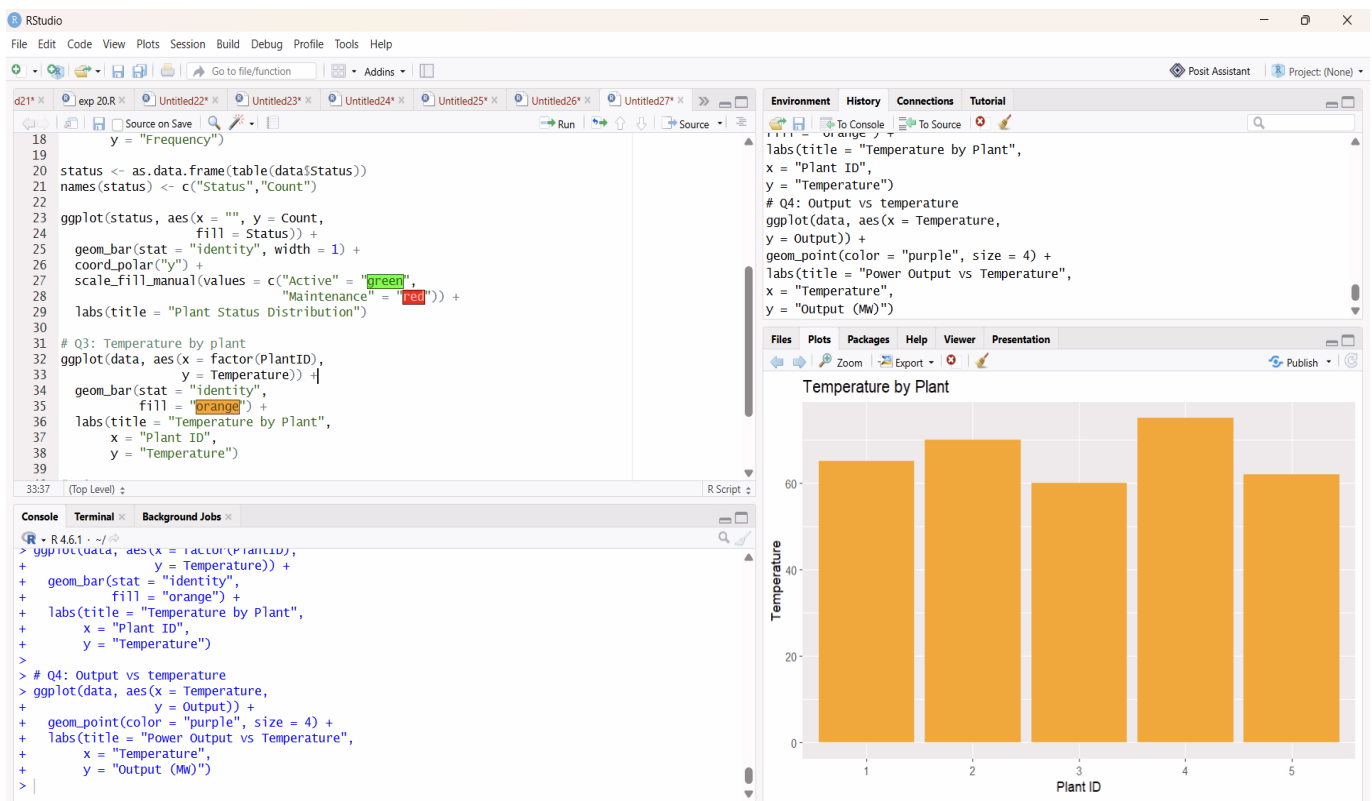
Aim:

To analyze power plant operational data using Tableau by visualizing power output, plant status, and temperature.

Algorithm:

1. Import the Power Plant Operations dataset into Tableau.
2. Verify the imported data.
3. Create a histogram showing power output.
4. Generate a pie chart representing plant status.
5. Build a bar chart displaying plant temperatures.
6. Create a scatter plot between power output and temperature.
7. Add titles, labels, legends, and formatting.
8. Save the Tableau workbook.





Result:

The visualizations successfully displayed power output, operational status, plant temperatures, and the relationship between output and temperature.

SET28.SocialMediaEngagement Analysis

Dataset:SocialMediaPosts

PostID	Likes	Comments	Shares
1	120	15	10
2	200	30	20
3	150	20	12
4	300	40	35
5	180	25	18

Questions

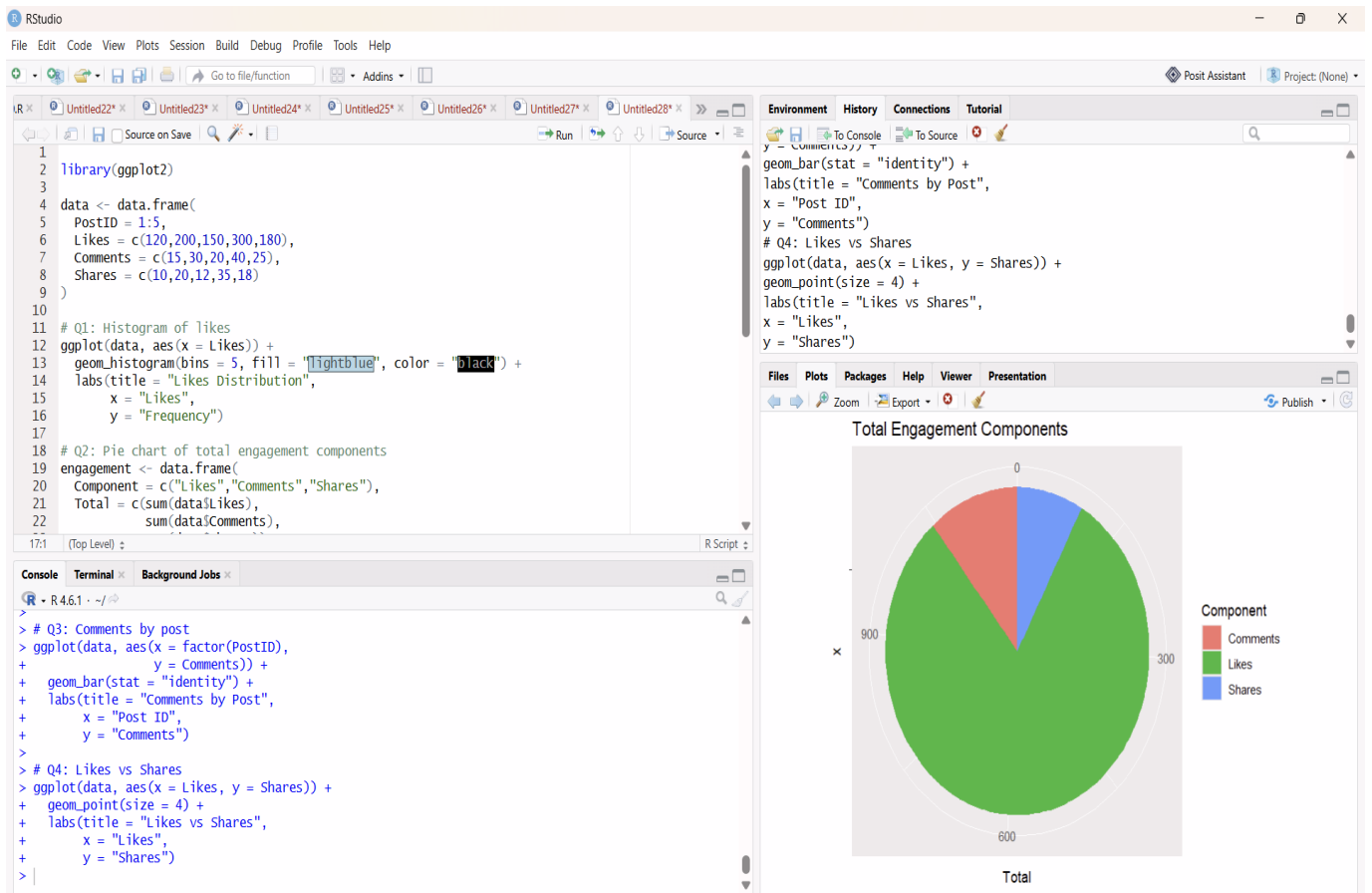
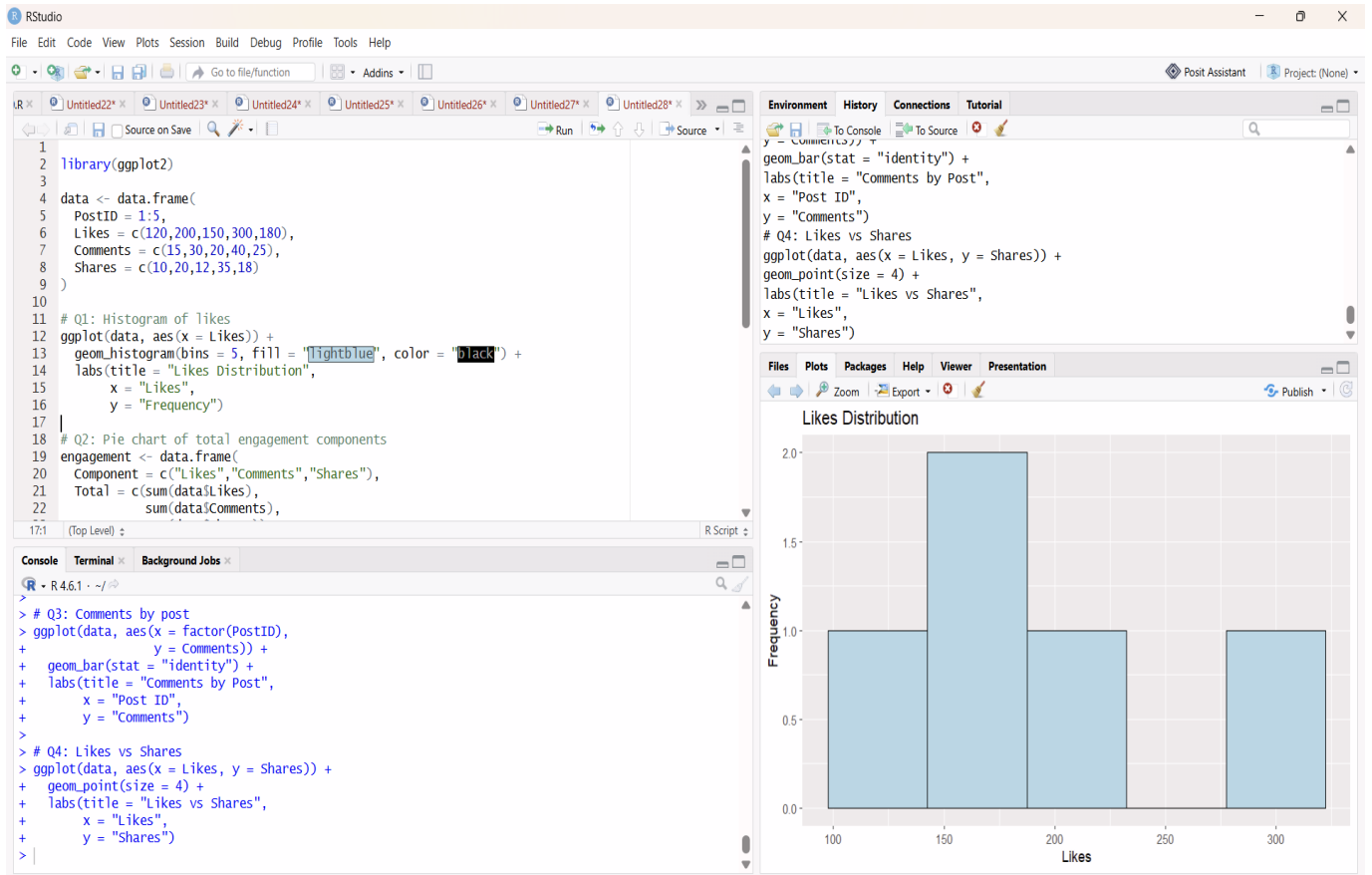
1. Createahistogramoflikes.
2. Generateapiechartoftotalengagementcomponents.
3. Buildabarchartofcommentsby post.
4. Createascatterplotoflikesvsshares.

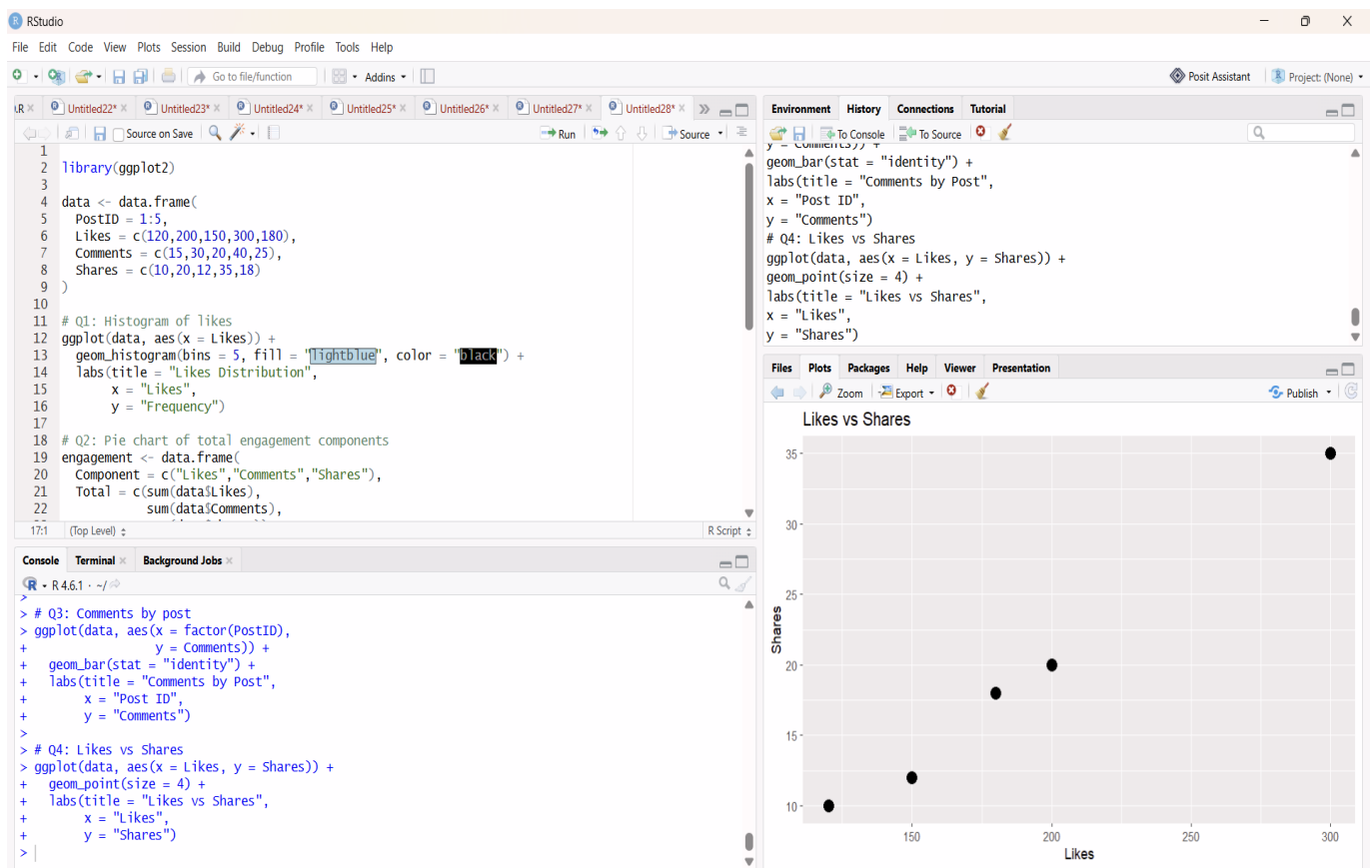
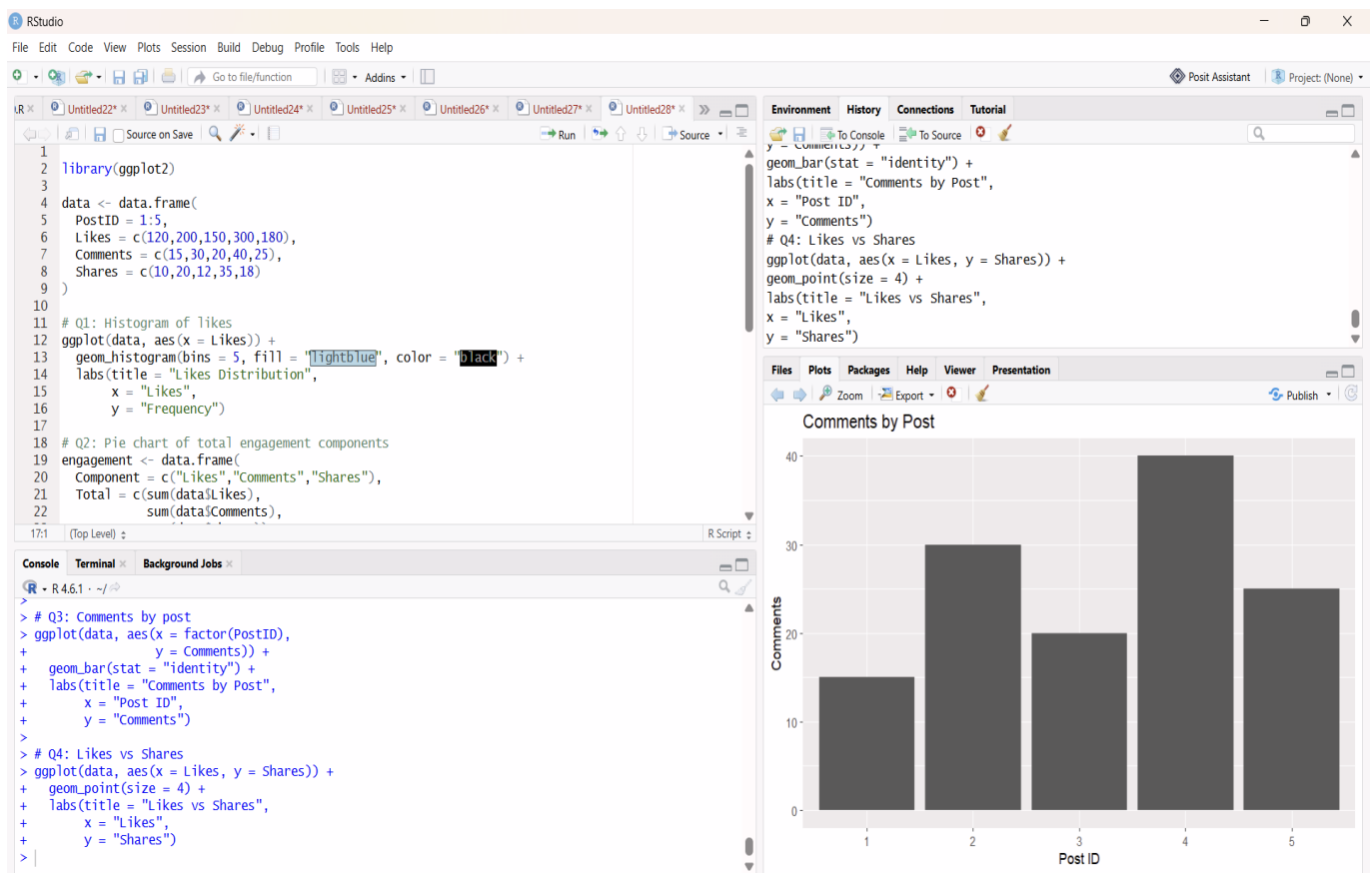
Aim:

To analyze social media engagement data using Tableau by visualizing likes, comments, shares, and overall engagement.

Algorithm:

1. Open Tableau and import the Social Media Posts dataset.
2. Verify the imported data fields.
3. Create a histogram showing likes.
4. Generate a pie chart representing engagement components.
5. Build a bar chart displaying comments by post.
6. Create a scatter plot between likes and shares.
7. Add appropriate titles, labels, legends, and formatting.
8. Save the Tableau workbook





Result:

The required visualizations were successfully created, providing insights into user engagement and the relationship between likes, comments, and shares.

SET29.WaterQualityMonitoring

Dataset: WaterSamples

SampleID	pH Level	Turbidity	Quality
1	7.2	3	Good
2	6.8	5	Fair
3	7.5	2	Good
4	6.5	7	Poor
5	7.1	4	Fair

Questions

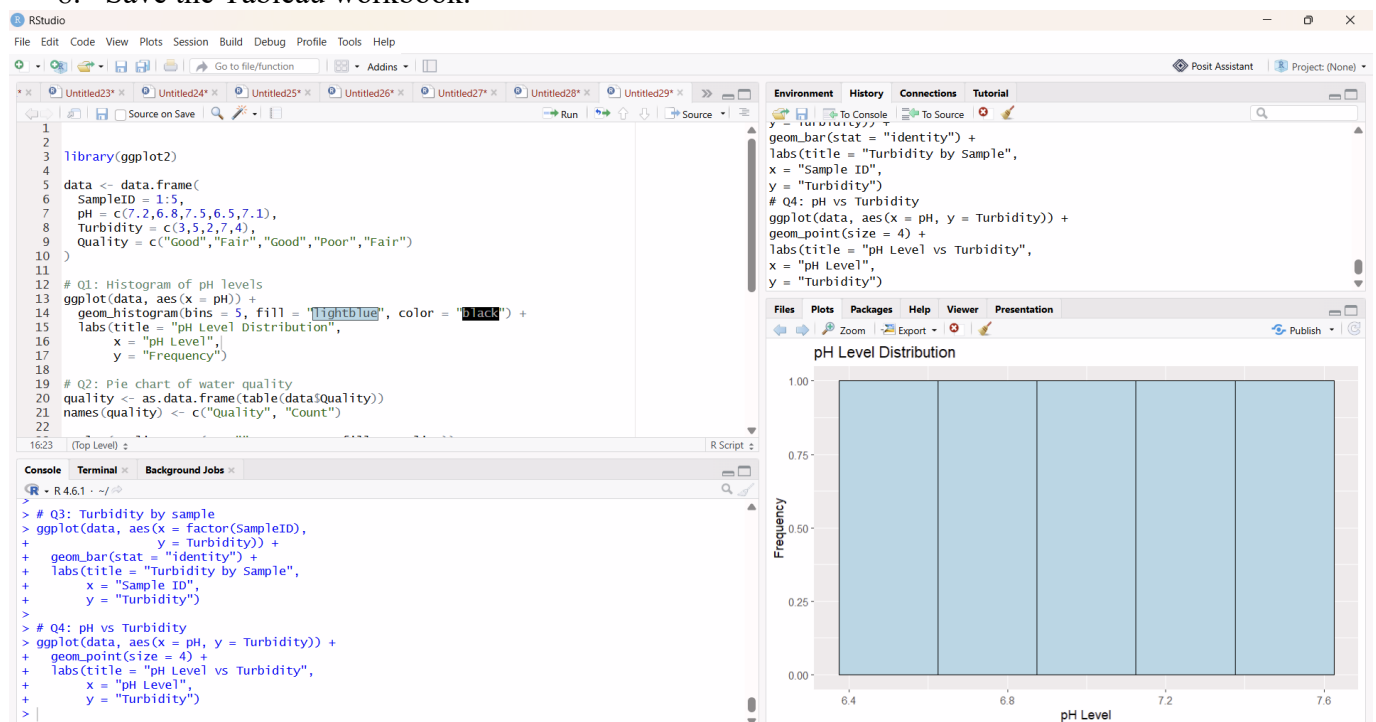
1. Create a histogram of pH levels.
2. Generate a pie chart of water quality categories.
3. Build a bar chart of turbidity by sample.
Create a scatter plot of pH level vs turbidity

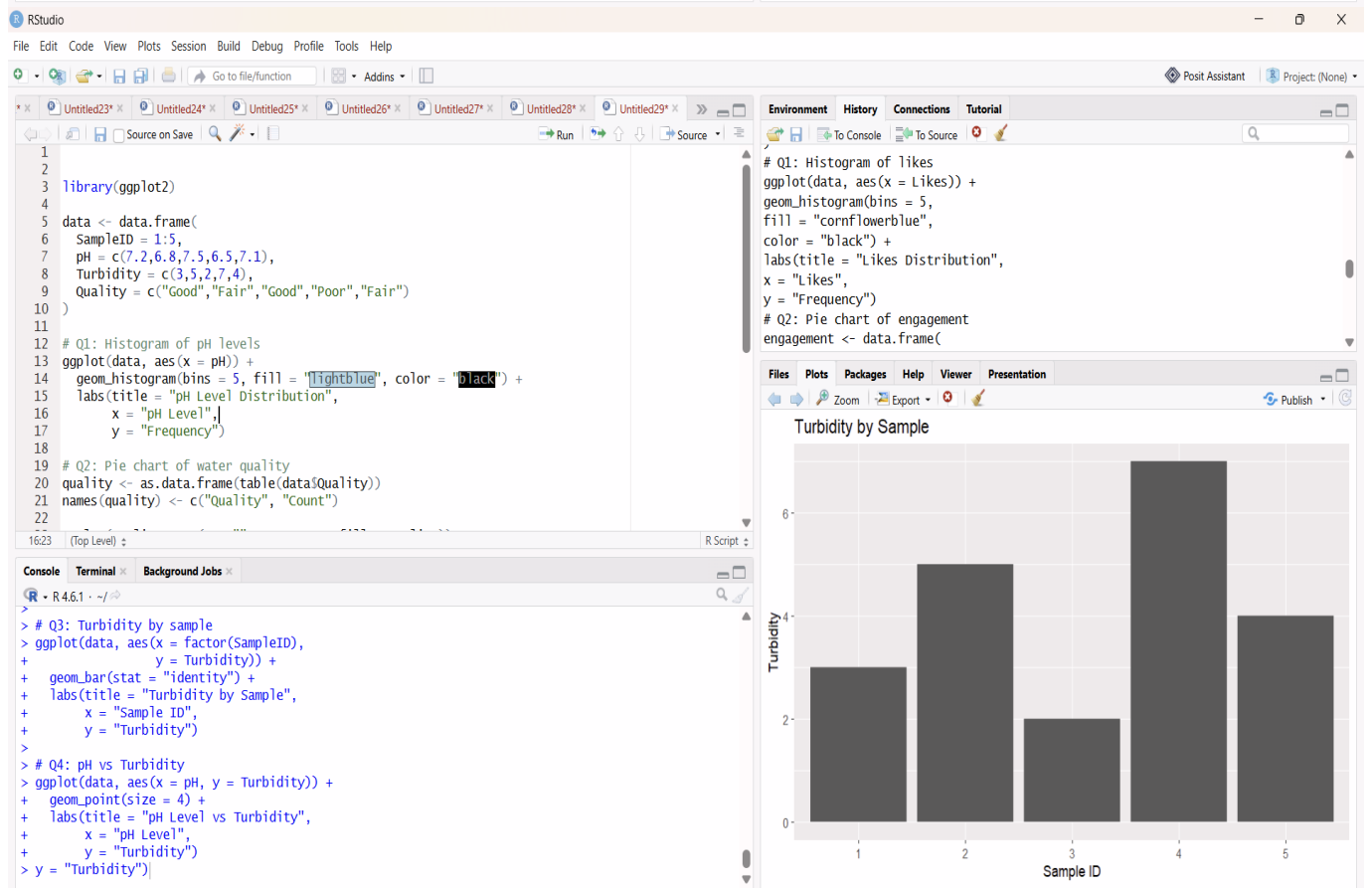
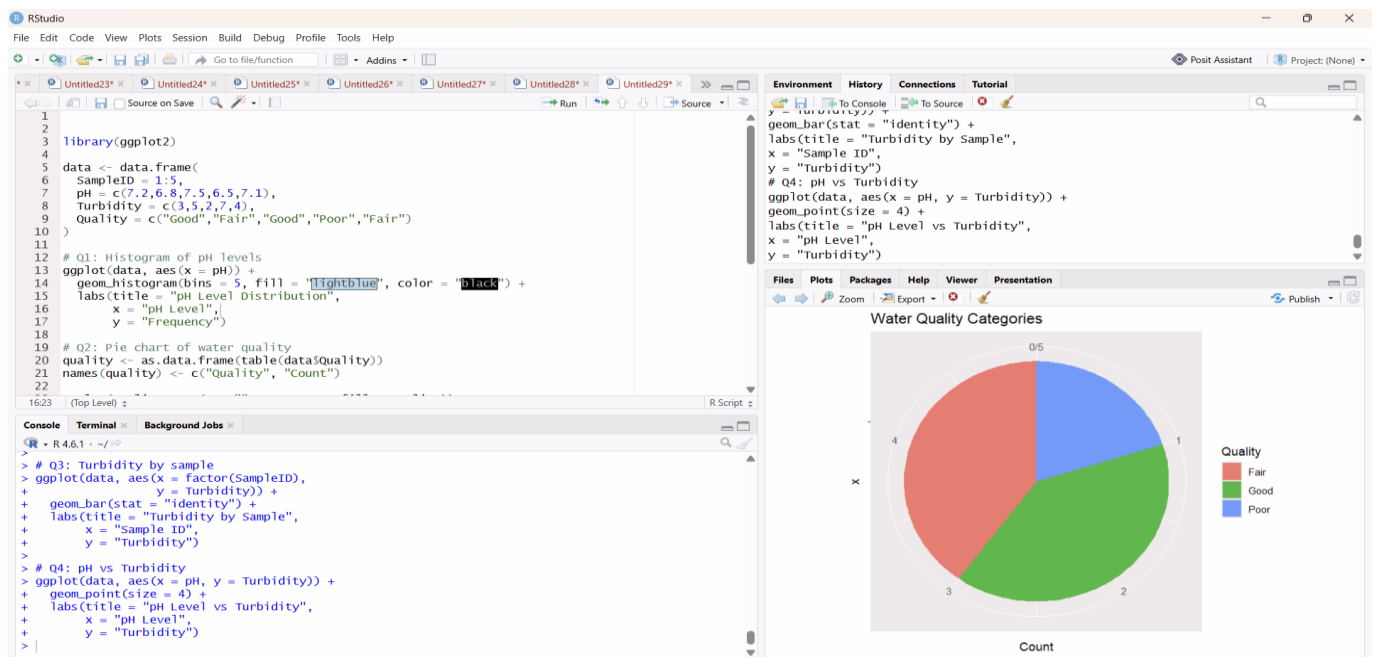
Aim:

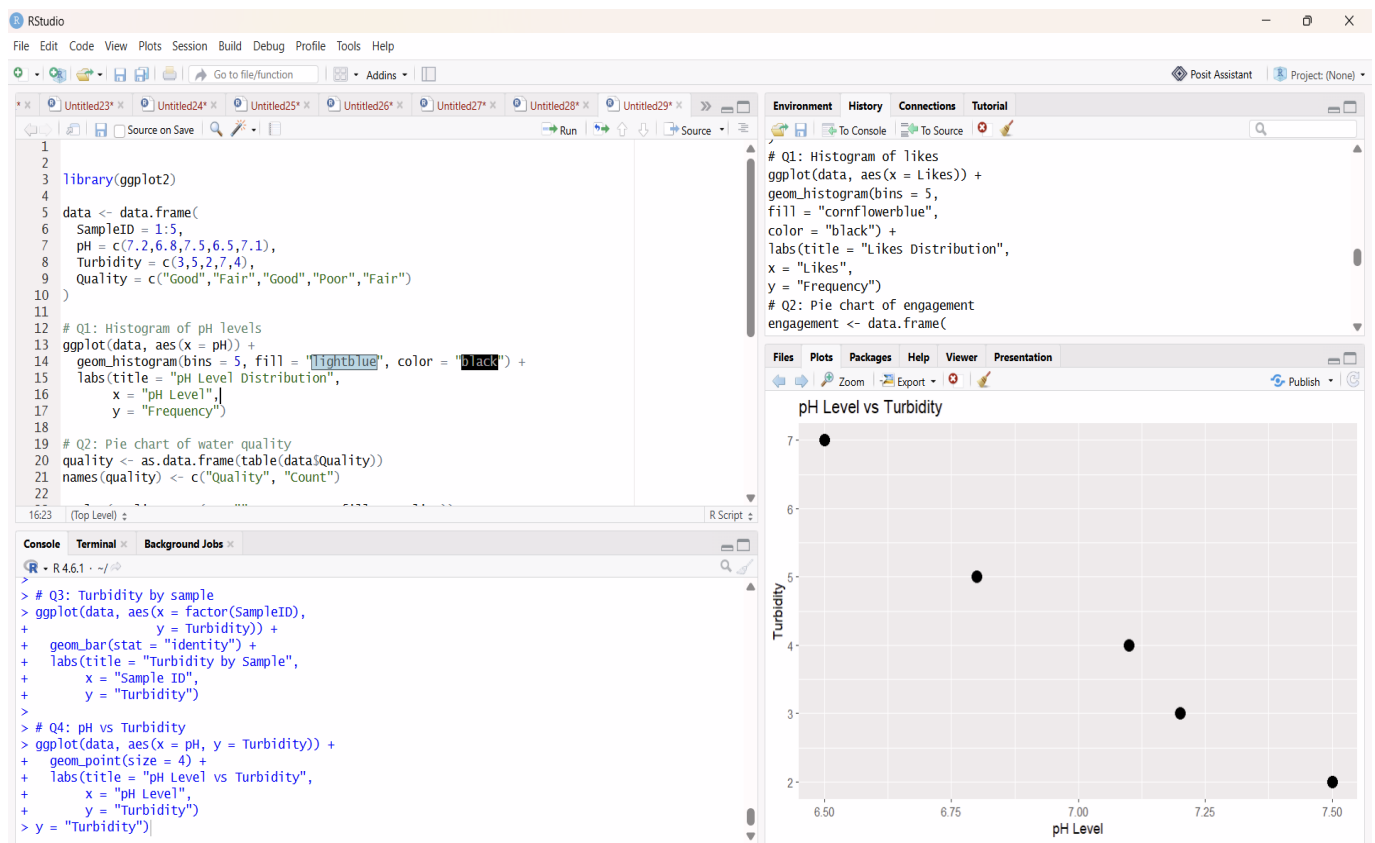
To analyze water quality data using Tableau by creating visualizations for pH levels, water quality categories, and turbidity.

Algorithm:

1. Import the Water Samples dataset into Tableau.
2. Verify the imported data.
3. Create a histogram of pH levels.
4. Generate a pie chart representing water quality categories.
5. Build a bar chart showing turbidity by sample.
6. Create a scatter plot between pH level and turbidity.
7. Add titles, labels, legends, and formatting.
8. Save the Tableau workbook.







Result:

The visualizations successfully represented water quality distribution, turbidity levels, and the relationship between pH level and turbidity.

SET30.MusicStreamingAnalysis

Dataset:MusicStreaming

SongID	Duration(Min)	Streams(000s)	Genre
1	3.5	150	Pop
2	4.2	200	Rock
3	3.8	180	Pop
4	5.0	250	Hip-Hop
5	4.1	170	Rock

Questions

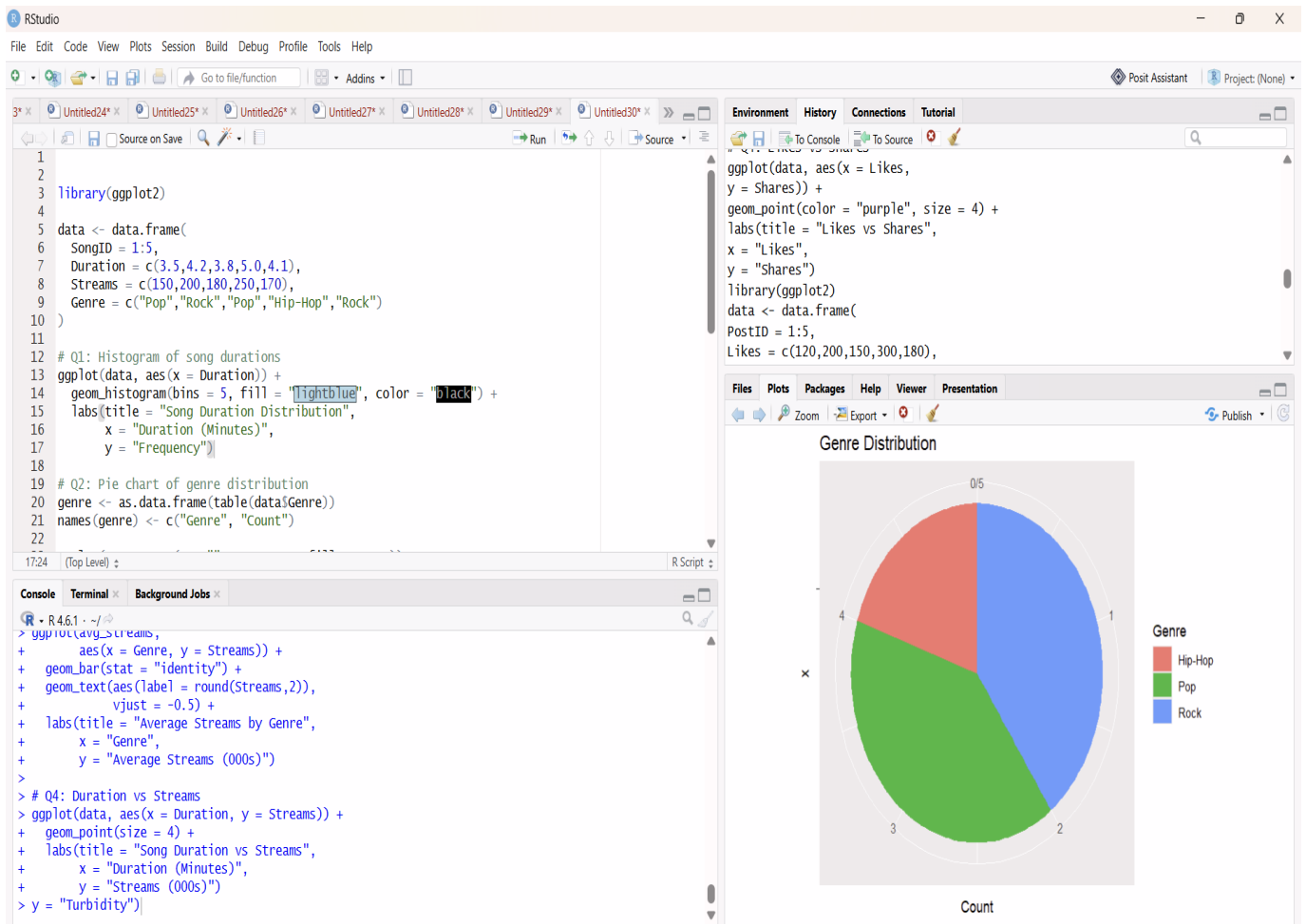
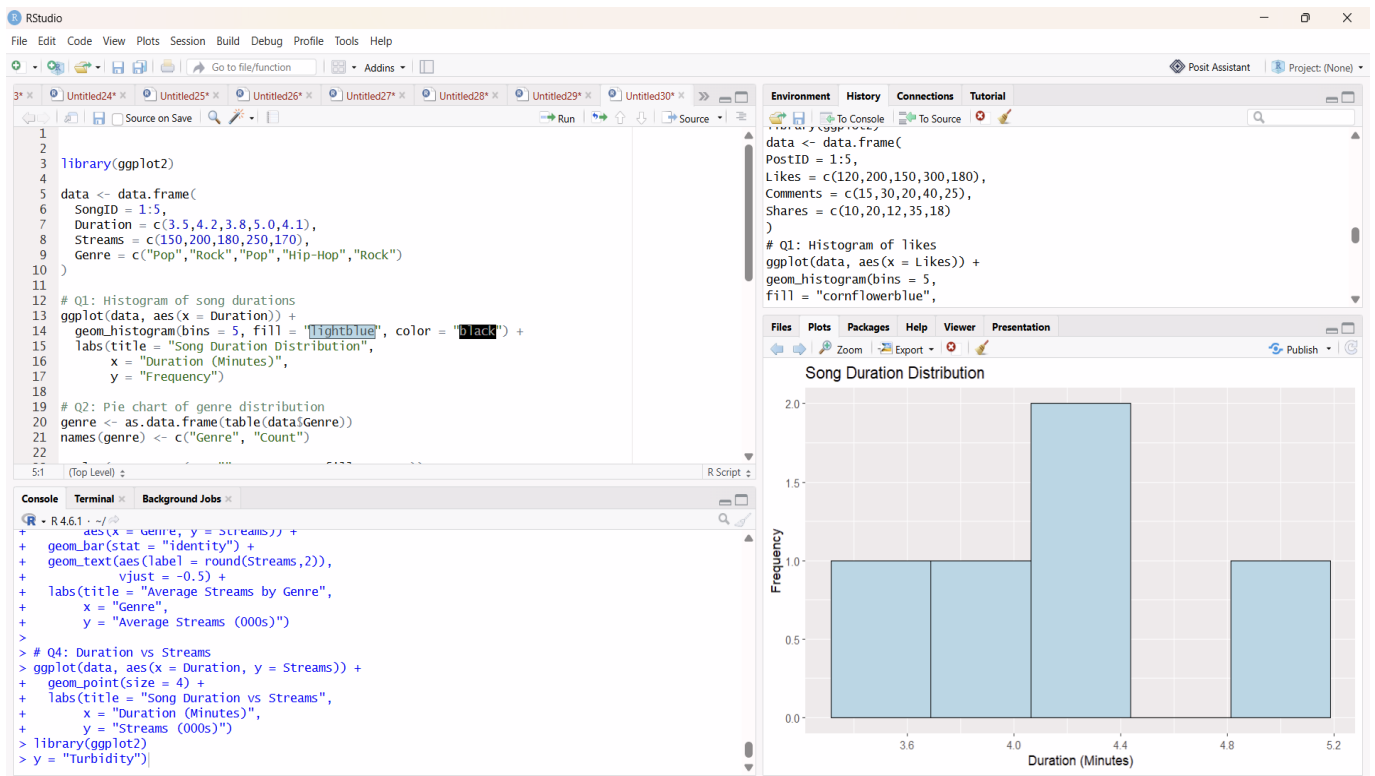
1. Createahistogramofsongdurations.
2. Generateapiechartofgenredistribution.
3. Buildabarchartofaveragestreamsbygenre.
4. Createascatterplotofdurationvsstreams.

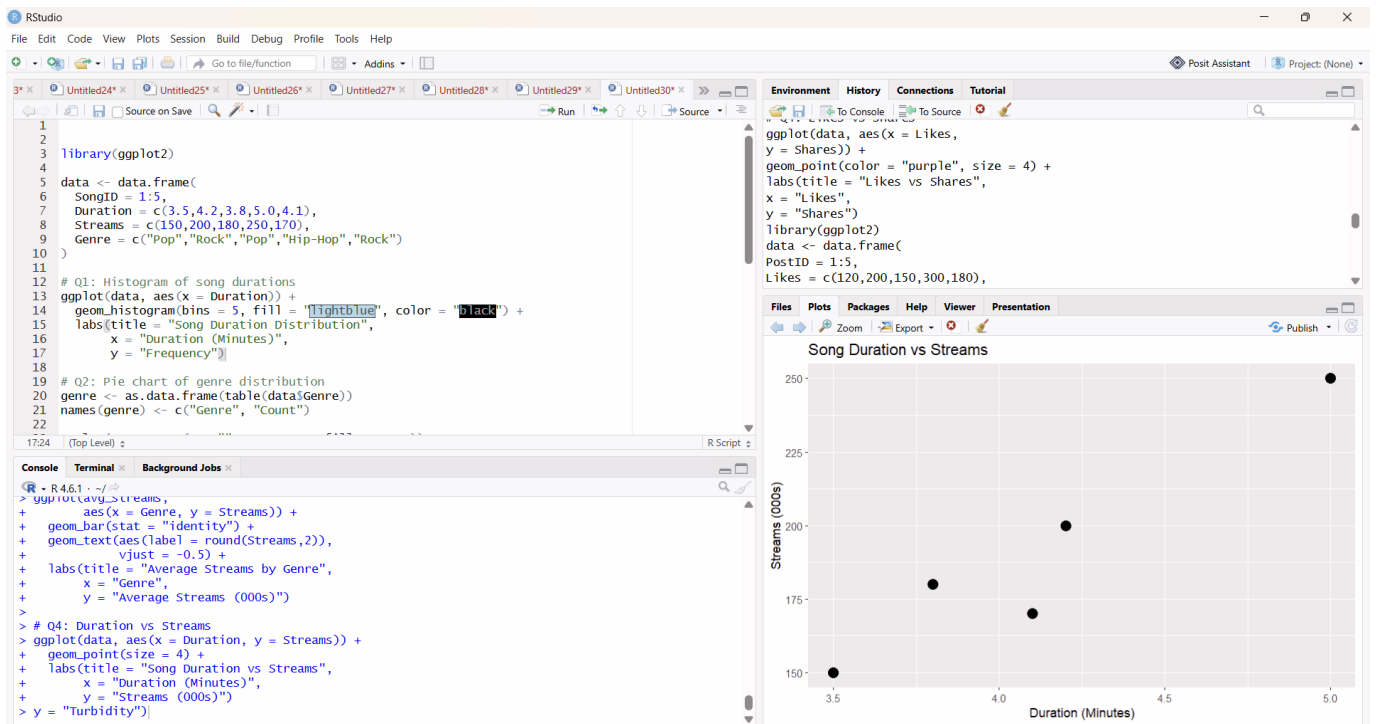
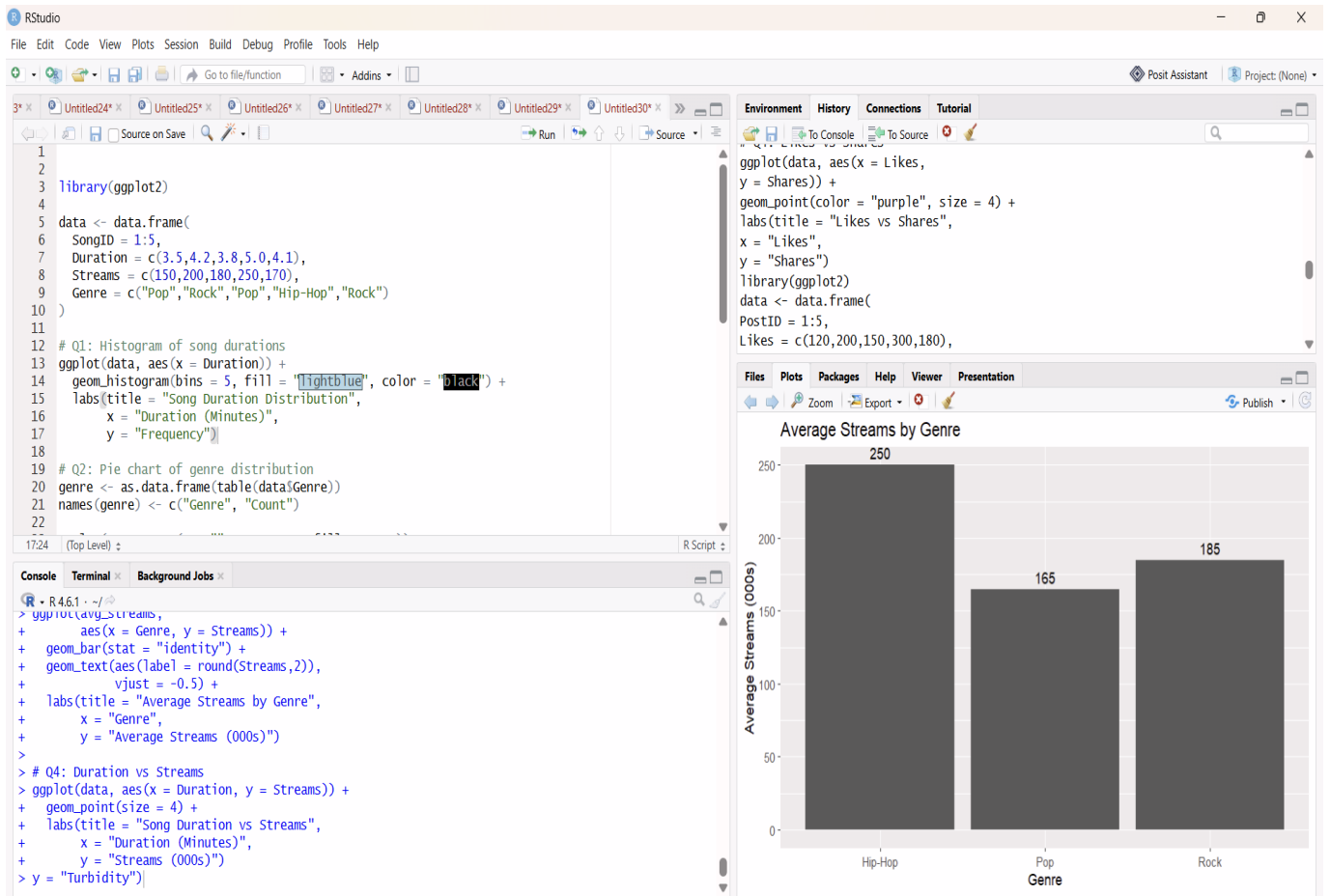
Aim:

To analyze music streaming data using Tableau by creating visualizations for song duration, genre distribution, and streaming performance.

Algorithm:

1. Open Tableau and import the Music Streaming dataset.
2. Verify the imported data fields.
3. Create a histogram showing song durations.
4. Generate a pie chart representing genre distribution.
5. Build a bar chart displaying average streams by genre.
6. Create a scatter plot between song duration and streams.
7. Add suitable titles, labels, legends, and formatting.
8. Save the Tableau workbook.





Result:

The required visualizations were successfully created, providing insights into song duration, genre popularity, average streams by genre, and the relationship between song duration and streaming count

